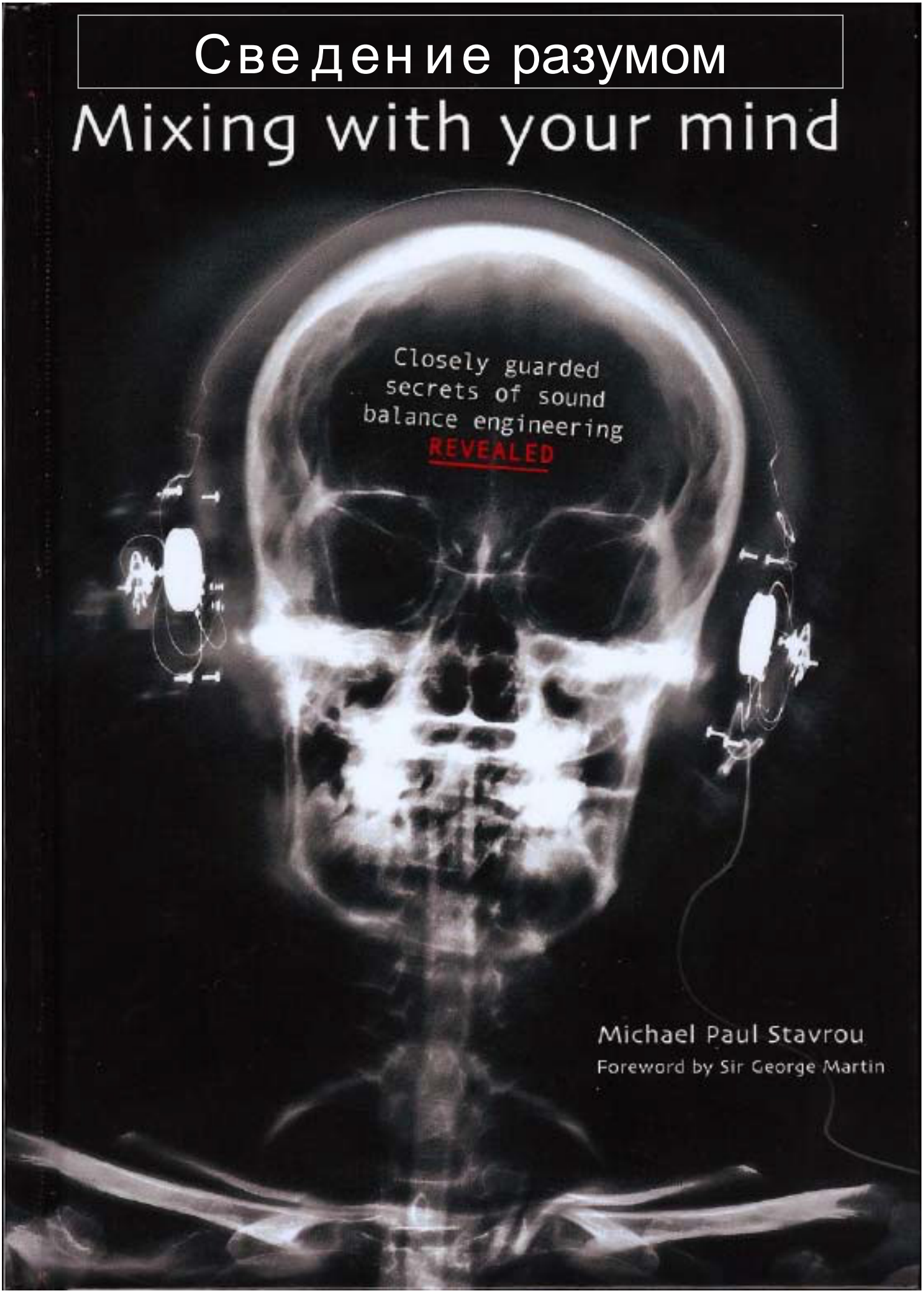
Сведение разумом



FOREWORD

Mike Stavrou first met at work a bunch of years ago in London, in the old Air Studios located on Oxford Circus. He was not one of those extraordinary breed of extroverted recording engineers that are always in demand in busy studios, and I was surprised by his calm and measured manner of work. Later, as I got to know him better, I began to understand his approach to recording and mixing. Mike thinks deeply about the music itself and the technologies that we use to create it. He was never afraid to experiment and helped me on a lot of sessions, achieving great sound without my help. Over time, he built up an impressive list of clients who, like me, appreciated his experience in creating great sound, both during the recording process and in the subsequent mixing. Ask the greatest classical guitarist John Williams who he would choose as a recording engineer and he would choose Mike Stavrou without hesitation.

This book is a significant contribution to the study of the most important device in the recording studio - the human brain. Forget your compressors and gates and all that stuff that seems to fill every corner of the control room. Mike rightly focuses on the mindset. The use of thought is the main way to improve results, and it shows you how you can improve your own results, no matter what level you are at the moment. I wish I had this book when I started out in the paraffin and record years.

Sir George Martin

(Sir George Martin)

October 2003.

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INTRODUCTION

Best from the best

When I first saw a professional sound engineer, I was completely shocked and mesmerized. He did everything easily and naturally, but the sound was amazing.

Why are *thousands of* engineers struggling to get a decent sound while a couple of *dozen* are creating breathtaking sounds while doing pretty much nothing? I could not understand how they create the illusion that all this is so easy and simple. Where do they do what I don't and where don't they do what I do?

Later, I realized later, most of the work happens in the head, which in itself is excellent - changing your own thinking is much easier (and less expensive) than reconfiguring studio equipment!

Path inspiration

The following chapters contain a unique collection of approaches and mind games that increase the diversity of the engineer's work in everything we deal with - speakers, microphones, drums, guitars, voice, phono, compressors, digital processing, release, etc. This book is organized as a journey through your studio and your mind.

We'll start by "turning on" your ears by adjusting the control room monitors. As you go through the steps to improve *the membrane effect* , you will understand how to present a mix with increased clarity and with much deeper and clearer space.

Next, we'll move straight to *advanced microphone techniques* . There, we'll see how a simple three-step mic positioning can help capture a richer, fuller sound from any source.

Next we will go to the sounds of the drums, which will be the first sounds that you will record - the basis of your rhythm track. We explore a very unusual approach that never fails, impressing drummers with its realism and gore.

While we are on the subject of phonograms, we are exploring *electric guitars* - how to record the good ones and recycle the bad ones. You'll learn to cruise through the endless textures of guitar sound without even touching the EQ.

Next, we move on to some quantitatively intuitive technique for recording phono. Sometimes one mic in the right place sounds better than two or three where everyone puts them all. On a case-by-case basis, we'll find out *why this microphone placement "works"* and then you can continue to develop yourself.

A good lead vocal can make a record sound more "imposing" and there are some little-known tricks here that you won't find anywhere else, I promise you. Every from them extremely simple V implementation. You Just obliged their

try - after this You be surprised How lived without this earlier. I Not I tend to look smug, but I've learned a thing or two in 20 years.

Now, perhaps, we will experiment with compressors. Even since some secrets became available to me, I still had doubts about what I could get the most out of the compressors.

We are now ready to dig into the real, still debatable, difference between analog and digital recording. Imagine how you would diversify your work if your film camera had zero dimensions!

After some excursion into the signals, we are ready to relax and let the mind deal with it all. We will begin this path by drawing an analogy between mixing and meditation. The real mixing starts after your "thinking" is turned off. How to achieve such a state? Read more…

We will go through three chapters *on the art of mixing.* We will take a look at all the tasks, go through them from one to another and transform them for perception by the mind. This will give us the ability to make the most complex mixes easily imaginable, and every right step will become easier.

For those who "demand more", the final section of the book explores other advanced approaches, like "like create reference CD, which will turn *you* into the most powerful diagnostic tool in town."

Completing the setup of monitor systems earlier, we give ourselves a chance more precisely - way to great sounds.

What follows are some advanced techniques for coping with the most powerful and most dynamic vocals on earth, plus some tips to help you you record a bunch of voices without headphones. It's a minefield of potential problems, but it's fun to see how a simple idea can turn a problem into a great outcome.

Now that you've got a taste for these oddities, you're probably ready for some background mixing. It's like hiring someone with more ability to mix your mix for free.

And even when you thought you already knew all this, we will reveal the exciting approach that has been lying in front of you all this time - the *time machine effect* shown to me by Cat Stevens (Cat Stevens).

And even when you thought you had control over all the forces that surround you, we will discover the biggest of them - *Gravity.* Does gravity really affect sound, and if so, can we somehow use it? You should definitely know about *gravity and its effect on music.*

Because I fully support your obsession with great sound for the rest of your days, here are a couple of tips on taking care of your ears to keep them healthy and strong :).

Now, if you still have a burning desire and do not consider homework to be any serious, there is one more difficult task that I do not I can hide, and which could potentially hold a common denominator under hit records. Sounds too good, I get it, and it's not easy, but it's worth a try if you're into really hit records.

Then one nostalgic chapter - how and why I moved to London. You can philosophize about any place you like to work.

AND, under end, compilation questions And answers from my personal email .

Thank you very much for ordering this book. Just remember that there will always be more of what we don't know than what we know. I really hope that the secrets revealed in this book will equip you on the way to creating your own great sounds, and your own secrets. Feel free to send them to me (only for verification, of course).

More Thank you, Mike Stavrou.

LET'S START WITH EMPTY ROOMS AND OPEN MIND

(And one speaker...)

Save distance

Loudspeakers - what is it? Just a piece of cardboard waving in a box, or a secret passage to another dimension? How you perceive your monitors determines what they give you.

In my very early days I worked at Air Studios in London and spent a lot of time in the main control room. Despite the fact that a huge number of hits were mixed in this room, the monitors did not impress me. The violins sounded wiry, the vocals were colored, the guitars and drums sounded flat and timid.

I once had the privilege of assisting the great Robin Geoffrey Cable on mixing. Although I was doing nothing more than rearranging the equipment, picking up sounds and careful observation made it daytime, which I am not forget never. IN process intelligence Robin put forward some drum faders, and I heard the normal sound of the monitors. They played for a couple of minutes, and the sound from the monitors slowly faded away. He then turned on solo acoustic guitar. The instrument was very clean, But painted sound monitors. Soon, How And previously, sound monitors gone to nowhere, leaving only a beautiful full-bodied guitar sound, floating in the air, taking its own place - I never even dreamed of anything like it. This wonderful day clouded my mind more and more. By the end of the evening, the monitors had evaporated, having been replaced by Robin's imagination and, like Dali's paintings, veritable marvels of acoustic physics.

From that day on, I stopped complaining about these monitors. I realized that the weak link was not the monitors, but my obsession with their shortcomings. You must learn to listen *through* the monitors, not themselves.

One way to help monitors open up is to improve the soundstage around them. To do this, we will have to learn the art of positioning monitors.

Let's start With empty rooms

Imagine the picture: you have a new studio space and the equipment has already arrived. Your instinctive reaction will be to start furnishing equipment like the interior of an apartment.

Beware: your first task is not to place the equipment where it will look best. Ergonomics is important, of course, but arranging equipment according to ergonomics will disturb the acoustic picture - although the room will look neat from the outside.

The most important component we need to position is invisible. It does not take up space, and at the same time takes up all the space. Of course, we are talking about the Sound Field. It is very important for your monitors to work in the sound field, because if the sound monitors fights off extraneous sound in the room, you will not be satisfied with the acoustics - in fact, this means that you will spend a disproportionate amount of money to correct it. Let's place the Sound in the room first, and then everything else around it!

So before you start setting up the equipment, play a little game with a friend. While the control room is empty, ask a friend to hold the monitor while maintaining a distance (2 or 3 meters) from you. To make it easier to keep your distance, pull the string or signal cable from the monitor (or the power cable for the active monitor).

The monitor can be of any size - mobility is more important here than the clarity of the response. The monitor does not have to be the same one that you plan to use in your work. you are listening to the room and how it interacts with the space around and around you. In other words, we are studying the sound response that is created by the monitor in the room, and whether you drop a healthy stone or a 20-carat diamond on the floor, the sound response will be the same.

We listen (IN searching optimum)

Now that you and your friend are connected by a constant distance, and he or she is slowly moving around you (while you are standing in the center of the room), what exactly do you need to hear? How the room maintains the sound field.

As the monitor moves around you, you will most likely notice more unevenness at lower frequencies than at others. Most studios trouble keeping bass "in phase" with room response. Determine by ear the positions where you feel a confident push from the bottom - where the bass of the monitors *connects* with the echo of the room (rest of the sound). Woo correct bass, and the rest of the sound will be place. Find positions where you get the feeling that the sound is fully developed (sound in fool bloom). And other places where the music seems to sound closer. Avoid places where the sound seems to disappear before it reaches you. Also avoid places where the sound becomes more "stereo" - we use only one speaker, and excessive stereo effect due to uncontrolled reflections, which is not good.

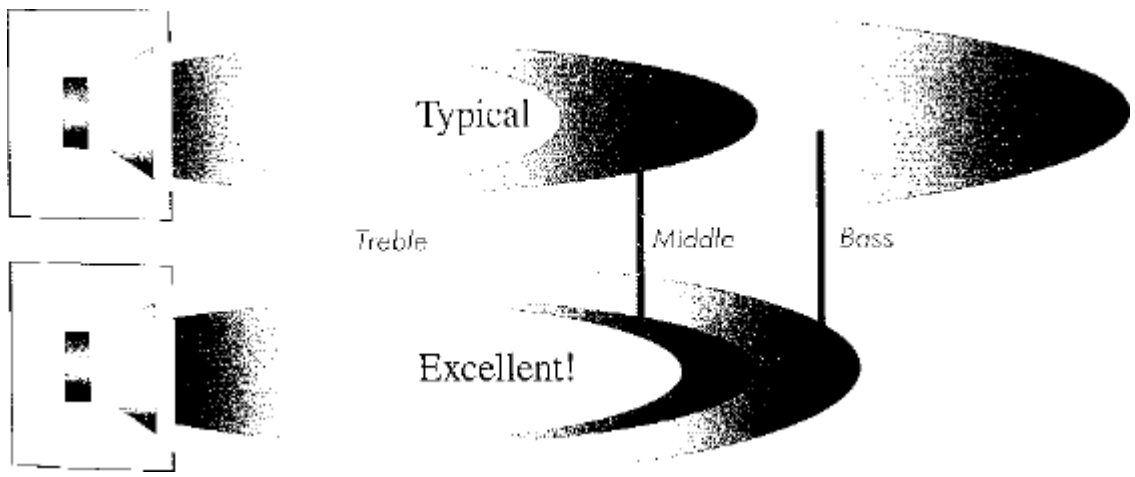
When your friend (who is already worthy of a decent reward) has done a few laps, you should already have noticed potentially good positions. Now place the monitors on the stands at the marked points. The best position is where the monitor sounds "closer" while being farther away - giving a solid, round sound.

An important lesson from this experiment is that differences heard cannot be recreated or corrected by equalizing monitor systems. **Remember,** if you compare the flow of sound energy in your room to the flow of a river, EQ will not change its direction, only its color.

Let's turn sound V flame (concept "Flames")

Every source of sound wants to be a Flame. The base of the Flame is near the speaker, and it shoots out towards you in an intense and focused way. Loud sound will be directly in front of you and possibly even be focused behind you, while soft sound may be focused at arm's length in front of you (this is one of the reasons I made you stay a constant distance from the monitor in the sound test) .

To feel the Flame, listen to the entire recording and move back and forth. You will feel that the focus of the overall sound changes with distance from the source. When you are too far away, it is blurry. When you're too close, even if the sound louder He determined Not So clearly, How V special point, which I call
"Tip of the Flame". As you move back and forth, you will also find that certain frequencies (or instruments) focus at different distances from the source (see Figure 1).



Rice. 1 - Not only myself source creates Flame, But And each separate frequency. Your head Not Maybe be in two places at the same time, so the closer the tips of the flames of different frequencies, the better.

This is an extremely important aspect. For example, usually high frequencies are focused at a shorter distance than low frequencies. In an ideal world, all frequencies are focused at the same distance from the speaker, but in reality this is never the case. No matter how much you then equalize your monitor system with high-tech equalizers, this relative difference between focal points and distances will not change - these are properties of room reflections and speaker geometry. Equalization can change the intensity of frequencies, but where they converge remains the same.

The results of this simple experiment - moving the monitor around the room - cannot be obtained with traditional measuring equipment. Pink noise and calibration microphones cannot measure the haze and resonances that your ears pick up so easily. In fact, your ears are much more sensitive to the by-products of the Flame than expensive and fancy equipment. So use your ears. Test equipment is well suited for eliminating acoustic problems And detection specific mistakes, But your ears

perceive texture, color, size, width, depth and magic - so drag your speaker around in a room and look for depth, integrity, richness, resonance.

REBUILDING MONITORS RELATED TO THE CENTER

You have already found the optimal position of your monitors, now you can arrange other equipment as you like. The next most important thing is to determine the distance *between* the monitors.

Not show to me money

There comes a point in everyone's life when they want to upgrade their monitors. Before investing in new speakers, you should get as close as possible to the best sounding monitors you already have to understand what they are really capable of. Try this simple trick and judge for yourself how much better your speakers will sound.

moving on from rough To accurate.

When creating a three-dimensional sculpture, you must start with rough shapes, gradually More and more clarifying the details. Monitor Tuning - That same most. Example coarse tuning is locating your speaker in the room (Chapter 1), and fine tuning is adjusting the angle of the tweeter. Somewhere in between is the often overlooked choice of monitor spacing.

I have noticed that many "experts", including some respected speakers, don't realize the importance of speaker spacing. However, a fact is a fact: if your monitors are too far apart, it will deprive you of panning options. Let me explain what's going on. As you know, the acoustic picture, located in the middle between your monitors, is a phantom, an illusion, a product of phase and amplitude matches between left and right monitor. But when sound is emitted from one source, the picture is always clearer - more tangible than ghostly.

Effect membranes

The distance between the speakers affects a lot. When the speakers are moved apart, the imaginary picture between them becomes more stretched. And more importantly, the patterns that should appear between the center and the edges - center-left and center-right - are strongly influenced and become unstable. The ideal distance between monitors is one that creates a uniform density throughout the stereo sound image. Let's take a look at the sound when we move the monitors apart.

Imagine a membrane that is stretched between monitors and forms a perceptual plane onto which sounds are projected.

Fig 1. When the monitors are too close, the imaginary membrane forms a thickening and the energy is V center stereo pictures. sounds, placed V center, will be louder how They on himself actually there is.

When the monitors are too close, the membrane bulges in the center. This bulge is immediately identified by excessive coloration (Figure 1). The test equipment does not recognize this problem.

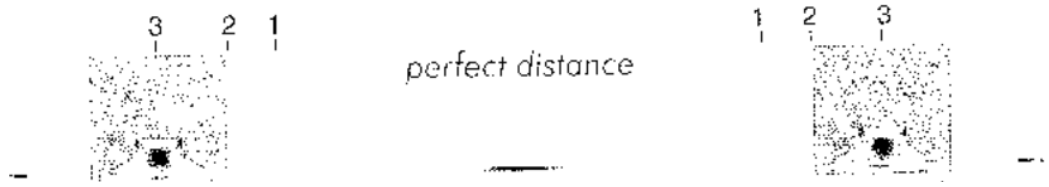
Rice 2. When monitors too much far Friend from friend, membrane will start stretch And sounds panned to the center are attenuated, causing a lack of coherence and presence.

When the monitors are too far apart, the membrane will begin to straighten out and the color will disappear. But with further expansion, the membrane will begin to stretch, the center will thin out, and the sound will lose depth and integrity (Fig. 2). Continuing to push them further apart will completely destroy the sound picture, to the point that you will get the feeling that you are at some kind of party with special surround effects. The ideal position is when the membrane is absolutely straight, without stretch marks, bulges or holes (Fig. 3).

Rice 3. When monitors on correct distance, imaginary membrane straight, And will have place a clear acoustic picture, regardless of sound panning.

Let's figure it out V physics (Practice settings)

Now that we know what we're trying to hear, here's the best way to choose the right monitor placement. The easiest way to move the monitors is by placing them on a board long enough to move them within the desired range. Two people stand at opposite ends of the board, moving their monitors towards each other until they touch. Now, at your command, we begin to pull them apart (leaving the frontal planes parallel) and move them all the way. In the process, you will hear a huge difference in sound. Have your assistants move the monitors in and out while you listen. Remember the location when the sound seems interesting to you. The next time they approach these positions, ask their stay And mark on blackboard the edges monitors. Designate these labels
"No. 1". Perhaps this is the place where the color disappears, or where the picture softens. When you have similarly marked the inner and outer limits of the monitors' movement, ask the assistants to move the monitors within these limits (Fig. 4). I do 4 or 5 marks maximum.



too narrow 3 good marks too prominent and colored 3 good marks too narrow

Fig. 4 - Location of zones on the board. Normal and bad zones.

When you've narrowed down the locations to two options, step back towards the back of the room and walk around in that area.

Use mono signal

You will be able to hear the membrane effect is much better if you feed a mono signal to the monitors. The stereo signal is too confusing - you're looking for breaks in the membrane, and the stereo mix already contains them on its own. Use a mono copy of a wide-range stereo mix that contains a lot of sustain textures - soft, not too percussive, multi-textured, with a wide range.

You discover one position, creating the best sound than rest. move your head from sides V side, back And forward. Painting will be constant.

Deprives possibilities panning?

So why is this solid membrane of sound so important? If your main monitor system It has gap V middle stereo paintings, she practically useless for information. If you try to mix with monitors too far away, which sometimes seems normal, your mix will contain too many instruments. V center. Why? Because What They will weaker bug, V

compared to those shifted to the left or right of the panorama, and when you notice this, you will raise their level to compensate for this. This mistake is catastrophic and destructive to the mix. That's why.

As you know, the best recordings have excellent mono compatibility. When you check the stereo mix in mono, you will find that the sounds located in the center will become louder than those located "side". That's why experienced engineers try not to pull out too much of the centered instruments when mixing in stereo. If you raise the level of the instruments in the center simply because your monitors are not coherent, you will do exactly the opposite of what you should have done! When switched to mono, sounds panned to the center become 3dB louder than those panned to the left or right. So, in order to keep the intended idea of the mix, you should reduce the number of sounds located in the center, and not increase them.

Most expensive measuring equipment on earth

Many books on monitor positioning refer to studies done in anechoic chambers and ideal listening conditions with calibrated microphones, spectrum analyzers, and other laboratory instruments. Luckily, you already have a more sensitive measuring device: your ears. Don't underestimate your ears, they are great "engineering" tools and don't cost any money. In fact, no measuring device is as sensitive to sound coloration as the ears. They are more sensitive to color than level or frequency. So if your ears are telling you something is wrong, double-check.

Confidence generates inspiration

I've spent years "feeling" loudspeakers and not knowing exactly what I'm doing. They were basic cause my frustrations And I ashamed recognize What usually blamed them for their failures. Ever since I discovered this welcome, I stopped blaming the monitors, because. this process allows them to produce sound as accurately as possible. As you take these steps, not only do you get closer to flawless sound, you (which perhaps more important) become more confident in your speakers. When you have confidence in your acoustic equipment, you create with fewer restrictions... Once you become "unlimited", you will be free to create creative mix.

For greater information O location loudspeakers cm. chapter 13 -
Door V sky. Create V their monitors door, leading V other measurements.

Method equilateral triangle.

We've all heard that the "ideal" speaker placement is one in which equal lengths of the sides of the triangle between the speakers are observed and engineer. This is perhaps the fastest and least problematic approach to equipment layout that can be done. The only problem is that this approach is not completely takes into account the reflections of the room. You probably be as surprised as And I, knowing What 80% sound comes from reflections from rooms, And only 20% -

directly from emitters. The perfect triangle method can only be correct in an anechoic chamber or in rooms with perfect acoustics, but it doesn't work in the real world. My suggestions?

Listen to the membrane. If at the end of the above techniques you get an equilateral triangle - congratulate yourself. But if not, then you can open a bottle of champagne for successfully compensating for the acoustics of your room with the placement of your monitors!

Advanced microphone technique

It would be nice to know right away how to remove any tool that is brought in the door? It's actually not that hard if you know where to start. Assuming you have already found a good place to place the instrument, you must choose the right microphone and diagram, find the Right Piece of Air, catch the Flame (the most interesting part), and if there are more than two instruments, turn it into an ambiance. Rigidity/Angle/Distance.

Choice microphone

I want you to forget about frequency response, distortion characteristics, and noise thresholds. Technical data will not help us here. We have a sensitive musician with an instrument. Be nice to the musician in every way, but in the meantime, listen super-inttently to the instrument. How "violent" is the sound? I don't mean "loud" or "bright" or "thick" or "pleasant" but, in terms of texture, try to appreciate "tough". You have to be there in the room listening to the instrument in order to do this. The bells can be stiffer than a Marshall amp and vice versa. Every sound has a harshness factor. It's not that hard to learn what I mean, but you'll have to practice. Once you catch it, you won't forget it.

Factor rigidity

Identify the instrument on a scale of hardness from H1 to H10, where H1 is the softest sound and H10 is the hardest. Microphones also have different rigidity. Like instruments, I mentally place microphones on the hardness scale. Neumann U87 is stiffer than U47 and Shure SM7 is tougher than Sennheiser 421 which is tougher than Neumann U87. The rainbow of microphone stiffness does not change. It is stable, like in the periodic table, you just need to open its elements. But memorizing the harshness of microphones is not very good, you need to hear it. Only then will you know for sure. When it comes time to choose the right microphone, start with the opposite principle. Once you have determined the hardness of the instrument, select the opposite microphone. rigidity. Soft sound H1 corresponds Very hard microphone with H10, and sound H2 - microphone H9.

Dot reference

Some studios have an impressive fleet of microphones. You, of course, do not there will be an opportunity to determine their sound by their appearance, and there will be no time to test every microphone on all the instruments that you will record. Solution? Before the session, place all microphones on the table in the control room. Plug in your microphone and headphones. Listen to the microphones, one by one, while you count loudly from 1 to 10. Mark each with a hardness icon based on the sound of your voice on the tape. Such an unusual exercise will save you a lot of time on mixing microphones later. **Side note:** my favorite number in this test is 4 (“four”), which is a nice meaty sound. (Over the course of a session, you'll soon be picking mics with the confidence and speed of a golf pro. "Here, play that one more time... I think it sounds like an H6... give me a fourth piece of iron!") .

Diagram focus

After choosing a microphone, I would like to say a couple of things about the directivity. First, the apparent distance between the microphone and the instrument varies with performance. An instrument will always sound more distant when written in a pie chart than in a cardioid, even if it stays in the same place. Just don't forget about it. It is for this reason that orchestral recordings from omni-mics mounted at the conductor's head usually sound so "far away". Secondly, do not forget about the features of figure-eight microphones. The dead zone of the characteristic of such a microphone is more dip than the back of any microphone with a cardioid. Need to record voice and acoustic guitar at the same time with a good mutual separation? Try use two microphone With
"eight". The dead part of the figure-of-eight characteristic is the wide rejection area displayed on the diagram. Direct the "dead" part of the diagram to the unwanted source. Mentally extend the plane of the diagram to the source of the unwanted sound and further, ad infinitum. Keeping this connection, twist and move the microphone in search of the best position of the sound source from the “live side”. Repeat this for the second microphone as well. Once you find two Pieces of Air for the live sides of the mics, the separation is great.

Rice 1. - Using two microphone With diagram focus "eight" write down vocals And guitar with almost perfect mutual isolation.

Correct Piece Air (Corner)

Finding the right piece of air is the meaning of everything. Remember, you are writing a piece of air the size of a nickel. It's like you're using a pipette instead of camera. The difference is huge. The lens collects light from Mars, but the pipette collects only what it comes in contact with. Understanding this is what separates a sound engineer from a filmmaker. A microphone is not a camera. Just because a microphone can "see" the entire guitar doesn't mean it will capture the whole guitar sound. Not important, Where are you put the microphone on He write down only the combination of sound waves that occurs on the microphone capsule.

Turn on your headphones and walk around with the microphone in search of the right piece of air. Every inch makes a huge difference. You won't be able to do this by placing the microphone where it looks good. I will repeat again. You must listen. Ask the performer to be quiet and turn on the microphone so that it sounds in the headphones louder than the source outside. Ask the performer to play softly so you can feed the microphone into the headphones. The harmonic composition is weakly dependent on loudness. This technique will help you find *the angle* that best captures the tone of the instrument you like.

IN searching flame (distance)

How Only you found right piece sound, you need find one the very tip of the flame. To do this, you need to return the source to its previous volume and look for a flame. I imagine that every source of sound is a source of flame. The size, location, volume of the instrument determines where the flame will focus. At the Tip of the Flame, the sound is more cohesive, resonant, and solid. All frequencies emitted by the instrument are more in phase at the Flame Tip than they are around.

Recording the sound of one instrument is easy, but when its waves cross with others, the sound will scatter if you don't catch it at the focal point. EQ won't be able to fix the issue. He can only raise some frequencies in order to "drag" through others. But if you catch the sound at the Flame Tip, it stays consistent throughout the entire work. When you use EQ, the sound stays consistent and the equalizer may seem more "powerful".

Sound Also endure more quantity conflicts at overlay, And stay focused until the very end. Once you've got a sound that can stand up to overdubs, you can do whatever you want with it! You can determine how loud the instrument should be in the mix, instead of having the instrument dictate it to you. If you lose the Flame, the sound in the mix will disappear.

So where to start looking Flame? Imagine an endless line that is drawn from a microphone capsule, mounted in the correct piece of air (as described above), straight to the instrument. The tip of the flame is somewhere on this line. Find the Flame. Listen to the instrument at full volume. Find your line at any volume, find it first! At high volumes, this line can go several meters behind the microphone. You can move your head along this line looking for the Flame Tip. When you find it, put a microphone in there and just make sure What He installed under corner, determined on previous step.

Turn leaks V ambient.

What There is "leaks" And What There is "ambiance"? Ambient makes sound things "larger" And "closer", V That time How "leakage" does All on turnover, And depletes the texture.

Question With picking: What create more "leakage"?

A) Loud orchestra B)

Silent orchestra B)

doesn't matter

Answer: If everything sounds loud, and all microphones are at the same level, then there will be no leakage. Typically, leaks appear when one of the signals is too quiet, or vice versa - too loud.

Imagine a roaring horn section next door to the polite sounds of double bass. No matter what you do, you're going to get horn sounds leaking into the double bass mic. It's just a nightmare. You can try to hide the bass behind sound-absorbing panels, but then you will destroy the contact of the players in the orchestra. Remember, their work is more difficult than ours. We just have to record what they do. If our recording gizmos are preventing them from doing their job, then we'll just ruin everything.

So whether we like it or not, we have the bass three meters away from the trumpets, trombones and saxophones. And they only play well next to each other. And what will you do about leaks? Turn them into ambient. The double bass mic picks up the horn section, right? If you're really smart, you'll turn those unwanted echoes from brass to ambient. **Remember:** ambient makes sounds bigger and closer, while leakage does the opposite.

You need to redo the bass pickup. First, find a good spot for the horn section's ambient mic. Then put our double bass in there and tune it with a piece of air and flame. Choose a mic with the right stiffness for double bass and look for Flame. Use a cardioid, put the microphone on a stand to properly position it and tilt it towards the flame. Now find a piece of air that contains good horn ambience, preferably somewhere where the bassist feels comfortable playing. Move the double bass mic and stand to this position, but do not make any adjustments to the angle or height of the stand. Position the microphone so that the horn ambience enters from the back of the cardioid. Now move the double bass V such position, at which microphone again will start catch tip his flames. Don't move the mic, move the bass!

Since the horn ambience comes from the back of the cardioid, it will be suppressed. This gives you an excuse to add high frequencies to balance out the ambient, while also helping the double bass sound brighter and clearer. Or vice versa - the EQ you use to boost the brightness of the double bass also boosts the ambient of the horns, reducing their leakage. Thus, the problem turned into an advantage.

The double-bass track contains "leftovers" of horns. On its own, it sounds stupid, but if you take this track out of the mix, the horn sound will crumble. Raise the bass level, and the horns will sound lively and reverently. The bass mic has a high stiffness to compensate for the softness of the bass, and it's placed at the tip of its Flame so the bass won't get lost in the mix, even next to the horns.

This brass/bass example is actually from the "Come in spinner" album featuring Grace Kight and Vince Jones, on ABC records (838 984-2). We had three trumpets, a trombone, and three saxophonists (with one understudy on clarinet or baritone). The trumpets had a hard sound of their own (H6), so I used a Neumann TLM170 (H4) microphone. This is one of Neumann's softest yet cleanest microphones. The trombones sounded softer (H4) and I used a Sennheiser MD421 (H7). Because saxophones played in the upper register, they had a very hard color (H8), and it seemed to me that they needed a soft but full-bodied Neumann U47 (H3). For double bass (H3) I used a Neumann U87 (H6).

MURDEROUS SOUND DRUM

Working with drums is interesting and exciting. drums are quite complex. They always sound great when you're standing next to a drummer, but how do you get that brightness out of a three-inch speaker? Most engineers start with close mics and then add distance mics. There is a certain logic to this: first we get a cool kick sound, then a good snare sound, add toms and hi-hats ... OK, some sounds are good, now we will add overheads for volume and - here it is your sound. Right? Or have you noticed that when you add overheads, everything seems to move away? With the addition of microphones, in theory, the sound should gradually become larger and better. But why isn't this happening?

Unraveling the problem of drum sound can take years of your life. Drums are a fount of problems: phase, dynamics, decay, frequency, color, pitch and size problems - Don't you love this?

Wouldn't it be great if you only had to do a couple of things, sequentially, in order to achieve the big, fat result that you dream of? Well, here he is. Follow the instructions in this chapter to get the Killer Drum Sound.

Maximum illusions / Minimum voltage

Your task, as a sound engineer, is to get the maximum illusion with a minimum of stress. In other words, we want to create a deep picture, without reference to the signal level. Recording the powerful sound of drums is hidden in the psychoacoustics of their sound.

Reel Positioning (Get it right from the start)

Getting the sound of the drums is not the recording of the direct sound of the drums as such. Drums so loud that record is much more related to sound, that they create around them. Every room gives reflections, and working with those reflections is this is the first direction.

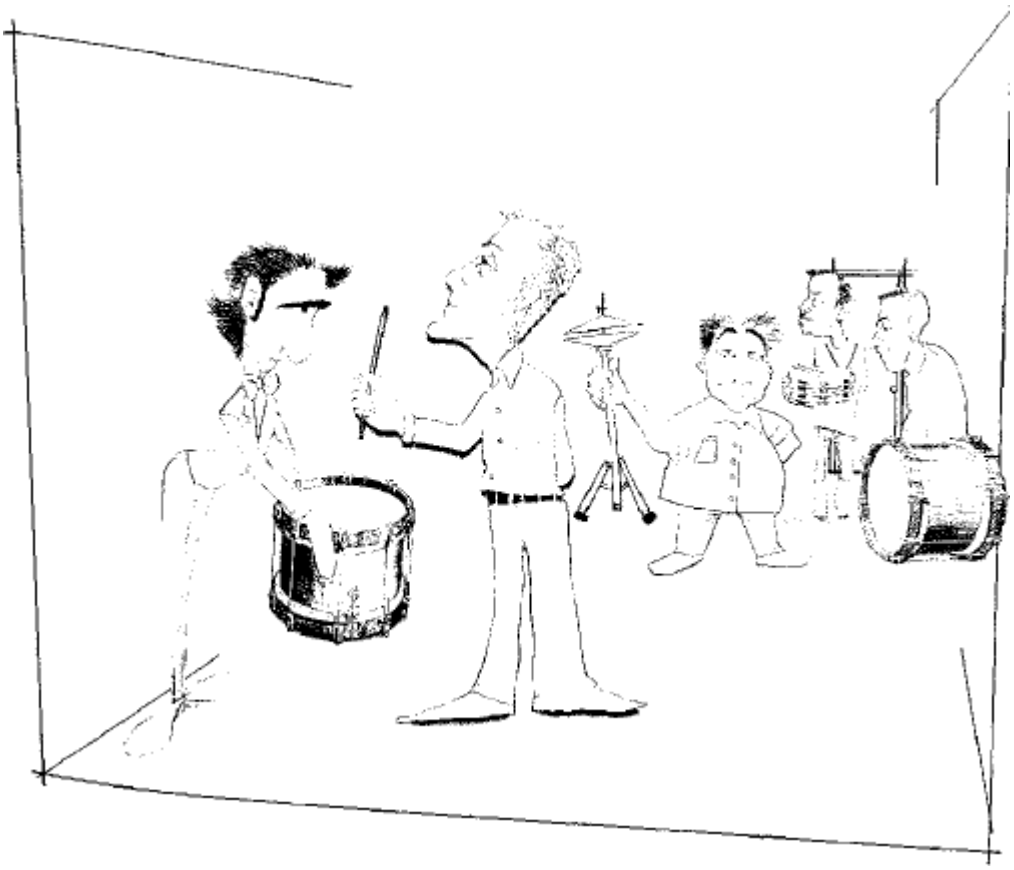
So before we pick up the mics and spend time honing the snare sound, we'll "hook" the drums into the room. And the drums join the room first of all ambient microphones. But How define, where to place overheads? The answer is hidden in the initial positioning of the drum kit inside the room.

The best way to do this is to first look at the instrument with the best and deepest resonance - the big tom. Because of the depth of resonance a large tom gives the best idea of how the overall drum set will sound in a given room. By experimenting with the location of the volume, you can find the best place to install and the first overhead.

If you put the drums where they just look best, there is chance, What You burn a volume With positions, where is it lower harmonic resonance will be completely suppressed. This is not uncommon. This depression is due to the standing waves, and they

each room has. After the sound is generated, it is combined with its own reflections from the walls. Depending on where the drum is in relation to the walls and your microphone, the waves add or subtract. It may happen that, by an unfortunate coincidence, the microphone turned out to be where the drum sound is out of phase with its reflection.

Fortunately, this problem can be avoided by moving the tom around the room in search of a better position for our first overhead. The main thing is to listen. Have the drummer beat the drum as you walk around and listen - up, down, closer, further. Then ask the drummer, or better still your assistant, drag the tom around the room while the drummer beats on it.



Rice. 1 - Not hurry up arrange all installation, Bye Not find the best place For big volumes.

The more you listen, the more you will hear. You will find positions where the drum sounds fatter, as well as positions where it is terribly poor due to phase subtractions. With reflections. IN dependencies from Togo, Where are you stand, And where the drum is, your head will be in the knot and or anti-knot of these wave interactions. You can explore and observe this phenomenon only by listening to the drum while moving.

Be patient when choosing the correct volume position. try to resist the desire to stop at the second or third position. I understand that dragging the volume the room can look silly after a few minutes, especially if the drummer Not saw engineers, which do such earlier. All What I Can say - I stopped

try not to look stupid for a few years now. There is no difference in the result until you find the Killer Drum Sound.

The first overhead will be positioned where you have found the most complete, resonant position for your head in the room. Always try to find position where this deep resonance is the widest, and position the microphone there. Try save from volumes distance How at least for elongated hand. You can define your experience with this technique by the distance from the volume. In proximity to find a resonance can Who whatever, But This too much close for overhead. Target - find a local point of deep resonance as far away from the volume as possible.

When you have found the point of greatest resonance for the tom and overhead, don't move the tom any more - let the drummer position himself around it. This job will take three to five minutes, but you must find this point before he begins to settle down.

What I have described may seem radical to some. In most cases, the idea to work on the location of overheads for engineers is belated. As a result, their overheads can end up at points with mutual subtraction of the lows, which has to be compensated by lifting the lows on the EQ, which in turn only leads to a cloudy sound of the installation. Moreover, adding LF directly contradicts our goal - instead of Maximum Illusion with Minimum Voltage, you get Maximum Voltage with a slight (if any) increase in Illusion.

When using corrective EQ, you simply shift the problem - one problem turns into another. Instead of adding an EQ, move the tom to a better place, which means moving the whole installations and - all from the beginning. Want to? More likely No. This too unpleasant - you was chance, And You lost. Maybe V next time. Or put it all down to the drummer. Or a bad studio. Sound familiar? Start with an overhead and a big volume and get it right from the start.

Why big volume?

It may seem to some that I pay too much attention to the big to that. After all that, maybe the drummer won't touch it during the recording. But a big volume is like the climax of a joke to me. When they walk and knock with him, and when he rings like a mad bell and shakes the room, know that it's worth it. Secondly, the same acoustic effect that is responsible for phase matching for the tom and resonance at its lower frequencies in the “piece of air” where your overhead is installed works for the rest of the drums. For example, if you place it where the tom is completely out of phase, I'm willing to bet that the small one will also sound flawed. With a volume, this is easy to detect, because it outputs a low frequency.

Now about overhead #2

Once you have found the best spot for the overhead, return to the control room and enter the overhead faders. So you can hear how the installation sounds right from the room. Small set too high? Do you have a better approach to setting everything up at once at listening overheads. This gives to you picture of room. This on actually gives to you the best performance O volume, What does All installation V in general, than

individual near-field microphones. This gives you a guide to what to aim for when setting up near-field mics - what gaps in the sound of individual drums you need to "fill" with them.

I think by now you should be getting the picture that overheads are the most important mics for getting Killer Drum Sounds. Some engineers only use them to "remove" cymbals. But the rest of the drums are so loud that - rest assured - a good deal of their energy goes into the overheads, and it is also subtracted/added to the signal of the near-field microphones. We must accept this fact and make it work for us.

Importance bass

Remember that high frequencies are easy to control. Getting the right bass is a much more difficult task. If the lows are correct and in phase, they become the solid foundation on which all other lower frequencies stand. This applies to all instruments. The lowest frequency becomes the base and greatly influences what is called "degree of tactility" - a factor that determines how your body feels the touch of a sound.

Murderous sound drums, Part II

Well, so far so good. We set up the drums in the right area of the room and found the right piece of air for the two overheads. Now it's time to take a closer look at the near-field microphones.

More fantasy, Please (less tension).

To sum it up: getting a killer drum sound doesn't mean recording the actual sound of the drum set. On the contrary, our goal is to capture the character we want, and try to fool ourselves into believing that everything really sounds like this (or even better!).

We are involved in this deception because, unlike other instruments, the drum kit produces sounds of very high intensity, i.e. voltage. Therefore, high voltages are a unique feature that characterizes the coloration of the sound of drums. Most illusions and emotions are not on the attack; she is in color, i.e. in the second part of the envelope.

The key to a killer drum sound is to get the overheads and near-field mics to work together to produce something truly excellent.

Coloring And general painting.

Putting together all the sounds from all the microphones of a drum kit piece by piece is like working on a huge jigsaw puzzle. Working as usual, from close to far, starting with the barrel and small - like putting together a puzzle without even looking at the picture on the box! And knowing the big picture from the beginning is very useful.

Let me give you another analogy: imagine a sheet stretched in space (calmly, stay with us, it will be interesting). Now rest your fist on it from behind to make a “slide”. This will pull the sheet even tighter - this is the impact of your fist. A failed phase is like a hole in the sheet, which will cause your fist to slip through - no good, the sheet loses tension. This often happens with the sound of drums: the near-field microphones stick out, but are only needed to make the whole sound (or the whole sheet, if you like) jittery and pathetic, while leaving individual sounds untouched. The result is a stretched, solid sound, in which the sounds of the near-field microphones serve to support the overall picture.

Choice location (more important how choice microphone)

You probably already have a list of your favorite mics for each drum, so I won't go into too much detail on this subject. Instead, let's test the location selection technique. Positioning when there are two or more microphones provides much more variety than the choice of microphones.

close location microphones

The hair on the back of your hand can become the hair on your neck! It would be nice if you could vanish into thin air and feel the vibrations generated by the drum to find the best mic positions, right? You will be surprised, but with a little practice, you can do something similar.

Most of you know the trick of using headphones (which just amplify the mic in your head) and moving the mic to find the sweetest acoustic guitar sound. This technique works great for quiet instruments, but is completely useless when you're dealing with loud instruments like drums. The level of reflections in the room goes off scale and interacts with what you hear from the headphones, while there is always a phase difference between the acoustic field and what the headphones give. In other words, we need some better way.

Hair on hand

The trick is to feel the sound with the skin on the back of the fingers and palm. I discovered this by accident, setting the microphone in the right position while the drummer beat on it - the hair on my arm twitched nicely. In fact, you can use this approach to find the best microphone position (Figure 2).

Can introduce, What your hand - This counter Geiger.

What do you really feel? Not just intensity - then it was possible I would just like to find a "hot spot" nearby with a drum, but that's not the goal. It is better to find a place where the vibrations remain strong at the maximum distance from the drum. With practice, you will find that different points in space also make the hair vibrate slower and longer. Here is what you were looking for. Slow and long vibrations on the back of the fingers and palm are equivalent to the maximum resonance and depth of the sound of your drums.

Rice. 2 - Hair similar counter Geiger at search resonance

Secret movement

The secret movement is shown in fig. 2. Place your palm on top of the microphone so that the back of your ring finger is aligned with the direction of the microphone. Once you have found the right point, all you have to do is lock the stance, and remove your palm - you're done!

All of this may seem very far-fetched, but this approach is really very fast and saves you from running from the microphone to the control room and back again 50 times. Don't worry if it looks silly - no one will notice your sleight of hand, I promise.

Job V zone (mixing microphones)

Mixing the sounds of the microphones is the final stage. Anyone can stick one microphone where one instrument sounds good. Unfortunately, as always, when you add drum mics, their overall sound tends to fall apart - it happens every time. The problem is that the sounds from two microphones cannot add up without changing the overall picture. When two sounds of the same volume from two microphones are combined, the result is blurred because signals arrive with varying delays. Since we recognize that the combined sound of microphones is blurred, we need to use this blur to our advantage. It will let us know what needs to be fixed.

Listening to the kit only through the overheads, ask the drummer to play all the drums slowly. You must accurately pinpoint the location of each reel.

Start with a big volume. While the tom is playing, listen to it only through overheads, mentally focus on it with your eyes - see it, feel it. Now slowly... enter the volume near-field mic fader. So slow. The moment you hear the change, press the phase rotation button. If the phase-reversed version is deeper, leave it.

As you raise the near-field mic fader, you will be able to hear the mix go through the "blur zone". When you pass it, he will again become clear. Keep in mind - this is not what we are striving for in the end. Ideally, we want to do his beautiful, clean, whole And deep in this range. If You pushed the near mic out of the fuzziness zone, making the sound clear, you end up having to do more work, and our experiment failed.

Zone vagueness

The main thing is to find this zone. If you define it, you will be in a better position to perform any operations with this drum in the mix. Conversely, if the sound from the near mic becomes blurry when compared in level to the overheads, then that drum will best sound played too loud, losing the color, cohesion, and resonance that you might have gotten from the overheads.

Working on the definition of the fuzzy zone requires attention. There will be a desire to turn on some EQ. Resist this temptation - EQ will not help you here, it will only mess up the phase picture. Your equalizer will prove to be more powerful when you correct *acoustic* causes . vagueness.

Instead, work on our zone by moving or rearranging the near zone microphone. Moving this microphone in any direction will change the phase interaction with the overhead. Even changing the direction of the microphone according to its diagram can give the expected effect.

The coincidence of the phases will make the tom in its ambient picture sound like it's glued in. Here's an interesting thing: when the close mic is in phase with the ambience from the overheads, the overall effect is magically transferred

V room listener, creating picture co sides loudspeakers, which in turn blends with the sound of the room.

Unpicked microphones - Not scary.

One of the best drum sounds I've been able to create was made in a low-budget studio with a bunch of obscure, unselected "noname" mics.

bad percussion installation (So Not It happens)

A very famous engineer once said to me, “Drums? If the drummer did not collect the installation, then you can not do anything. I have never encountered a bad drum kit before.” A very inspiring and arrogant statement. Unfortunately, he was right. Never blame the drummer any setup has the potential for great results. In other words: there are at least 100 ways to make a drum sound terrible. Let's try to avoid them.

It always seems amazing how the most absurd set of microphones gives you the richest sound when the hardness factor is used (see Advanced Microphone Technique, Chapter 3). Simple explanation: put on your headphones and speak into each microphone. Label each microphone with a hardness factor from 1 to 10. Evaluate the sound of each drum and match it with the appropriate microphone. Soft as a pudding, the sound of a drum with a hardness factor of 2 deserves a mic with a factor of 8. A drum with a factor of 6 deserves a mic with a factor of 4, and so on.

Naturally, there is always a place for a matched pair of microphones. If you're writing a great drum kit that deserves a proper pickup, maybe you should use matched pairs and the same mics everywhere. But remember, reality is rarely what you want it to be, and most attitudes require you to get as much character out of yourself as you can manage. Try to use as many **unpaired** mics as possible for the overheads and a variety of mics elsewhere. Each mic will add its own tint and the overall result can be unexpectedly colored or depleted, creating a kaleidoscope of hues.

Double-meeking (near And further microphones)

Each sound creates a Flame of Energy (see Advanced Microphone Technique). Good results have place, when the microphone installed in tip of the flame. The flames are much longer and further away from the drums than where engineers usually put their mics. This is mainly due to the fact that we need more separation of sound than distant microphones can give us. How can we achieve separation without losing the benefits that the Flame gives us?

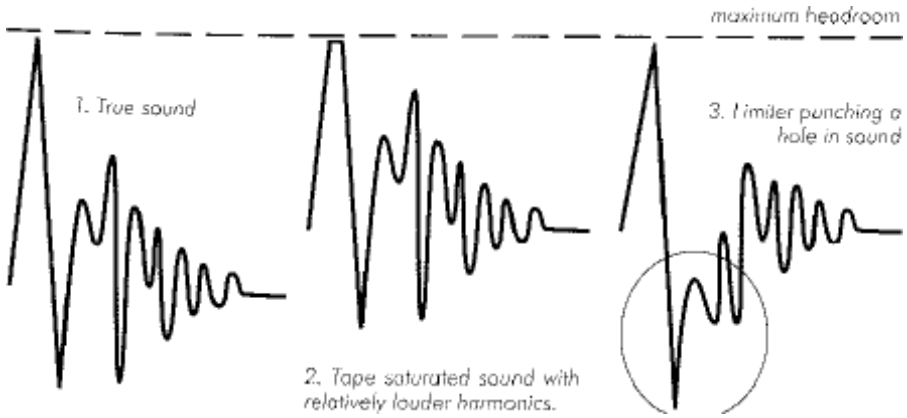
The answer is to use two mics per drum - one at the tip of the flame (far) A another - Very close (used only like a signal discoveries noise gate). This works well on toms and snare drums, and not just because the close mic catches highly concentrated energy from tom/snare, But And That's why,

that he picks up this sound a couple of milliseconds earlier than the microphone at the tip of the flame. This gives the gate more time to open. As the gate begins to open just before the powerful sound wave reaches the far microphone, it feels like the sound is crashing down on you - bang!

Film Saturation vs Clipping peaks

Why production engineers love film For sound drums?

Peak clipping introduces "holes" into the sound, while tape saturation is like a covering cloud that absorbs small transient vibrations.



Schedule 3. - Attack limiter V transitional process shifts the whole signal, including harmonics And other, when your imagination wants more. The rest of the drum's envelope rises to boost the weaker parts of the drum's resonances and harmonics. Like that.

Force dynamic processing

Now that we've created a wide, stable, cohesive, vibrant image, we can spice it up even more with a couple of tricks.

Barrel

To get to the root texture of the kick, let's turn it "out" using 2 mics with a significant difference in stiffness and out of phase with each other. By mixing them in different proportions, you will get more shades and more quickly than by switching between deluxe EQs. Also try moving these mics a few millimeters apart. Let me remind you: the stiffness factor from H1 (soft) before H10 (hard). This double out of phase the microphone applied to the kick should consist of a pair of H2 and H9, otherwise, it will simply be lost in the mix.

Small drums

They never sound as loud and wide on the record as they actually sound. Try using two microphones next to each other to capture more square sound waves. Save their phase, And, For the best result

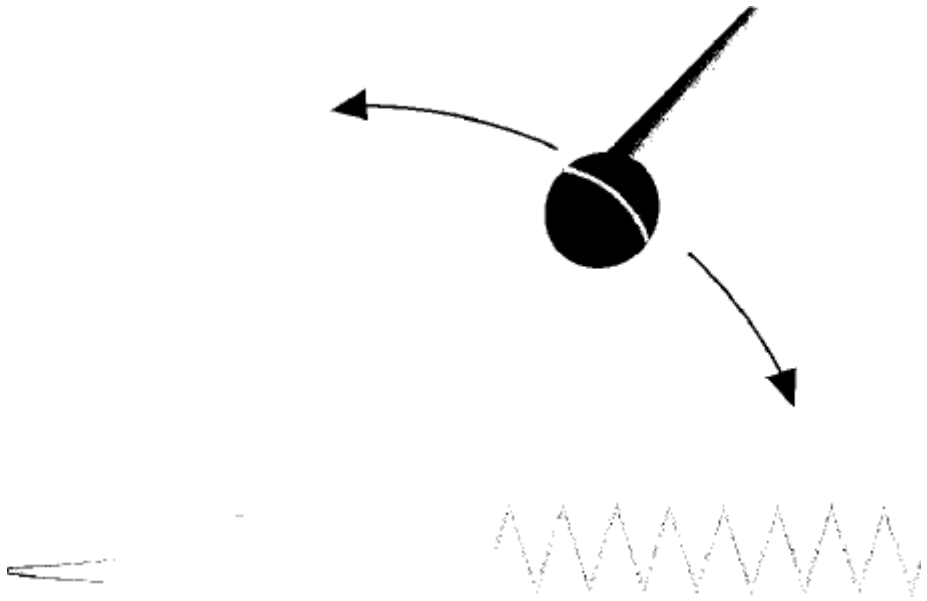
(Big secret, no one else will tell you about this), use two microphones with significantly different sensitivity (but not the hardness factor). Each microphone individually responds to sound pressure, and their combination will give a deeper coloration to the wide dynamic range of the snare drum sound. Offsetting one microphone relative to the other in the direction away from the source gives you another plane to experiment with.

Dishes (balance even And odd harmonics)

The sound produced by the cymbals is rich in harmonics. Odd harmonics (such as thirds and fifths) give the overall timbre a harsher sound, while even harmonics (such as second and fourth) give softness and roundness.

The key to a good iron sound is to record these even and odd harmonics in the right proportion.

Why? The direction of even and odd harmonics from the dish is predictable. Even harmonics always rise up (perpendicular to the plane of the cymbal), while odd harmonics always rise along the plane.



Rice. 3 - When have microphone above plate, remember: corner his inclination - This tone. moving microphone in an arc, you will hear a smooth transition from fat to thin sound.

With the movement of the microphone in an arc, the tone of the sound of the cymbal will change very much (more than from EQ, believe it or not). By moving the microphone downwards in an arc towards the edge of the cymbal (Fig. 3), the sound becomes brighter (but thin), and upwards - more round (less bright). Don't waste time with the EQ if the problem can be corrected with a slight twist of the microphone.

These cymbal manipulations should be on the first shelf of your subconscious mind when you hit the first overhead. As a final touch on the drum alignment, I adjust the overall sound of the cymbals by moving and turning the cymbals themselves on the stands while listening to them on headphones.

I know drummers don't really like it when they move their cymbals, but if you do it carefully and with worthwhile results, they will let you move them next time. If you wear white gloves, they will know that you respect them.

Checking Maximum Illusion (Auratone test)

To determine that you have reached the maximum illusion, switch your monitoring system to a mono 3" Auratone and check the result.

The tension will drop, but the texture should be preserved. If not, we return to the drawing board again.

Magical hi- hat

When you've got the hi-hat sound you want by setting the mic at the right angle and distance to catch the Flame, you're probably worried about the snare leaking into the hi-hat mic. Below - reliable way to avoid this.

Use controlled muting on the hi-hat channel, but control this muting with the mic closest to the snare drum. Make the gate attack minimal (super fast) and the release time just long enough to recover before the next hi-hat hit. Range - -10…-20dB.

In probably ask yourself “If you crush every hat that falls with a small one, will this hat be lost?”. Pretty weird, but no! It's like sleight of hand. The downbeat on the low has such energy that you won't lose a single rhythmic component, while at the same time it will give you limitless control over the level of the hat. Don't be afraid!

Then we

Those who love to play will love this tom effect. When recording drums on tape, don't dump the toms into a single stereo track. Put them on separate tracks. Then, when you have enough personal time, record each track, one at a time, in slow motion onto the final stereo track. Use this opportunity to save attack, level, pan, and dynamics to make each drum part special. Switches, fade-ins, fade-outs, pan can mess up listener perception moments enough time to create magical effects with volumes. The result should be a completely clean track with toms, from which you can later make a crazy echo effect, not affected by the rest of the drums.

Psychoacoustically speaking, we recognize the size by the strength of the fundamental tone and the set of harmonics. For example, if the high tom and the middle tom are tuned to the same tone, our ear will still distinguish the higher volume due to the strength of the fundamental relative to the higher harmonics. Task: find the fundamental tone and raise the frequencies of the upper harmonics (if size matters).

Remember: the maximum energy of the reels is not the most valuable. The key is resonance and texture. Peak is our enemy. Texture is a friend.

Notes on memory

Our target - reach maximum illusions at minimum voltage. The human ear is more sensitive to texture than voltage;

IN texture - All, texture - This All!

First focus on lower frequencies And resonances; The upper frequencies will easily fall into place;

VU = power And volume. peak = facet extremes. Focus on VU, limit peaks.

electric guitar magic

I love guitars because, when recorded properly, any cabinet produces an infinite amount of sounds and textures.

I once recorded with Laurence Juber (former Wings guitarist), his old jazz thing called “Four Brothers”. The arrangement featured a large horn section with saxophones, trumpets, trombones and clarinets. In the end, we played the record to the session pros, who praised us for the quality of the horns. We laughed and let them in on a little secret - every brass part was recorded on the guitar.

So a guitar amp is potentially a very rich palette, and the trick is to find a way to find what you need among this variety as quickly as possible. Wouldn't it be nice to turn an electric guitar into something perfect, the guitar sound of your dreams, with simple operations, step by step? Read on.

rule #1: size - Not The main thing :).

The power of the amplifier and the size of the stack in the studio practically does not matter, because, in ultimately it's still just an illusion. Remember, no matter how loud and big the cabinet is, your recording will still be done at 0 VU. Some of the most aggressive guitar sounds were actually squeezed out of the classic Pingose, a small, five-inch box.

rule #2 (Magic or garbage?)

I approach recording guitar amps in two ways. The first is the Magical Approach. I've concluded that the amp sounds magical and I'm trying to capture that magic. The second is the Bad Approach - the sound is characterless, noisy, without resonance or texture, and I'm trying to make the best out of a bad situation.

Comment. If you're tired of trying to capture the magic sound of an amplifier with Plan A, you have a Plan B - treat this task like garbage.

Magical an approach.

You are now well acquainted with the technique of capturing the Flame. With a loud electric guitar, your goals are the same. First determine the hardness factor of the amplifier, and choose a microphone of the opposite hardness. Explore the sound field with your ears and body.

Warning. Do not explore the sound of a loud amplifier without earplugs. The occasional overload will put your ears on an unscheduled rest for the rest of the day. I know it's not cool, but it's true. Use regular cheap earplugs, and walk around the sound field at full power for as long as you like (earplugs will give you a more accurate picture of what the mic is hearing, anyway).

First, look for the optimal distance. While the artist is playing at a comfortable volume, move your head along the central axis, quickly enough away from the cabinet And before opposite to him walls, And constantly listen. IN this process We

trying to find a balance between extreme lows, grit, resonance and tone. Flame capture is primarily related to distance. Moving from near to far, we first have sand, then resonance, then sensations of focus, as all the speakers are phased, and then the opening of the bass. The bass I'm talking about is so low you can't hear it - but with earplugs it's easier to feel it resonate in your body - stomach and lungs - as you move through the field.

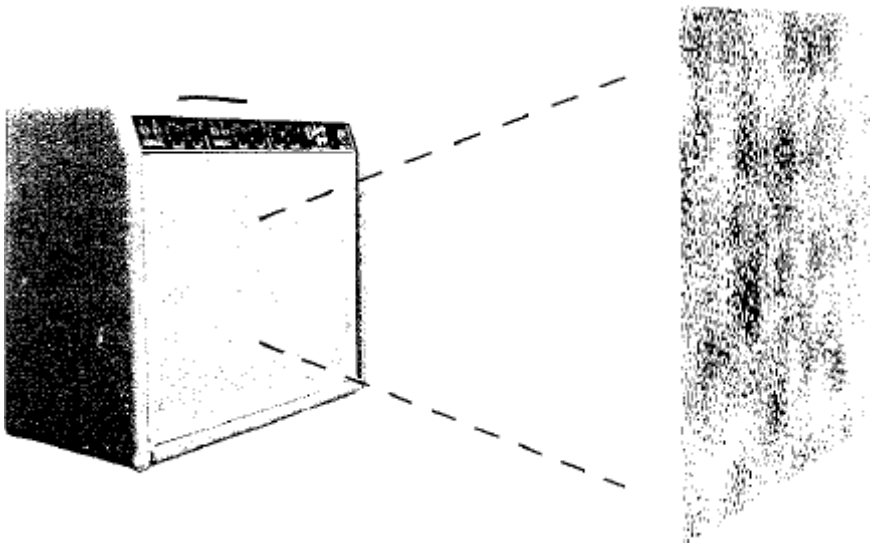
With your mouth closed and your lungs half full, your body will resonate like a drum. Have the guitarist play the low end and power chords to help you find the right spot. Positioning the microphone at this location will raise the Tactility Degree of the guitar (see Glossary). Mark this spot on the floor for future reference. Adding bottoms is not the same. Positioning the mic in the extra-bass zone will pick up near DC frequencies (frequencies that we feel more than we hear), which are like being touched by someone – in this case, the guitar is touching you.

The best distance is one where you feel immersed in the bass and resonance of the instrument.

We found the best distance from the office, now we need to move up and down and to the sides relative to the tip of the flame. To do this, we need to apply pink noise to the cabinet. Take a look at fig. 1 - it shows something like threads pulled out of the cabinet in different directions. The guitar cabinet flame is a large energy field that radiates from the speaker like a candle flame. Inside this flame concentrated huge a bunch of "hot points", "holes" "peaks"
"gaps" - places with a non-uniform frequency distribution of energy. They are brighter than your equalizer. It would be great to see them as clearly in real life as in this picture, but that's not possible. With every note played by the guitarist they come and go But matrix, according to which They are born, remains unchanged V air, and waits for sound to fill it.

This creates a bit of unpredictability when we try to place the microphone as accurately as possible in the "right place". By sending pink noise into the cabinet, we create a static picture of all audible frequencies at the same level and without standing waves.

Another problem that we have is that often some areas of interest to us are only 1-2 inches in size, and tracking the picture with the ears fails to get a fairly accurate microphone location. So you have to use a microphone and headphones. To this the method worked sound level V headphones should be more than the external sound from the cabinet.



Rice. 1 - guitar cabinet radiates mass hot And cold spots.

This is where pink noise comes into play. Run it into the cabinet at a low volume so you can hear mostly what's coming from the headphones. You will hear a whole kaleidoscope of colors as you move the microphone through the acoustic field. Some zones will be bright, some will be dim, some will be average, some will have a 7kHz peak, a dip at 80Hz, and a hump at 90Hz! Once you have found the point of interest, fix the microphone, reduce the sensitivity, go to the control room and listen to the guitar from there. The guitar will acquire the properties and form of this special zone. If you want the guitar to sound wider, more signature, smaller, thicker, thinner, then: turn off your guitar, put on pink noise, lower the volume, put on pink noise, and listen carefully. (**Remember:** pink noise won't sound like a guitar, but unlike a guitar, it contains the entire spectrum of frequencies.) Now slowly move the microphone in search of what you want.

The location of hotspots and low frequency nodes does not change when the guitarist is playing, as their location is a property of the cabinet and the acoustic space around.

No generator pink noise?

If you don't have such an oscillator, have the guitarist lightly touch the tip of the jack to create an appropriate background. Make the background quieter and increase the microphone sensitivity. You already know that the goal is to save the guitar when it's shrunk into the mix.

defective an approach

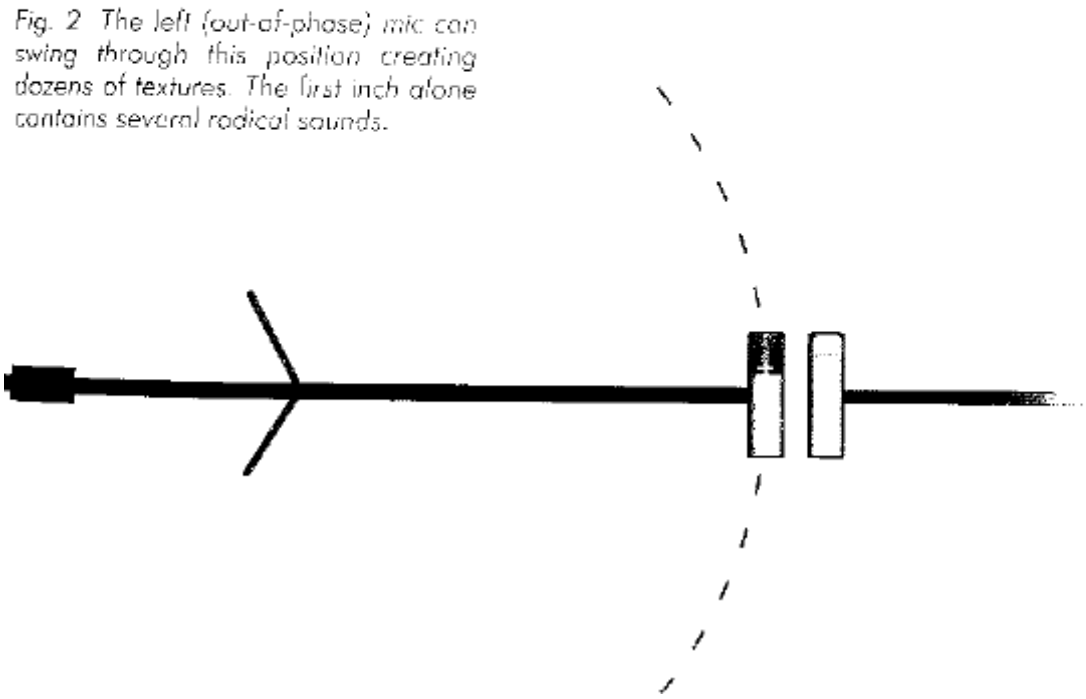
Are you out of luck and your amp is not in the top ten? When the real sound of an amp isn't worth recording as it is, it's time for Plan B - The Bad Approach. This approach doesn't require dexterity, so it's time to get to the point!

Let's leave aside some of the rules already learned, and take two microphones. Place one directly on the flame (not on the tip). Select the second microphone opposite To first. If first - H4, That second - H6. Put second

microphone behind the first on the stand, perpendicular to the axis dynamics, How shown in the picture below..

Have an assistant (see note at the end of the chapter) slowly move this microphone towards and away from the cabinet, meanwhile turn on the microphones in anti-phase. Each time you hear a second microphone (movable) pass over a point of interest, have an assistant mark the position of the microphone on the floor. Keep going until you have some interesting shades to choose from .

This process has the effect of flipping the sound "inversely", making a weak sound into something completely different. The parameters you can change are arc distance (balance between two microphones) and the phase button. Among the variety of these parameters, you will find many radically different sounds in just a couple of seconds or minutes - this is much more powerful than hourly experiments with EQ. Another advantage that this setup has is the ease with which you can return to the previous settings - with just one movement!



Rice. 2 - Left (anti-phase) microphone Maybe move relatively specified positions, creating dozens of shades. Even one inch contains some radical sounds.

When Not you can decide With choice option sound - lay down their V mix in a small speaker and compare these options at the same VU level. I choose the one that creates the most illusion with the least VU.

Not By axes? Wonderful!

The worst thing that can be done with any microphone is to place it directly in front of the sound source. It's hard to believe, but it's another secret to the Magic of Electric Guitar Sound. Positioning the microphone in a hotspot creates a much cheaper, high-contrast sound. It took me 20 years to realize this. it is counter-intuitive (ergonomically, logically, emotionally, musically, spiritually) to place the microphone in a different direction from the instrument. This phenomenon is exactly what I love about sound recording: psychoacoustically sound can behave very differently than you logically expect. I'll explain later.

Let's try to create a big guitar sound with a lot of air around it. To hear the "air", we need to add some high frequencies. Problem: The "air" is not very loud, unlike the amplifier. How can you add highs to emphasize the "air" without making the whole guitar sound brighter? If we turn the microphone off axis, it will make the guitar sound poorer. Now if we add through EQ to correct for this poverty, we will thus increase the purity and saturation of the "air" around the amplifier!

Example. Let's say we rotated the microphone 45 degrees off axis, which caused a 10dB reduction in treble at 10kHz. The microphone now looks into empty space - into the vibrating "air" of the cabinet. If we now add 10dB at 10kHz with the EQ, the guitar will sound normal again, just with the extra sensitivity to vibration, openness, and gain. **Note:** When rotating any microphone for this purpose, remember that you are rotating the microphone pattern to ensure that the pattern is capturing the “chunk of air” you need.

There are several positive reasons for turning the microphone off axis. This is analogous to achieving the correct exposure for the general scene. Think of your microphone's cardioid as a filter on a camera lens that darkens everything around the edges and brightens what's in the center. Now imagine that you need to photograph a scene with a bright object in the center (like the sun), with darker objects on the sides (trees, shadows, rocks, etc.). It would be silly to point the lens directly at the sun, right? Likewise, it would be foolish to point the microphone at the brightest point of an acoustic source. You never even point your ears directly to the source unless you are deaf and the source is not too quiet. If you get too bright a sound and compensate for the treble boost with the EQ, remember that you've also cut down on the "air factor" treble. It's like throwing water on a red-hot sound. You may have corrected the guitar, but the air was crushed. I prefer to leave
"air" is clean, clear, and alive. If the guitar is too bright, try moving the microphone further away.

Imagine that the microphone is a point source of light. Will you shine it in dark corners or in lighted ones? In the dark, of course!

Use this technique for all acoustic sounds. If you want something to sound brighter without generally adding high frequencies, rotate the microphone 45 degrees and add highs with an EQ. You will feel the brightness without changing the overall spectrum.

Strengthening Feel

When deciding how far to place the mic, don't chase volume: it will eventually come down to 0 VU. **Remember,** you're looking for a sense of amplification, not attenuation.

Electrical Bas-guitar

Have you noticed that when you combine bass DI line recording and his mic sound, the two of them never sound the way you expect and hope? You want a straight, clean, convex, crisp, linear sound. You love the warm, deep and round character of the sound from the amp. But when you mix them you also get a zone of vagueness - something is lost - the sound becomes somehow soft and is lost in the mix. You can spend hours and hours tweaking the EQ, but it won't work on the root of the problem. The main problem lies in the phases of the signals, not their equalization. The line input receives a signal about 2ms earlier than the microphone. The time it takes for the signal to travel through the guitar amp, move the speaker coil, and make its way through a short section of air to the microphone (1 foot behind millisecond), He late relatively linear signal on
1..4ms. Equalizer can't fix it. The only way out is to introduce a very short delay. My favorite type of delay is the one with an instant update so that you can hear all the changes at once. Such a delay provides a very powerful tool for varying between different tones and shades of sound. You will be surprised at the difference. When you find the right delay, you will notice that the signals bring together the best of both worlds. When you bring them together, the blurry zone disappears completely, replaced by a dense and solid bottom. Experiment with the phase button to double the fun. Hear how different notes in the bass line become dominant or fade out with different delay settings. Listen to how some settings extend the low end beyond the apparent frequency ranges of the sources. Explore these changes on large monitors and small Auratone.

Separation

There is one key concept to keep in mind when determining the technique by which you are going to achieve separation between loud instruments. leaks You get only Then, When one source strongly louder another. So that if two loud guitars are side by side, the leakage between them will be the same as if they were quiet. Loud guitars force you to turn down the sensitivity of the mics, which ultimately reduces leakage. You'd be surprised at how little soundproofing you need when you're playing two guitars at the same volume. One simple isolating screen between the microphones, preventing direct sound between them - and that's enough. Many electric guitar recordings have been ruined by engineers when recording with mics too close to prevent leaks. Give them to breathe. Do not be afraid.

Three-minute call signs

If you need to record a good guitar sound in a big hurry, here's how I usually do it. Leaving a faithful assistant at the console, and roughly placing several microphones in front of the cabinet, I go into the studio, put on the guitarist's headphones, tell him not to move or play anything, and ask the assistant to raise the sensitivity of the microphones by 40dB so that they can be heard in the headphones natural noise. This way I can hear the various noises from the speakers connected to the amplifier. Even if there is no source of pink noise, I am happy to use some kind of broadband background or crackle. Move the microphone like a Geiger counter while listening to this noise. Define the point of pleasant, soft, broadband noise, fix the microphone there and go to the control room. After three seconds of mic movement - nine times out of ten - this place is a perfect piece of air for this guitar. You will be very surprised at how much more effective such a search by ear is than just placing a microphone where it looks beautiful.

* **Your assistant:** I sometimes mention the assistant. If you don't have it, then get it. You deserve it and he or she will gladly help you. The world would be a better place if everyone had an assistant. It's good for you, for him, and for the ensemble - for everyone.

lucky hunting on this piece air.

tricks records piano

Elusive heart And soul track.

For many, the Sacred Chalice of the art of sound engineering is the recording of rich and transparent phono sound. Here are two of my favorite approaches to this.

The piano is one of the most intriguing and exciting acoustic recording instruments. Physically, the phono is so stunning with its sound in all directions that recording this instrument is like trying to catch rain in a cup - no matter what you do, you will lose most of the sound of the phono.

For live work, we prefer good source isolation, and the old mic-in-the-hole technique works on hooray. In a studio, where we have more opportunity "to play With room", We Can yourself allow diversify search
"piece air."

Key To phono

Before diving into the question of microphone positioning, it's worth familiarizing yourself with the philosophy behind the two approaches to picking up sound. Many engineers place the microphone where the sound is warmest and brightest. You need to avoid this instinctive desire, because. this can give a "small" sound. Large, resonant sounds come from various points. So, instead of "hot and bright", I would like you to think about *deep harmonic resonance.*

Deep harmonic resonance

Suppose for a moment that the piano is full of harmonicas. The vast majority of these harmonicas have a higher pitch than the fundamental and are much quieter. It would be quite justified to raise the upper frequencies on the equalizer in order to pull these harmonics into the light of God. If the microphone is already positioned where there are a lot of these harmonics, and you turn up the upper frequencies, what happens? Answer: hard, crunchy, small sound with reduced amount of bass (deterioration of internal sound balance). On the other hand, if you position the mic where the sound is deep, dark, and resonates well, it will give you a good chance to turn up the highs and hear more of those harmonics. This will reveal the resonance and improve the transparency of the low frequencies. In this case, the EQ also serves to improve the internal balance of the notes.

Also, by remembering what was said in the previous chapter, by adding highs to a dull sound, we make the air around it sound brighter - as if we are making the molecules move faster. It also makes the sound more transparent. And vice versa - lowering the highs of a bright sound floods the air molecules around it with water, creating a more crushed, strained sound.

So what way this new model thinking affect on microphone setup ?

If you place the microphone closer to the lower register of the phono (so that the notes and upper register keys are much quieter), and then raise the highs to bring out their back, phono will become balanced And bright, at this Not becoming

sharp. AND vice versa - placing the microphone so to All notes were V balance sheet And raising then high ... get what? The top notes will become very sharp, and the bottom notes will be sloppy. This is perhaps the main mistake made by sound engineers when removing the phono.

So, to summarize: since we are all We like to add highs to the phono sound, we have to place the microphone where the sound is relatively darker and not rich in highs, otherwise we end up with a narrow sound. Moreover, if you try to place the microphone where the sound is the brightest, you will inevitably come to the need to boost the low frequencies. What's bad about it? If you add low end to a thin phono sound, it can easily become muddy, fuzzy, blurry, as although you boost the lower frequencies, you do not add harmonics in this range. Essentially, you're just raising the voltage of the fuzzy sound. Enhancing high frequencies from a thick and rich lower phono register is much easier and more efficient than the inverse problem.

Maximum illusions / minimum voltage

What I Understand under voltage fuzzy sound? WITH points vision chapters "Max Illusion/Low Voltage" It is important to distinguish between two kinds of voltages: the one you hear (which stimulates your imagination) and the one that doesn't, although it makes the indicators twitch. If the indicator jerks up, then it is better for it to contain some interesting, impulsive, expressive information, otherwise you lose tension, which simply uselessly clogs the free dynamic space of your project, for example, leading to distortion or forcing the limiter to reduce the volume of the entire project . If the mic is in a bad position for the bass - the piece of air doesn't contain a coherent low frequency spectrum of the sound wave - adding bass beyond the limit will not restore the integrity of the wave, and will not be able to correct the shape or envelope of the low frequencies, as they are physically distributed elsewhere.

Here's an analogy: adding 20dB at 80Hz to white noise will boost that frequency range, but not turn it into a sine. It won't make your hair shake or your neck or your soul 80 times a second. But by placing the mic where the strong 80Hz section is, you will make the world tremble at that frequency. Boosting the treble will also boost the 160, 320, 640 Hz harmonics, making the bass feel more pronounced in small speakers. So let your microphone contribute to that deep resonance.

There are two different approaches to using microphones for you to choose from. Which one you choose depends on the mics you have and whether you're in the mood to experiment at night. Take a good look at the figure, where AA and BB just represent these two alternatives.

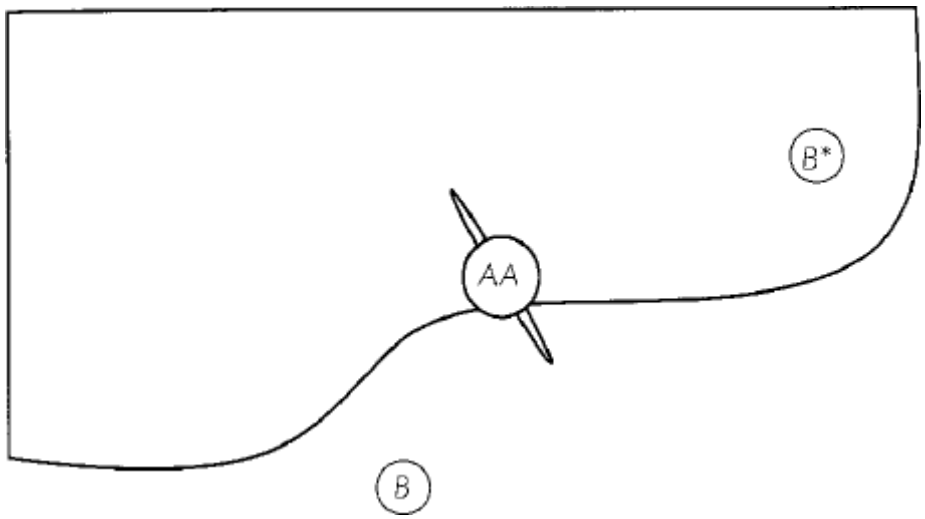


Fig.1 - Sight With heights avian flight on my favorite options (AA, BB*)

Reception A

This is my favorite technique using a source with excellent phase coherence. "Sum and Difference" - otherwise known as the MS (Mid-Side) technique, using a combination of microphones with omni and figure-of-eight patterns. It delivers a very natural, transparent phono sound with the added bonus of always-on “thickness control” via a figure-of-eight microphone. If you know how to set up a pair of MSs, great, if not, I'll explain it in more detail later in this chapter.

In fact, using the MS technique, we are looking for one piece of air for the entire phono. Point the cutoff plane of the figure-of-eight microphone directly to the center of the phono (where the sound is strongest), and point the more sensitive part of the figure-of-eight toward the bass. There is a nuance here: I have never seen a figure-of-eight microphone, in which the sensitivity would be the same for both halves. But it's okay. Just make sure the more sensitive side is facing the bass (Fig. 1).

Reception IN

I understand that not everyone can have two microphones with the right directivity for Receive A. Fortunately, there is a second trick - Technique B, which can solve the problem with two cardioid microphones. This method gives a huge number of options for shades and spaces. **Warning:** This method is less reliable, but has the potential to give excellent results. Here's what you need to do. Position the first cardioid microphone closer to the plane of the strings (soundboard) in the lower phono region (position B* in Fig. 1 and Fig. 2).

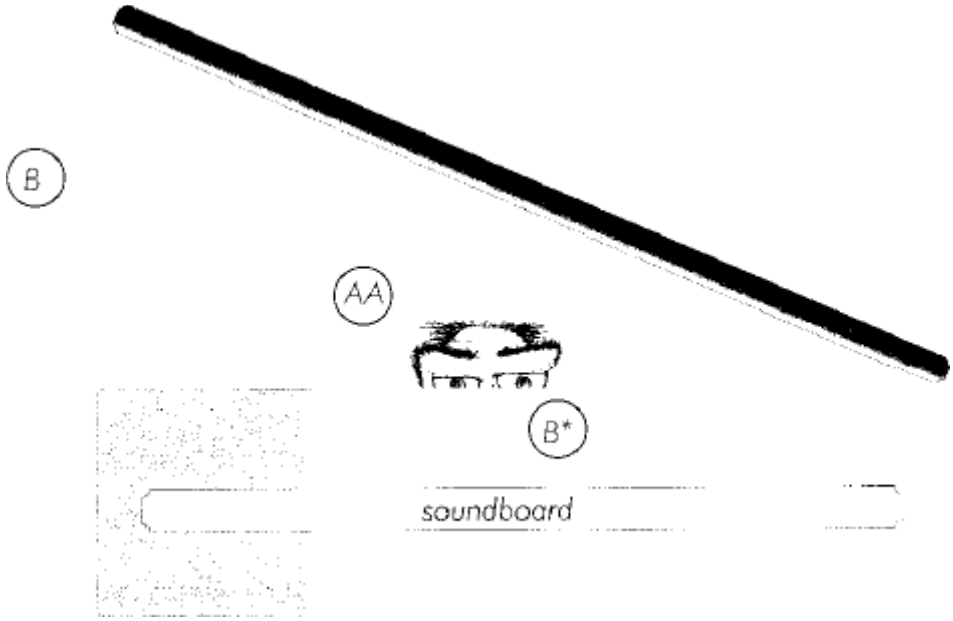


Fig.2 - Horizontal view Togo same location microphones.

Position the second microphone in the air outside of the phono. When these the microphones are equal in level and spread out in pan, they give a huge depth of phono sounding - as if the whole phono is deployed by all 45 degrees, the lower notes sound close, and the upper ones recede deep into the track. This effect sounds surprisingly natural, giving the feeling that when you stand next to the phono the lows seem close and the highs seem far away.

Only one microphone

This marvelous, How deep Can to know nature sound, When turned out to be face to face limitations - say, using just one microphone. It's much easier (and more fun) to turn a good mono phono recording into stereo with special effects than it is to make a bad stereo recording sound good. Since it's all about getting a single mic in the right place, adding a second mic for stereo becomes easy and fun.

Note: I am ashamed to admit it, but sometimes, at the request of the producer, I removed an additional track of thin air, i.e. source (B), leaving only (B*) in the final mix. Of course, this is only in case when the recording was made by technology
"IN". Here How must sound Fine recorded source (B*).

This trick works just as well when you have a couple of unmatched mics. For example, if one microphone is stiffer than the other. In general: use a harder mic on the lows and a softer mic on the highs. (Except if a hard microphone is harsh because it lacks low frequencies.)

To help you understand the location of the microphones, in Fig. 2 shows a horizontal view - from the lower register to the pianist.

More about mid-side (Sum And Difference)

Reception B can work well with any pair of mics, but Reception A requires you to set up the MS pair correctly. Given this, I will give a couple of practical tips on how to do it right and what you need for this:

- one omni microphone And one microphone With orientation "eight".

- remote controller, which You use, must have button switching phases V channel.

- bilateral scotch - to glue microphones together. When You their maximum brought together, film tightens design, And additionally prevents the formation of earth loops from the contact of the hulls.

Adjustment Sums And Differences

Connect omni and eight to adjacent channels of the remote control. Make an omni panorama in the center. Pan the figure eight to the left while sending the signal to a free bus (say, channel 24). Send the output from bus 24 to a free channel line input. Pan this channel (a copy of the figure-eight signal) to the right and invert the phase. Ready.

Adjust the sub channel volume to equalize regarding the first two. I usually include the sine source in the pre-fader insert of the original (left) channel to check if the levels are identical. If you don't have an oscillator, mute the omni, temporarily pan the right channel to the left, and adjust the fader so that the signal disappears completely - this happens when out-of-phase signals are level matched. Now tape the right channel fader to the console so that it doesn't move accidentally.

From now on, you only need to move the first two faders - the omni and the original eight - to adjust the level and/or width of the stereo image.

Note: The fastest way to flip the entire stereo image is to press the phase button on the omni or *original* 8 channel.

breathable stereo space

Don't forget about the ability to move the faders during the recording process, not only to control the overall level, but also to bring the stereo picture to life. When I record phono With orchestra, I I always regulate phono width, So What it narrows when the orchestra gets louder and more powerful, and vice versa.

Hard/Soft

Regulation level microphone eights Also affects on rigidity sound phono.

Eight quiet - sound already and tougher;

Eight louder - sound wider And softer.

Cardioid against omni V couple MS

Some people will find that using a cardioid instead of an omni would be appropriate. No, it will not. This is a really bad idea and I'll tell you why. The stereo picture is created from the sum of the omni signals and the two halves of the figure eight.

Using a cardioid will result in the left and right edges of the picture being composed of a nice and smooth figure-of-eight response and an incredibly dull off-axis cardioid sound. The result is a stereo image that loses reality towards the edges. Moreover - when you switch the picture to mono mode, you will lose the clarity of the sound of the sides - the picture will become narrow and unrealistic.

The main idea of MS in that in mono mode the whole picture is saved, and if you use a cardioid, then you will hear only the central part of it.

Eliminate Problems Sums And Differences

If the stereo level is skewed to the left or right, do not touch the third fader. He is engaged their deed. Instead of this, turn figure eight microphone axis to compensate for the weak side. Usually, in three approaches to the microphone, the problem can be solved. But before you move the microphone, have someone put their fingers on it so that you don't confuse which is the left and which is the right side.

Advantages Sums And Differences (And one flaw)

Benefits: In mono compatibility mode, you will never lose the top. This is due to the subtraction of the phases between the two microphones - in mono the only signal that remains is omni, therefore, everything is in order with the phase. The MS is intended to be the only valid mono/stereo pickup (it was designed by the Air Force for this).

MS Also Wonderful manages co softening too much hard tool.

Disadvantage: This is such a good way to bring out the instrument's harmonics that if the phono is a little out of tune, it can sound extremely depressing. Sometimes I I don’t even notice that the phono is out of tune until I turn on the figure-eight microphone - this is such a revealing method. If it sounds delicious, then it is delicious. And if depressing - look for the source of the problems.

More one comment (two axes games)

Not forget What there are two axes along which you can rotate the microphones with to correct the balance of the left and right channels. Imagine that the bottom mount crosses the diagram. Rotating the chart counterclockwise around this axis will lower the right channel and raise the left. This action will raise the level on the indicator, and the balance of the stereo picture will shift to the right.

And, of course, the vertical axis: turn the microphone to the left or right, or somewhere in the middle.

If you make the mistake of removing the tape from the third fader and moving it instead of moving the microphone, the picture will lose its integrity in the stereo field. One side will become more “ghostly” than the other, plus the 8 will stop disappearing in mono compatibility mode, leaving you with an uneven balance. Moreover, if you are caught doing this, the MS-police will come and take you away…

Illusion disguise phono And voltage

Don't overdo it when trying to add phono presence, it may not take the lead in the mix. Why? The phono is so rich in harmonics that it tends to mask other instruments, perhaps more so than any other sound. This means that in the final mix the voltage from the phono will not move the meters, so you won't need to compress and equalize it due to the illusion/voltage ratio.

Fono Man without masks.

VOCAL SUPERSTARS

The lead vocal sound is the main sound in your recording. The human ear is extremely picky when it comes to the sound of vocals, and we respond to a good voice sound more emotionally than any other sound. That's why vocals deserve special attention.

Most people don't like the sound of their own voice on a recording - no wonder, considering how many bad ways there are to record it! Here are 15 secret ingredients we'll put together to make vocals sound better.

Philosophy……. our mission

Atmosphere…….. create it

path …….. path signal to freedom

Path signal V headphones .... then What You hear, This That What They

receive! Headphone mix..... main mix of the day

Voice vocalist……. What here at us?

Choice of microphone…. in accordance with the

voice Piece of air…… arrange microphone

Acoustic control….up and down = high and low Catching a

flame…….. positioning for volume

Voice……... y everyone faces its architecture

EQ……... less - This more, focus on architecture Compression

…… your guardian angel

We manage levels……..large risk = big sound (or big shame)

Movements……... get into their rhythm

Let's start preparing the first four ingredients in the studio before the arrival of our superstar. This will add magic.

1. Philosophy

Years later, I came to the conclusion that we should have an approach to any artist, as like he already superstar. Especially This applies to the vocalist. I thought about this... it's bad enough that they might not be as talented as real superstars, even without being treated badly. Rest assured, and real superstars are not laid out to the maximum, in case of a bad approach to them. (Yes, they all went!) Because focus on newcomer And on their work? At them the most heavy Job V studios.

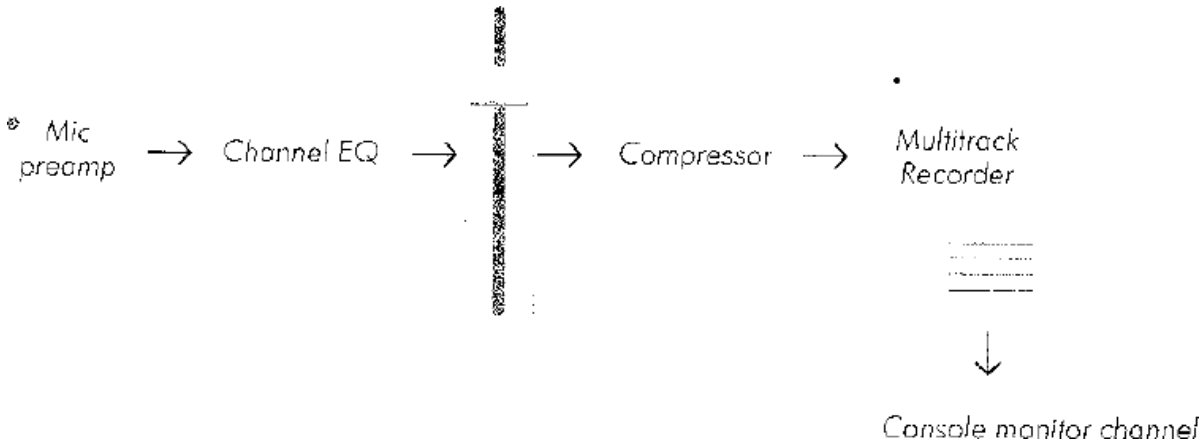
They have to do the magic, and we have to write it down. In short, we have no other no choice but to approach everyone like a star.

2. Atmosphere

Take him somewhere. Create a dedicated area in the studio for the vocalist. Give him the illusion of space and privacy. Don't push it against the glass between vocal and control. Better move aside, prepare a table with water, an ashtray, a candle. Adjust lighting. Make it inspiring. (Caution! Do not change the lighting during the performance, as this sometimes affects on their sense of tone).

3. Path signal

Here scheme For records super vocals, which I I suggest:



Microphone > Cord > Your Darling preamp > Input channel V remote control > Equalizer > Fader > Compressor > Entrance V multitrack > Exit from multitrack > Exit > Monitors.

Did you notice that the compressor is between the channel fader and the multitrack input? Usually engineers include the compressor in the pre-fade. This is understandable from a switching point of view, but in reality introduces more noise and distortion than post-fader compression. Assuming you're using fader automation, we're avoiding the problems of overcompression on loud sections and excess compressor noise on quiet sections. Always use monitoring via multitrack.

4. Path signal To headphones

The headphones in your control room should be powered by the same amplifier as the musicians' headphones in the studio. This allows you to hear the same balance as they do. With toy same volume And those same distortion. If signal too much loud, them high notes will be heard sharp and bass notes flat. And if they try to compensate for that, it will backfire on the record. It's a psychoacoustic effect Related to reaction internal ear (detailed review of this phenomena are not included in the purpose of the book).

The most important mix of the day is the one in the vocalist's ears. The mix in the control monitors is in second place.

5. Mix V headphones

Make an independent mix for the vocalist's headphones. Don't just send the signal from your main monitors to them. Sound waves travel differently in headphones and speakers, so the same mix won't work for either of you. You need to create a custom mix for the vocalist using the same model of headphones connected to the same amp to make the mix complete and cohesive. Give the excess of the bass guitar (the basis of any chord), the dry click of the kick (the basis of the rhythm). Keep a couple of vocal effects handy to create an effect of space. A plate reverb and a 100ms delay are fine. Try to spend the entire vocal session without removing your headphones to control everything.

6. Voice vocalist

Determine the harshness factor of the vocalist's voice as before. Ideally, we try to record the first takes with the best microphone. Next step may cause some embarrassment, but it's worth it. When the vocalist is in the test, ask him to sing a little so that you can hear his voice in real life. It's amazing how many engineers only listen to sounds through their microphones. When the vocalist gives the right dynamics, try to have time to place your ears in front of his mouth. At you must leave Total couple of seconds For definitions factor a rigidity. Remember, we don't want the vocalist to get tired. When you have determined the hardness factor, stop recording and move on to the next step. Don't even try to record your voice without listening to his previously live - the only way defining it rigidity. I understand that this may be a little shameless, but these two seconds can replace many hours of experimenting with microphones and equalizers.

7. Choice microphone

You already know - according to the harshness of the vocals. In Chapter 3, we already discussed how H4 sound requires a H6 microphone. Voice with a factor H8 requires microphone H2. Having practiced You become expert V microphone selection area for vocals by hardness. Finally, put three mics in for a quick comparison. Ask the vocalist to sing a little (5 seconds) each, make a decision and move on. Listen to the flesh and soul. Feel which microphone will capture the soul.

Rice. 2 - All sing only one side mouth, even If at all A little

8. Piece air

Let's get to the fun part - we place the microphone in the right piece of air. Here we need to decide on three things: *side* , *height* , and *distance* . The properties of the human face, which are responsible for the acoustic propagation of the voice, make us first decide on *the side* , then on *the height* , and then on *the distance* from the vocalist.

Those. at first move the microphone left and right Then up down, And Then - back and forth, in that order. If we do it differently, we will get a less accurate result.

Side note: There are some obvious advantages to having the microphone off-axis, such as avoiding "p" and "s". But besides this, with age, people's faces change asymmetrically. Because of this, everyone sings a little away from axes mouth. Never Not put microphone, Bye Not determine V what side sings

person (see Fig. 2). I discovered this while recording one distinctive voice. Everything was fine, but when the performer entered the role, he made a face like a pirate, and the sounds he gave out were carried away from the microphone. And the voice lost all its presence and drowned in the track. Further more. I asked each vocalist to stand straight, make their mouth round and sing one note of their choice for 20 seconds. While they sang their long “Aaaah!” I walked from side to side listening to the voice. Lowering the head helps to see and hear the distribution of highs from the side of the palate. Each side is always individual. One, for example, is brighter. If the inner voice is dull, then you may need this side. If the voice itself is thin, then perhaps you will choose more
"meaty" side. It will not work to turn the left side into the right side by equalization due to the initially different composition of the harmonics.

9. Control acoustics

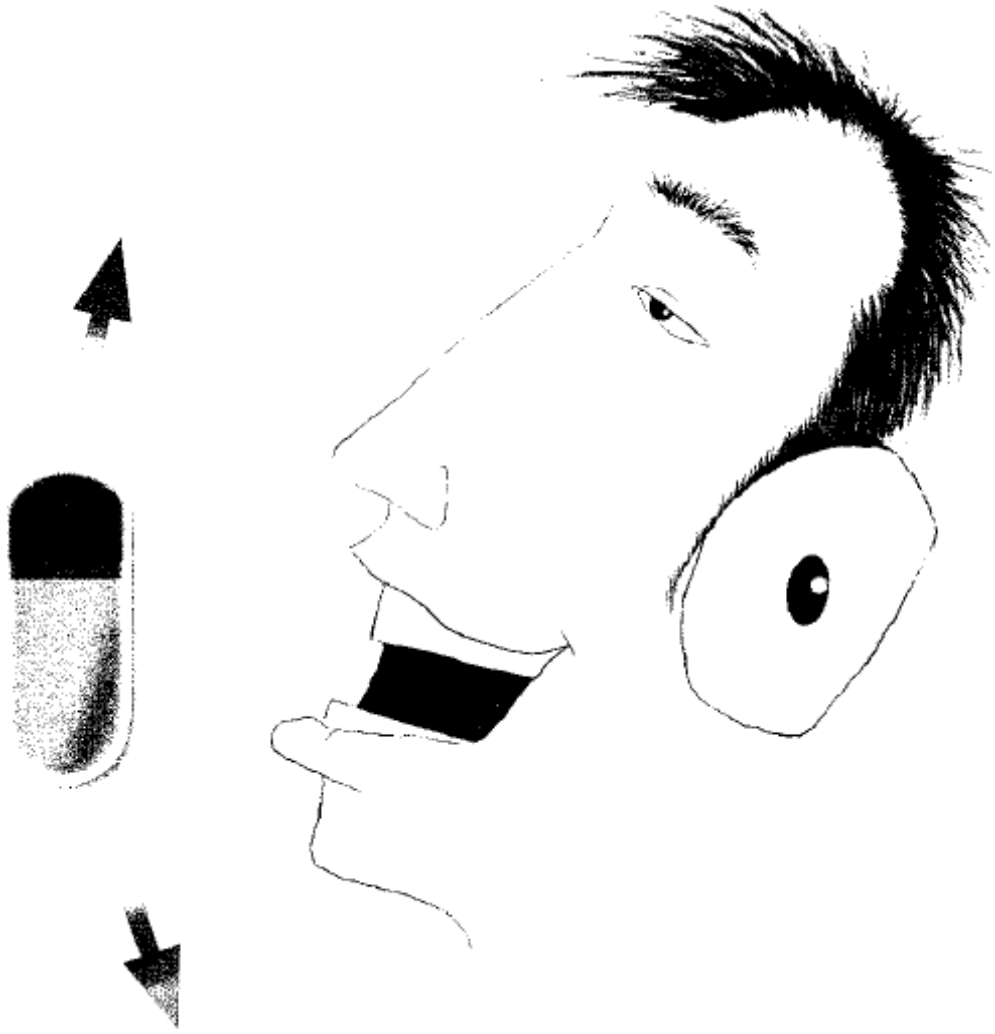
Having found the right side of the vocalist, you need to find the right height for the microphone. By lowering the microphone in relation to your mouth, you make your voice brighter.

When the microphone is raised, the sound becomes more muddy (Fig. 3). You can hear it with your own ears or with a microphone and headphones.

You don't have to sing a song for you to explore all of this. Playing regular textured sounds in front of your face will reveal these acoustic nuances on its own. Moving the microphone up and down is similar to frequency control: up is low, down is high.

I thought, that moving the mic down should give more bass, because the microphone is closer to the performer's chest, but experiments have shown the opposite. The sound balance at the top of the mouth contains higher frequencies. Usually this effect disappears above the lips, closer to the chest - more high, and the microphone catches them. The sound coming out of the nose is something directional, also completing this whole "south".

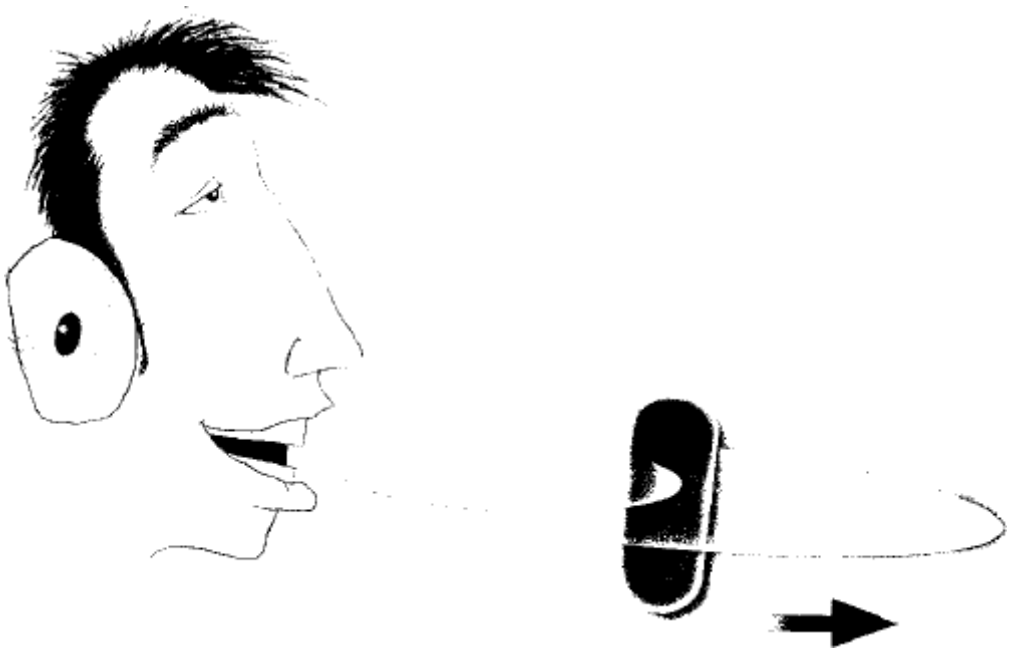
In general, if the vocalist produces too many bright harmonics or is nasal, move the microphone to the “north” for a fuller, smoother sound. If the sound of the voice is dull, move the microphone below the teeth.



Rice. 3 - Tonal control

10. We catch flame

Usually, the louder the vocalist sings, the farther the microphone is. Catching the flame of a voice can be tricky. Skillful use of a microphone is rare and comes with experience. It is not at all necessary to make it a rule to rearrange the microphone by you. Sometimes I would stand in the studio with the singer and vary the distance from the microphone quite widely to compensate for the singer's dynamics (Fig. 4). The effect was very powerful and required a lot of trust on the part of the singer (belief that I wouldn't knock his teeth out).



Rice. 4 - We catch fire.

11. Fresh voice

Each face has its own unique architecture, just as each room has its own acoustics. We correct the response of the room by introducing pink noise and adjusting the reference signal. When this is done, even the music sounds good. Before the performer begins to spend energy on the song, it would not be superfluous for him to make a few reference sounds, relative to which you can rebuild the equipment, and thereby delve into the architecture of the voice.

The vocalist's first takes are usually more precise and vivid than subsequent ones. Capturing a vocalist's first attempts at making his mark is important for several reasons. Usually on the first take, the voice is less tight and fresher, with a little " Bolan's Itch".

That, What I I call Itchy Bolana* - Wonderful works on the radio.

12. Equalization

Wouldn't it be nice to be able to gently equalize the voice without forcing the vocalist to sing? It's sad to wear out your voice for 5 minutes just to tweak the equalizer. Every *take* wears out the voice, and you want to have it at its best on tape. With fatigue, the voice loses its brilliantness. When this happens, raising the highs only backfires. Fatigue also leads to loss of control. dynamics and your compressors may lose efficiency.

One Alternative - equalize the voice on sounds that do not load the vocal cords. Ask the vocalist to say soft "A", "Bo Ba Blue", count to 20, whisper, hiss, etc. This will give you the opportunity to equalize while listening to the best, What Maybe extradite architecture his vote. listen carefully V frequency,

that bring you closer to the back of his throat, or lips, or other physical features. These same frequencies will give expression to the voice in the song that he is going to sing.

The purpose of equalization is to create a unique envelope for the voice of a given vocalist. By listening to the many sounds it makes (not singing), you will be able to understand what frequencies give the voice its unique qualities. Better EQ will enable them to sound as good as possible. The louder you sing, the more your voice becomes like someone else's. Fundamental harmonics increase with increasing volume with a greater multiplicity than higher ones. Higher harmonics distinguish you from other people like a fingerprint. And the fundamental ones are just the notes in the melody.

13. Compression.

Why compress? Everyone does it, but for what purpose? To make the most of it? No. To do "more"? Sure, but compression rarely makes things sound louder. There are two things you need to know about compression: firstly, if you compress audio today at a ratio of 1:5, and then again tomorrow, say, when mixing, at a ratio of 1:10, what is the final ratio?

A 1:10 ?

B 5:1 ?

IN 15:1 ?

G 50:1 ?

Answer: 50:1!!!

Many People And Not suspect What compression multiplies A doesn't add up.

Second, beware of overcompression. High compression psychoacoustically sounds "less" than low compression. Once the mastering engineer and the radios have run your mix through, you'll be at a 100:1 bar on vocals.

Let's get back to the purpose of compression: move closer to the juicy component, push back the loud elements, and move back to the measured, sexy line. Right? The best way to do this is to move the fader towards the tape.

14. We manage levels

This is a truly exciting and educational stage! The stage at which you will feel like a pro. Controlling the level on the way to the tape is infinitely better than doing it during the mix, for a number of reasons:

Incredibly improve signal / noise; We destroy

distortion or reduce their;

Compressor never Not squeeze, because He costs after fader

The mix in the vocalist's headphones is more subtle, because. he hears every nuance of his Superstar Vocal performance.

Intelligent sound. By manually controlling the fader, you do things that a compressor can't. Dynamic anticipation gives the voice focus and dynamics that the best compressors can't. Fader control while recording is risky (you will miss something), but with experience you will start to create sound Superstar Vocal right away.

15. magical movements

When you are involved in the rhythm of the vocalist, it becomes relatively easy to pick up his punctuation, muffle his breaths, draw out beautiful endings and fragments that require it. Either way, your favorite compressor is the guardian angel at the fader output, catching the bad bits that come from your belated moves. Practice, practice, practice on vocalists who haven't become superstars yet.

Remember: a great team recording with a cheap voice can make the overall sound cheap. And a Superstar Vocal Sound with a mediocre crew can make a whole track very serious.

We open compressors

On writing this chapters me inspired e-mail from engineer from Melbourne:

Stav, I'm a desperate guy and I have a fundamental problem that haunts me. I've been doing a good job for years, but to this day I can't use Attack and Release on compressors properly. I guess, I listen, and I pray.

I know I'm not the only one with this problem, because when I recently got a client's material out of mastering, the mastering engineer got it wrong as well. He installed the compressor release incorrectly. Not necessary to say that the thing turned out to be crazy pumped up (pumped), and now I can’t even listen to it. Assuming that I know what attack and release times are, I want to ask how to make them friends with the signal they process? Snare, Guitars, Vocals, Overheads, etc.

What I'm missing is the "concept". The answer to the question "why". If I knew this, I would solve many problems for myself.

At me There is 13 compressors, And, Mike, I tired of guessing.

I know how this engineer feels. I spent years listening to the difference between attack, compression, release, and couldn't figure out what I should be thinking when listening to these settings. When it finally dawned on me, the task of compressing went from intimidating and embarrassing to one of the most interesting and creative.

This All equals, What hack safe.

The compressor has four basic settings, and the path to great compression is easier depending on the order in which you fit those settings. When the sequence is correct, you will hear the effect of each one more clearly - thus arriving at a more accurate and musical tuning. The compressor combination key contains 4 handles. Turning them in the correct sequence prevents you from returning to their original positions. you are not pissed off when you're happy with a Release but struggle with an Attack? They affect each other when tuned randomly or out of sequence. Catching the result by the tail - now it is a thing of the past for you. It's like opening any encrypted lock: when the handle is in the correct position, you don't need to touch it anymore. Each step is a critical step.

Getting Started (temporary settings)

To crack a secret combination, you need to put all the settings in a temporary position while you concentrate on one parameter. When the first one is in the correct position, the other three remain. Focus on the next, listen, adjust, and - the second is also in place, and so on. Approaching this method of cracking safes in a different order will produce different results.

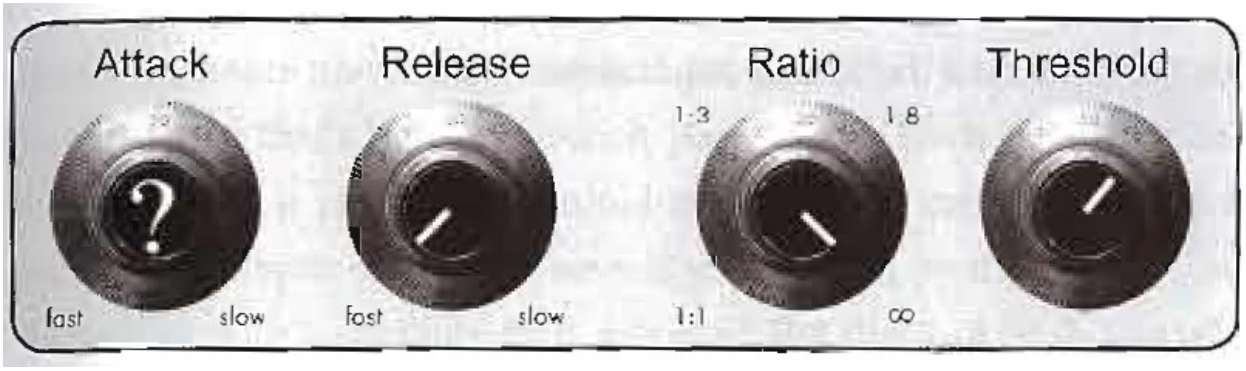
Attack - How whatever

Release - at a minimum Ratio -

on maximum Threshold -

sensitive

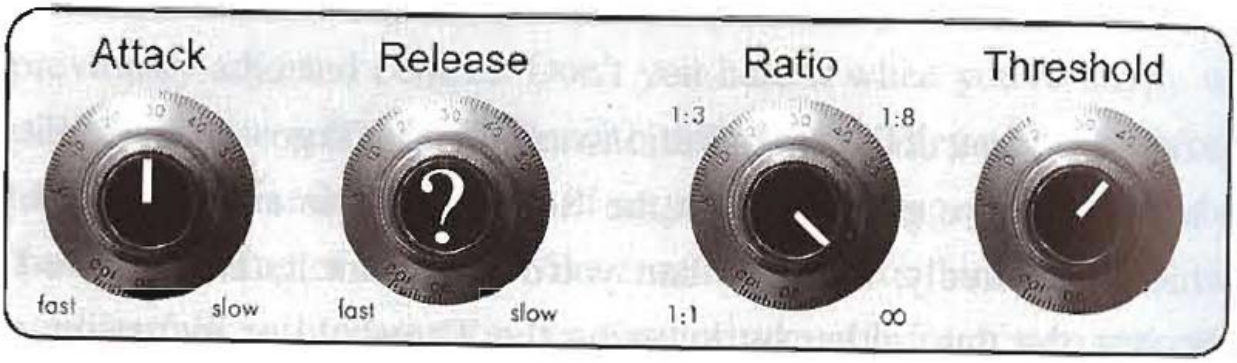
1. Attack



The first thing to do is to set the ratio as large as possible (20:1, infinity, etc.) - in short, the maximum. Next is the release to the fastest option, which is likely to be faster than you need. Next, feed the audio signal to the module, either by lowering the threshold or by increasing the input gain (depending on the compressor design), and listen while adjusting only the attack. Listen for the attack - the rising edge of the signal - as you turn the attack knob. Try to ignore the horrendous dips caused by the release's post-effects - listen only to the attack. (A super-fast release gives you the opportunity to hear many more individual attacks than a slow one). Listen for the rising edge of the signal. Determine how the attack affects the size of the burst. So if you compress the snare and attack at minimum time, it's like leaving only the bare snap of a drumstick.

Conversely, if the attack is on a slow tick, it is like a more powerful strike. Conversely, if you are playing an acoustic guitar and you attack in fast mode, you will only hear the touch of the nail on the string at the moment of impact, and with a slow attack, you will hear the whole pinch - the whole transition process will skipped compressor. So what, forget about post-processing, listen only to the duration of the attack until it becomes "delicious" - Maybe she to you need more "thick" think only about aesthetics. Thanks to the fast release and high compression you There is opportunity hear attack more clearly than on other settings. This approach effectively "turns on your ears" to improve your ability to distinguish an attack.

2. Release



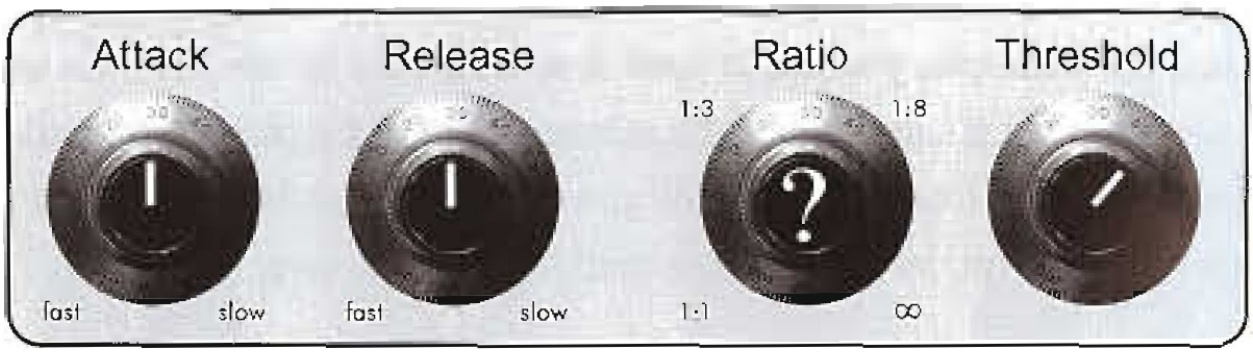
The second step is to play with the release. Release is the amount of time that the sound will be muted when you exit the compression mode. The trick is to make it the musical component of the sound. You may ask "Do you mean the release time according to the style of the music?" or “In fast styles, should the release be shorter than in ballads?”. Maybe just don't think "I want to make this time shorter to clean up the dirt" - don't do it. On the contrary, make the release as long as possible, so so that the compression envelope "fits in", supporting and/or creating a groove in the music. Remember that any change in dynamics in the music affects the groove, and compressors/limiters are no exception. (When the singer is getting closer or further away from the microphone, or you yank the fader like crazy, or the compressor chatters the sound level - the groove is in danger of being destroyed due to such dynamics). So don't make a quick release just to hear something buried behind the sound. Forget about it. There is a bigger fish here, and we will fry it. You're already compressing a bit, so those background sounds will come out anyway. Instead, you're probably thinking "How slow can I make it without losing control?" as the energy of the groove moves more slowly, it's subconsciously still a powerful wave - and it's not a fast attack that crunches the sound. Even in an extremely fast tune, the slower, subconscious undercurrents carry the bulk of the energy. For example, you can make the release so slow that by the time of the next hit, the release is not yet complete. But there is nothing wrong with that. The formulaic approach can logically tell you that the release for the next beat should be completely done, but it's more math than groove.

Listen to the release. Feel how it slips away and comes back to you again, and there will be a point where you feel that this disappearance and return becomes like a hesitation - like someone is swinging on a rope in a groove with music. It doesn't really matter if it's on time, because the groove - as anyone who teaches music will tell you - has to bide its time, but it doesn't have to be on time. Never play with a metronome. Never play with a conductor's baton. Don't release in a quarter beat or whatever - catch the groove, that's your release time. Breakthrough the release of a musical component that pushes you to the next beat without first emptying the previous one. Give the musicians back "push" you V moment, When pressure continues grow, instead of Togo, to give the compressor complete This swing - dead air - lifeless moment… Nothing

good. Let the compressor push the sound forward while the music prepares for the next beat.

If, as usual, you're all worried about maximum volume (no matter how harmful it is to the groove), then ignore this last paragraph and put the release in "maximum annoyance" mode. But I must add that if you are trying to make the product appealing (with the maximum groove, for example), the listeners themselves will surely raise the volume of your track, which will give a greater effect than any hard compressor.

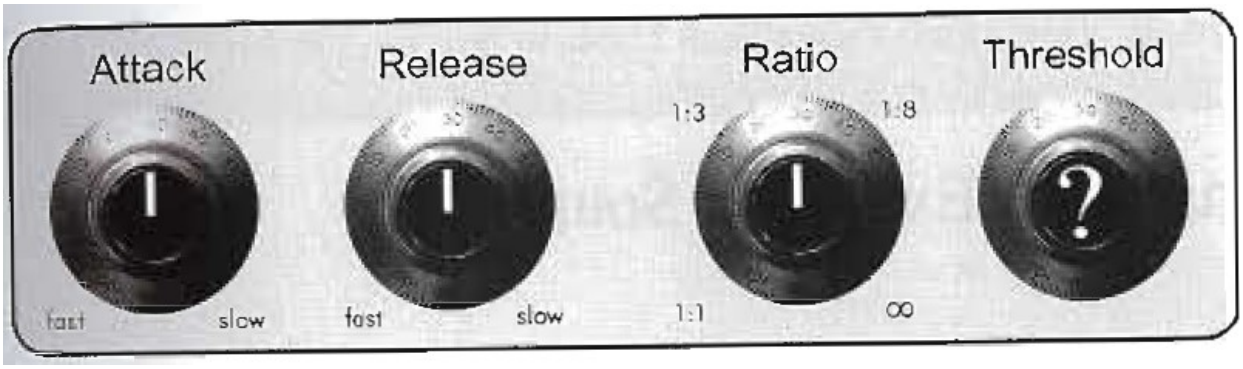
3. Attitude (Ratio)



At the moment, the ratio is set to the maximum, so everything sounds overcompressed. Next target - lower his maximum So, to the effects created by the attack and release are preserved.

Think of the ratio control like the lenses of a telephoto lens - the larger the ratio, the smaller the sound, although it may be more controllable. Reducing the ratio to 2:1 (while delivering the same voltage) is aesthetically similar to a larger picture. The smaller the ratio, the larger it is - with the risk of going beyond the acceptable limits. And vice versa - with the growth of the relationship, it becomes smaller, but within a stricter framework. The idea is usually to try to make it bigger, but keep it within the proper limits. So lower the ratio until you can't hear the effects you like - the thickness of the click, the groove created by the release - you can raise the ratio a bit by constantly controlling the size of the picture. At this point, don't think about attack and release, and don't think about release in terms of numbers - think about the size and signature of the sound. Did you count how many times I mentioned "brand" and "hardness factor"? So, raising the ratio, the sound becomes more firm (and smaller), and when lowering - softer (but more). So you might be thinking "How branded do I want to make this?" instead of "How big do I want to make this?".

4. Threshold (threshold)



The last thing you'll adjust is the threshold. It's important to set the threshold so to compressor Not squeezed sound constantly. correct installation threshold lets you see the movement of the dynamics taking place at the right moments - otherwise you'll end up with a flatter, lifeless sound.

Uncompressed fragments of sound that appear from the processor at the right moments give the sound contrast and color. For example, the ability to fade dynamics processing into quiet moments gives those moments the chance to achieve a 1:1 temporal presence and avoid being pulled towards the listener from unwanted noise. It's rather unfortunate that short quiet moments remain small without being overwhelmed by large ratio values. Each time sound approaches air, it acquires a sense of reality - a ratio of 1:1.

Warning! What many engineers don't realize is that compression ratios are multiplied, not added. If you compress a mix 10:1 and then the mastering engineer compresses it 10:1, you end up with 100:1 instead of 20:1. Consider yourself forewarned. This applies to any compression. If you press 10:1 and then 4:1, you don't have 14:1, but 40:1. The next time you hit a mix, consider the ratios commonly used on radio stations as a last resort. Ask yourself "how small can I create a sound to send to ether?".

This Very Expensive sound

If you follow these steps, set up your compressor as shown, and you're good to go. way yellow Pens, To the moment When You get there before this places V paragraph, you'll have a big, lively sound with a smooth groove. Or, in other words, "more expensive sound".

I can promise you - whether you're compressing a single guitar, voice, drums, saxophone, or an overall mix, these steps will get you there quickly and decisively - by training your ears to hear what you need, you'll get away from working blindly. It's like growing big ears because you hear the effects more clearly; and a big brain, because confusion is replaced by purity of thought.

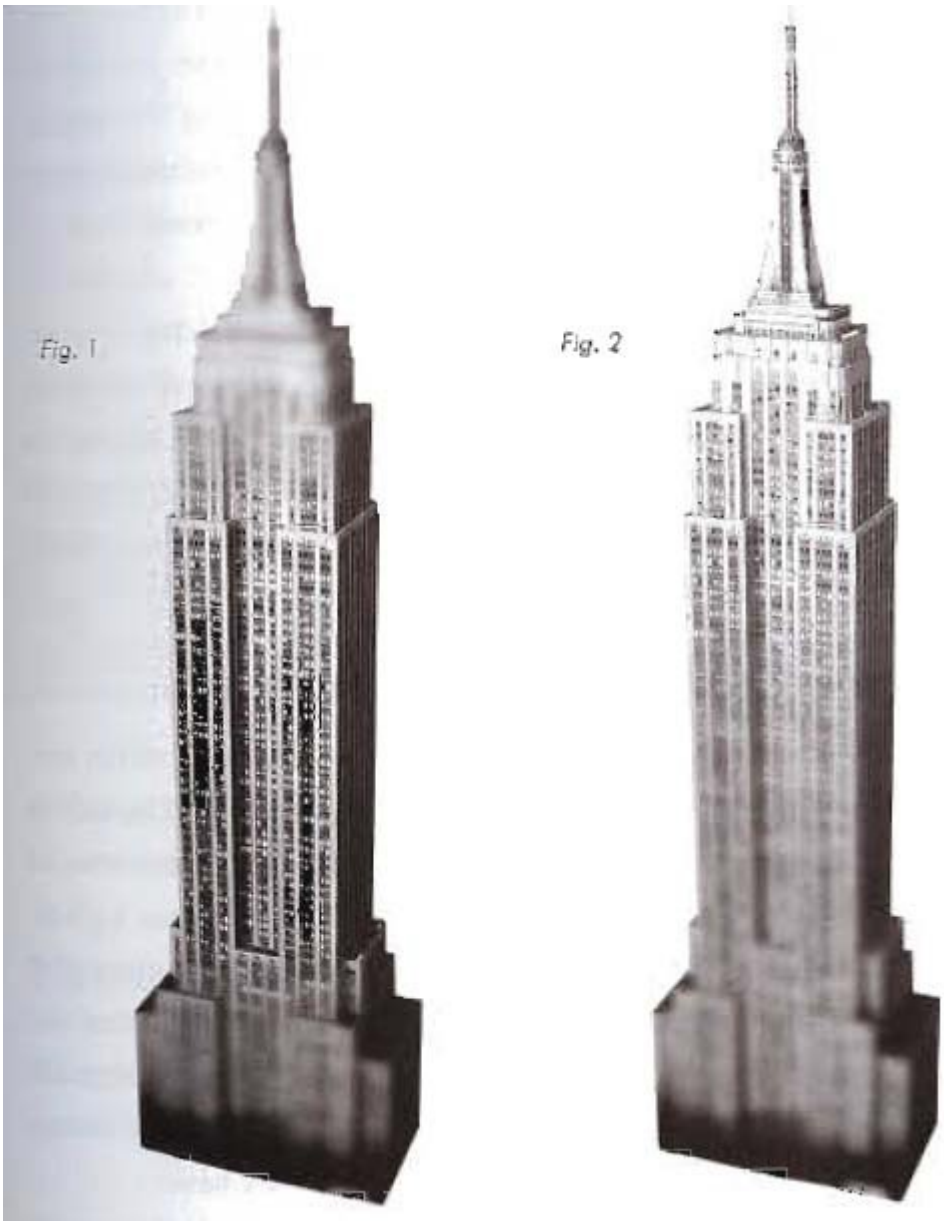
NUMBER AGAINST ANALOGUE

The difference between digital and analog recording is much bigger than you think. It's all a matter of focus. Each wearer adjusts their lenses for different levels of music dynamics.

This chapter explores an interesting analogy by evaluating the difference between these two forms of notation.

I am by no means an expert in digital processing, but would like to share my perspective on how I see and hear the difference between digital and analog recording. If I'm right, we should stop ridiculing digital in order to get the most out of our digital records.

You know what I always say: “audio behavior is often the complete opposite of what we logically expect from it”? Digital is no exception. For example, in analog recording, lowering the level always reduces distortion. In the digital world, the opposite is true. What's going on and how does it affect the music?



Rice. 1 - Analog focuses on the middle of the skyscraper - all the juice will be in the middle. The top and bottom of the building are V weak region lenses. But This Not scary, because resonance, body, heart And soul sound located in the middle of the dynamic range.

Rice. 2 - Number, vice versa, guides maximum focus peak building, gradually losing "permission" with a drop in signal strength. Oscillations at the top of the signal envelope are transmitted with maximum fidelity, and lower fragments suffer. In the first place, a high-contrast limit signal, and medium sounds, with warmth, resonance, remain in second place.

Analogy skyscraper

As an analogy, I would compare the dynamics of a recording to photographing a large skyscraper. As you understand, due to the huge size, it is very problematic to capture the entire building in normal focus. This is where Analog and Digital differ - limiting the focus to different parts of the building. In the case of the analogue, the focus falls on the center of the building (where the maximum movement occurs). Building sections at ground level (foyer And the shops, analogue noise threshold) blurred And unclear A top building

(radio antennas and upper floors - for example, bursts of drum beats) are also out of focus and slightly blurred. This is partly acceptable, because. the lower floors are usually the noise of the film, and the upper floors are only fleeting bursts.

A skyscraper is analogous to the volume of the sounds we record. Analog recording "remembers" the most important parts of the sound's dynamic range with the highest fidelity, while the low and high points are less critical and are recorded more rough.

Digital skyscraper

The digital representation of a skyscraper is somewhat different (see Figure 2). In digital, the focus of the camera is constantly directed to the top of the building. The upper floors and the antenna are extremely clear, while the rest of the building is gradually out of focus from top to bottom. Why? Because the top of the building is the only part played (and recorded) by all the bits involved in the recording.

pruning at mastering

To extend the analogy, it's sad that once you've finished mixing digitally - with the cleanest elements at the top of the dynamic range - what usually happens when you send them out for mastering? The best bits are surgically cut out to raise the overall level!

So, unlike analogue, where the worst elements are removed, here (in the digital world) the best, most focused elements disappear, leaving us with grainy midrange sounds. And we did not notice their graininess while they were quieter. Now they're louder on 10dB, and graininess became noticeable. Wanted loud record for cheap? And how did we get to this point?

Story

How did we not notice this in the first 10 years of digital recording history? The reason is that we forced our ears to focus on the weaknesses of analog recording in order to exalt the virtues of digital. Seems logical. When digital burst onto the scene, the first thing we heard was "the top of the skyscraper" with breathtaking accuracy. It seemed like magic that the peaks of the signal envelope (which always sound bad in analog) are finally reproduced accurately. It seemed inconceivable that the simplest and quieter parts of the signal could be affected. But it happened.

Optimal level For work.

We believe peak signal the best For any audio processor, This "optimal level of work", because the signal does not suffer from crosstalk or noise and loss of detail due to low recording levels. "Optimal Level" is traditionally well below peak distortion and above the noise floor. This goal is clear and simple.

classical guitar

When the understanding of numbers came to me, it gave me an explanation of an old riddle from the time of my recordings of classical guitarist John Williams (we recorded 11 albums together). We recorded separately digital and analog (quarter inch Dolby SR). John has great ears for our business, which opened the way for me to listen to the quality of our recordings, produced separately for analog and digital in during double-blind tests (even after copying to audio cassettes!). John noted the discouraging disappearance of certain sounds from the digital recording. He noticed that the sound of his fingers bouncing off the strings was always absent on digital recordings, but always present on analog tape. I can only assume that this disappearance from the digital recording is caused by the location of these sounds below the center of the skyscraper. - those., Maybe because of playback only six active bits.

living space

Each write space has an "optimal" write level, where shows itself from the best side. Everything that is above this level, we call living space.

Q: What is the size of the living space of your favorite equipment? For example, I can say that most good analog consoles have space V within 20..30dB. Imagine, What you have case With equipment, which has no such space at all. Guessed: the living space in the digital record is equal to zero.

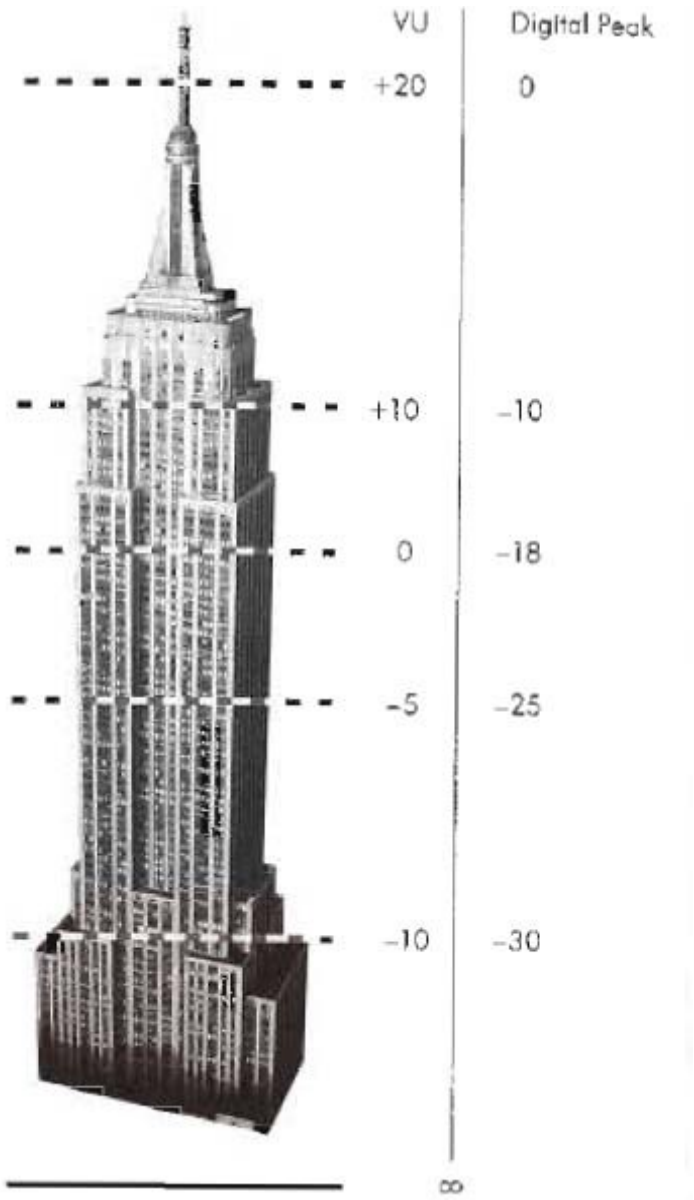
What it's me I have V mind?

The maximum level is the optimal level for the digit. It would be more accurate to say that there is no living space in digital recording devices.

Naturally, you can create a pseudo-live space by lowering the recording below the optimal level, say 18dB. But you must take into account that in some degree worsen signal, Not writing down his on optimal level - those. you have *a smaller number of* significant bits that describe the most critical elements of the recording - namely, the musical information, which is significantly below the peaks of the envelope.

If by some miracle you hear audio components that have full bit resolution in the digital recording, you won't hear any harmonics. You will only hear the fundamental frequencies and some of the wobble-related pops and clicks that push the VU meter to its maximum.

Once you understand what it is about, you will be able to approach issues in different ways.



Rice. 3 - Dot V digital records, With maximum purity, minimum noise And distortion (0dBFS) - it's a point with 0dB of live space above it - you just can't rise above without limitation and distortion. Those. to please from an almost perfect sound to a distorted one, even 1dB is enough.

Recording on digital multitrack (carefully)

The beauty of digital multi-channel recorders is that they don't have noise. You can include all 32 tracks and not hear anything unwanted. The ergonomic interface of such systems, however, encourages the user to make compromises in sound in exchange for easier control.

When writing to a digital multitrack, don't be fooled by the super-low noise floor and think you've got a usable dynamic range. A false sense of security pushes you to play all tracks at the same volume while holding the digital recorder at just the right level to properly lay signal V phonogram, at this exposing All fader "V zero". This,

publicly recognized most comfortable, an approach - one from the worst of things, which can be done with sound quality.

People who use hard disk recorders often do this because it seems natural. (You only need to work with half of the faders). Such systems are often too difficult to adjust as usual - first on the input level, and then operate the output fader to create balance. It's much easier to leave all the faders at the same position, and only control the input gain to balance the track with the rest. Fader in half! This is very fast, but very bad for quality.

This really necessary - lift up level signal, Bye He Not reaches the digital maximum before adjusting the output fader level of the recorder to achieve monitor balance. (This is a basic theory, I know, but it's too often ignored in the digital world). If the interface were faster, you probably would, but today's choice of digital remotes is an even weaker link to sound quality than bitrate (but that's another story).

Fooled bounce track

One of the reasons we rarely see signal degradation when working this way is that during recording you hear the balanced original analog signal before the conversion. The damage done becomes audible after being played back by a digital recorder. But during this playback, we focus more on the musical performance or some other problem, and also lose sight of the problem. The study of signal quality should be done at the time of recording, and not later listening (until the problem crawled into the mix).

Недооценка шипения

Мы договорились понимать шипение, как акустический компонент, который маскирует тихие гармоники. Это действительно так, в любом случае. Однако его отсутствие не говорит о том, что гармоники вне опасности. Если на аналоговом плёночном устройстве шумовой порог предельно мал – без шипения – тихие звуки прекрасно читаются ввиду недостатка шума. На цифровом устройстве записи, однако, неверно будет предположить возникновение такого эффекта. Всё происходит иначе. Не смотря на то, что шума нет, низкоуровневый сигнал описывается меньшим количеством эффективных бит, чем высокоуровневый, т.е., не смотря на отсутствие шума, сигнал чище не становится.

Structure Geina (Analog-Digital conversion)

The next time you need to transport an analog multitrack to a digital system, try a trick that can be done with any high end 520 nWb/m2 tape like Quantegy GP9.

If you tune the digital recorder's perception of the 0VU reference signal from a 520th tape to a peak level of -14dB, the maximum signal that can be transferred film will become equal +0.01dB For numbers, those. We we get

operating level without overload. Beyond this level, the analog film has an already absorbed and latent energy, as if the clouds had descended obscuring the Sydney skyline. It's worth experimenting to find the saturation point of your favorite device and calculate the gain needed to synchronize to the maximum level of the digital recorder.

Adjust my structure Geina

To calculate the equivalent for your tape device, record some extremely powerful signals on blank tape. Point them to a digital drive - it has more accurate meters. Adjust the recording level as close as possible to the overload limit. Now, without making any further adjustments, play back the VU reference signal from this tape and read the signal level on the digital level meter. In the future, tune your 0VU signals to the same digital level. (This technique works more accurately on quality films, as they saturate faster).

Not panic! (we control drifts)

Any racer driver knows What Not important With Which speed You you drive A what matters is how you control the skid. When you optimize the level when converting analog to figure, an overload should be expected; but thanks to the hardware designers - you will learn how to use their red LED as a sign that you have reached the limit - do not cross it. Practice will put everything in place, the race will be won in the zone skid.

Level against clarity

Analog - the loudest instruments are blurred, the mid-range instruments are the clearest, the quieter instruments are in noise.

Number - the most loud tools cleaner Total, medium - V compromise quiet
- grainy And lost.

Ideal copies No - If You listen to V CD, which reissued - again and again - notice that only the loudest instruments sound clear and round, and the rest have lost their integrity, retaining only outlines. Analog copies of copies degrade V reverse manner. middle survives A on borders increases blur and depletion.

Expression - in musical history, as the volume of the instrument increased, its intensity and passion became more noticeable, bright, hot, intense, emotional - all these words suggest overcoming the barrier and the inevitability explosion. Indeed, in the case of analogue equipment, including the human ear, room acoustics, musical instruments, everything gradually becomes less soft, more harsh, blurred and distorted as the upper limit of sonic expression is reached. This language of dynamic expression takes place in long passages several minutes long, in one phrase or word that fits in a few seconds. For the first time in history, we have a technology that emphasizes this expression in exactly the opposite way. The loudest signals become soft, and the quiet ones become blurry.

AND You wonder Why There is feeling, What something Not So? Our meters And spectrum analyzers say that everything is OK. trust your feelings.

Mixing V figure (pre- mastering)

How often have you given the mastering engineer a great mix and got a grainy one back? And crackling master? Not blame in everything mastering engineer. IN In the process of mixing, three main mistakes may have converged to cause this effect:

You established entrance V DAT So, What 0VU was equal to -18dB. Big error!

Because the at digitizing you did not receive overload DAT, You assumed that the dynamic range is all right.

You have been monitoring at too high a volume, forcing amplifiers, monitors and ears "swallow" many bursts, giving you a false sense of their space. These unnoticed spikes will also give you a false impression of the perceived loudness of the product. Sharp bursts, limited in the mastering, will cut the volume of resonant components in your mix by tens of decibels, making the whole product quieter, but you will never hear those sharp bursts listening to material at 110dB SPL! The mastering engineer will do a real check and suppress these bursts in the process.

Again let's think O your "digital OVU"

If you don't have a VU meter in your studio, skip this paragraph. If you have there is a VU meter on the output of the mix, the big question is: when do you send 1000 Hz signal at 0VU, what should happen on digital peak meters?

It's reasonable to assume that the lower you map 0VU to a digital input, the freer and less controllable the dynamic range can become. In other words, the lower the bar, the more living space for recording. (You will eventually fall into this trap.) I propose to raise this bar. Don't worry, raising the 0VU/peak equivalent doesn't reduce usable dynamic range, but it does reduce uncontrolled and random.

If your 0VU is set to -18dB peak, you can afford not to worry about dynamic range, but you can pay for it at the mastering stage when someone wants to compress your material. A level of -18dB means that even a sloppy engineer is unlikely to overdrive. On the other hand, if your 0VUs are equivalent to -12dB, you should definitely keep an eye on the level and work extremely hard on your psychoacoustic goal of achieving Maximum Illusion at Minimum Voltage (our goal in life, remember?). Setting zero to -12dB is like taking the game to the next level. When your gain settings are done like this, you'll quickly learn to control those annoying distortions that prevent your instruments from recording at their best at maximum digital resolution.

Start by feeding a 0 VU signal from the mix output (0VU = +4dbm=1.228V) to bring the DAT to -14dB. Ideally, tune the DAT output (using variables resistors, If necessary) on That same level (+4dB=1.228V), to achieve

unity of levels in the chain. If you don't have a VU meter, purchase and read this again.

Useful tools (Exact measurements)

A combination of VU meters and peak meters is required. In the digital space you want a higher VU to peak ratio than analog. If you manage to increase the VU without increasing the peaks - that's cool! This means that you have achieved more effective volume. If You received climb peaks without increase in VU - This just a waste of energy. Over time, you will be able to raise your 0VU bar from -18dB to -12dB! By comparison, in the past, when using analog recorders, a skilled engineer would tend to increase the ratio of the highest peak to the 0VU level. The peak could penetrate the blurry area of focus without distorting the VU-sensitive components. We know that analog systems (such as film) absorb peak bursts by half, so it took a lot of work to create powerful bursts. In digital studios, this approach backfires. VU ratio to the peak is difficult to measure without VU meters and peaks. Even the most experienced engineer will benefit from both of these tools.

It's only "trendy" manufacturers who resist the use of VU meters. Might be cheaper without them. I have seen good meters on the screen - so the output for manufacturers may be the release of the corresponding software instead of
"iron" equivalent.

Compression (limiting mix)

Compression is not as saving as you might think. The beauty of analog tape compression is that half of the peak spikes are absorbed, leaving the sub-threshold envelope intact. However, when using a peak limiter, the signal curve below the clipping curve is also distorted. Why?

Black background

There is a little-known psychoacoustic effect called "black background". It can be seen while listening and colorographic visualization of the mix. Imagine that you "see" the mix - the space behind the farthest instruments may appear to you as a gray or black background. I associate the mix with a black background because blackness neatly outlines the contours of all sounds, giving them individuality and distinguishability - "The Dark Side of the Moon" gr. Pink Floyd is the best example. Engineers create this effect by selectively filtering sounds so that their harmonics do not interfere with the weaker harmonics of neighboring sounds. The digital to some extent independently creates this effect, suppressing weak harmonics (not selectively).

Peaks And Job limiter

Reaction limiter (knee-jerk) is V suppression records generally, A Not only peak. A peak is just a command to the limiter to reduce the overall volume. The limiter doesn't shorten the peak, it leaves it the same, but makes it lower in level like everything else. All constituents envelope signal, having place V moment peak,

exposed strong distortion form knee And timing limiter. Whole peak "pushed" into the mix, taking the sound with it, simultaneously pushing that sound into the low-res dynamic range zone.

This creates holes in your mix. You should process these bursts independently of other sounds, as long as you have them on separate tracks. Forcing a mastering engineer to work on such a problem is fraught with trouble. The best recordings are practically not processed during mastering. It's not because the mastering engineer can mess something up. This is because 99.9% of the work is done before mastering.

Bass - our Friend

In analogue, the bass is our enemy. In numbers, a friend. Bass is a major contributor to over-throwing the signal and bringing the signal closer to a level where you could break a vinyl record. Bass stronger affects on vu, how high, because human ear less sensitive To bass, and you have to create a large RMS at low frequencies to compensate. Since digital transmits peaks, bass is no longer a problem - extra bass cannot cause a CD to burn at a lower volume. You can add a lot of bass to the mix without moving the peak meters. If you have had experience with the analog-to-vinyl puzzle in the past, beware that, perhaps out of habit, you will start processing the mix in in a manner opposite to that required by the figure when it is necessary to achieve "maximum illusion with a minimum of stress." Don't be afraid of the bass - save the fear for the peaks.

Number (doing All Right)

It's taken a lot of time for audio engineers to squeeze the best and sweetest sound out of the world of analog tape, and it shouldn't come as a surprise that we're still moving towards rules that will help get the most out of digital. Realizing that there is a fundamental difference between analog and digital is a good platform to jump-start.

Moreover, the digital world is still in a state of constant change, with changing standards, new media appearing here and there, PCM vs. DSD… phew, it's hard to stay on top of it all. Whatever happens, don't forget the basis of good engineering practice in all this craziness. No carrier will be able to independently answer all questions regarding the reproduction of what happened in your tone room. Work on the compromises of the digital world so that your sound doesn't compromise. The point is precisely this.

And above all, listen. Regardless of your hardware specifications, if you feel like you're hearing something "fun" going on, you might be right - only you won't be able to read somewhere about the inconsistency until the manufacturer fixes it...years later. Trust your instinct as I have found that when you listen too closely to the difference between the two systems, it often becomes impossible to understand exactly what is going on with the sound. Perhaps this is something that defies logic. Just avoid the "slightly annoying" stuff and that's the right starting point!

DAT/CDR Conspiracy? (determine the calibration of your output)

We all like to hear the same volume when switching between the mix output and the tape signal. The DAT makers know this and covertly change the recording level by calibrating the output gain. Because digital has zero live space, digital device manufacturers set the output level to around 4dB, which in turn forces us to lower the level by 4dB. This protects And manufacturers And us from exit on surface unpleasant distortion from overload when reaching the upper limit.

More something O figure

This phenomenon does not work in favor of numbers. In analog, this is not so scary - it will simply lead to some blurring and distortion of the loudest instruments, because, by their nature, they are always audible than others. Secondly, if the loudest sounds are distorted, this corresponds to the natural effect of loudness. You will never intentionally use an effect that distorts only the quieter sounds in a mix, leaving the louder ones clear, but that's essentially what digital does.

Effect digital trading stamps

Imagine a mix where you're manipulating levels - pushing instruments up a level and then back quickly. In digital, the focus is on the tools in front of you, and with the distance, the distinctness decreases. Listen again and see if you can hear what I'm talking about. As the level and intensity of the sound rises, they become clearer. If it's quieter become more distorted.

Exceptions - if your CD starts to distort at high volume, it may be caused by installing a poor quality (cheap) op amp on output of the D/A converter. Perhaps the manufacturer has set the internal gain so that the operating level is, say, -18dB, but the op-amp range is 10dB. This often happens.

Swedish cheese

Remember, if you don't control the dynamics, the mastering engineer will take care of it, but He will more ruthless And Not That's why, What He angry, A That's why, What him on thirty fewer channels to influence the process. When the mastering process raises the level of your mix to the optimum digital level, the peaks will stop protruding as start making holes underneath. The signal remaining between the peaks, being at low resolution, will become louder, and you will hear its granularity more clearly. But it will be too late.

Prevention is better than correction and better than punishment.

The solution could be to limit yourself when mixing and work with these problems, And So - tool behind instrument, instead of Togo to wait miracle from

mastering. Working with the instruments individually prevents holes from being "drilled" into your mastering mix.

Mixing And meditation

As a prologue to the chapter on the art of mixing, a word or two about the reality of diving into your creative subconscious.

Last year I was present at an experiment where a sound engineer was connected to an EEG device. We've looked at what happens when it's immersed in sound to add a little echo and pan some space around. An experienced EEG operator suddenly stopped the picture and said, “Wow, look at on this! Your Alpha and Beta waves have completely shut down, and the Theta wave has grown like a bubble! Are you sure you are not in meditation? Only Tibetan monks are capable of this!” "No," said the engineer, "I've never meditated in my life - but I love mixing."

We have all heard about transcendental meditation and the benefits of studying such disciplines. Each meditation session seeks to clear the mind of all the rubbish that accumulates throughout the day, week, month - the trick is to regularly throw out all this rubbish so that it does not accumulate. Most people who have never delved into this discipline and were not interested in such a pastime question the expediency and even sometimes criticize people who do this, talking about a waste of time. Meanwhile, those who are carried away by immersion in this relaxing silence think of others as not controlling their body and mind enough to achieve balance.

This experiment with the sound engineer is true. I know this because I was there, otherwise I would not have believed it. The EEG operator had studied many brains, and was particularly interested in looking into minds in altered states of consciousness. He noted that practitioners of meditation are able to easily reduce the activity of certain areas of the brain, and their most impressive ability is to control both hemispheres symmetrically. He could, by the state of the brain regions and their symmetry, determine where the subject was immersed in significant forms of meditation.

How could this rock and roll engineer fool an experienced EEG operator? The point is that he didn't fool anyone. He meditated for many years without even knowing it! Perhaps all of us who compose music and gaze into the cosmos, achieve inspiration by isolating ourselves from this world, are actually engaged in a similar form of self-healing or mental support - like Tibetan monks. Have you noticed how distant and irritable you become when you haven't been mixing for a long time?

One gets the impression that the human mind is designed for the creative process - painting, poetry, music - focusing on something that will block the piling up of everything unnecessary. Even monks don't think about "nothing". They often focus on their breathing, which seems like "nothing". Well, I prefer mixing music. A happy sound engineer is one who, being undistracted by people and all sorts of devices, sinking deep V mix And acquires big clarity V mind.

sound engineers - strange creation

They are often mistakenly considered techies. And this is due to being surrounded by sparkling light bulbs And digital technologies. All So, There is heap technical

tasks, But moments concentration - When going created them magic - This moments of pure imagination. Naturally, something will certainly be distracting, but the trick in is to achieve and maintain a state where everything is clear and the goals are simple.

clash face To face With illusion or ghost between speakers - the most healthy thing For intelligence And us themselves Same.

Distract?

Because every mix deserves a fresh approach, organizing some of the well-known elements of a mix can spur creativity in unknown areas.

Don't just dive into the mix - don't jump right into the drums, for example - because once you get into creative things (right brain), you will inevitably run into a number of technical problems later on (left brain). Instead, solve all technical problems first (signal paths, cables, failures).

Secondly, if you have not yet formed an accurate picture of the sound of the mix (few people know how), in the preparatory phase you can fill the subconscious with meaningful information about the specifics of this case, catching inspiration at various levels .

Your meditative, creative state is the most important component of the progress of the mix. So, next time before mixing, be ruthless...cut all these problems (kick the producer and musicians out of the control room!)

Art intelligence (Part I)

Break down the mixing process into manageable chunks and allow yourself to focus on the art, not the science of mixing. And here's how.

Mixing is my favorite time. I love it because it's complicated. It's like playing chess on 10 boards at the same time. The solution is how to simplify the process so that you spend most of your time in your creative head area - known as the Zone - i.e. in the right hemisphere. Even if you haven't heard the description in those terms, I think everyone knows the feeling when you close your eyes and see sounds...like ghosts between the speakers.

It doesn't matter if you're mixing 8 or 88 channels, the mixing process is exactly the same most. You can do hundreds of things with every mix. Some of them are from the right hemisphere (creative), some are from the left (technical). Switching from one hemisphere to another interrupts the force that drives thought. Our goal is to support and deepen the creative beginning, without technical, from the left hemisphere, problems. How can you control a machine complex with a thousand handles without letting the technical needs chase the ghosts between the speakers? Answer? Try to separate the functions of the left and right hemispheres so that each gets what it needs from a concentrated stream of attention.

To introduce you to this concept, I'll list the various tasks that everyone will need to complete in order to complete, for example, a 60-channel mix. The list is in special order. Note that a hemisphere is assigned to each task. Remember that the left brain processes technical data and the right brain processes artistic data and is responsible for the Art of Mixing.

Mixing - This control above mind

... and not above the console. Your brain craves creative space. Most of the time working with technology blocks us from even reaching the true state of being creative. Stop beating around the bush, forget about the console and give way to the mind.

The Key to Mix Control in the complex - this is mind control, not equipment. A detailed list of tasks will help you maintain concentration in both hemispheres. By grouping tasks, you get away from having to switch between hemispheres.

Task	Hemisphere	Target
Approximate outline	Right	General picture
Compound cables	Left	We connect spaghetti
Structure Geina	Left	Technical questions
Reduction lines	Left and Right	Training subconscious
Settings panoramas	Right	Composition

Equalization	Right	Setting imagination
Compression	Right	Content magic
Balance	Right	Glue And attraction
Reverberation And echo	Right	room, space, groove
Movement faders	Right	We express myself
take	Left and Right	Only Not smash

1. Approximate outline

First of all, sketch out the structure of the console. With a track list in hand and sounding song, familiarize yourself with the subject of the work. Now your task is to describe where *you want* to have the faders of the channels you need, and not as they are now - by chance. It is extremely important to place the instruments where it would be convenient for you to manipulate them, like a pianist playing the phono.

Multitrack tracks should be located under the fingers so that it is possible to simultaneously control instruments with due dexterity. For example, you may find that if you position the section of copper pipes correctly, your little finger will be very conveniently above the flute. Another approach is to group instruments that sound like a group. Need to think ahead - I always have an empty channel after behind voice For his subsequent echo And other processing. Approach the issue ergonomically, the time to put everything in its place is now, not later. IN after all, a multitrack never endows an engineer with good ergonomics, but after grouping the whole mix starts to look like easier, So What Further You you can Just "play". Use a few extra wires, make yourself happy. Your right brain will thank you very much.

By the way, avoid using "words" on the sticky tape of the remote. Instead, use your own symbols. The brain perceives symbols faster than words, especially when you yourself have come up with these symbols. It's more interesting to make each track look the way it should sound.

Here some my hieroglyphs, But to you costs come up with their.

Fartsuzsky Horn Vocal Tambourine Saxophone Electric guitar Ac. Guitar

Another benefit of using symbols is that they can be recognized from anywhere. rooms, V difference from letters. Large groups tools should designate

big symbols. One long squiggle for five violin tracks makes it easy to visualize the mix. And most importantly: only you know who is who :).

Connections cables (infrastructural stage)

In the process of mixing, we are dealing with two things: musical instruments (sources of sound) and infrastructure (pipes through which these sounds pass). Let's deal with the infrastructure. This includes all signal paths. pave the way To And from effects, Not forget about devices mastering - devices records stereo, or
5.1. Now get it their sketches.

So far so good. So far, everything is interesting and not problematic, and you are getting closer and closer to the material without having made a single creative decision yet. This is important because On this stage, you are still working with the left hemisphere, and any creative now will not grow together with the spirit of setting up equipment.

While you're in this "infrastructure mode", test all FX returns and their signals with an oscillator (preferably pink noise). You don't want to be distracted by solving echo return issues when you're in the midst of a ghost hunt. Plug in all the reverbs, delays, phasers, etc. you can use as you see fit. Then send on each of them pink noise to check the connections, balance the sensitivity and volume, detuning from noise and overload.

The best structure Geina For reverbs.

Feed pink noise at operating level into the reverb and turn it up output so that the VU meter on the console reads -7dB. This is 7dB below 0VU or 7dB below the equivalent target peaks. (i.e. a target signal of -14dB would correspond to -21dB). I always keep one fader to connect the pink noise generator so that I can quickly feed it to any processing channel or bus. Sending pink noise from a channel is better than directly connecting the oscillator to the effect, as this detects any problems in the signal path.

Pink noise works better than a sine wave because the latter creates standing waves in the reverb, making it almost impossible to balance its output.

You are almost ready to start creative. But first, double-check your mastering devices. They should have been enabled before, but check anyway. There's nothing worse than plugging in a stereo recorder *after* you've listened to the first take, six hours from now.

Structure Geina

Send a 0VU signal from your mix buses to get -14dB on the recorder. On the monitor panel, select "TAPE" to see that the return from the recorder corresponds to 0VU. The same gain is important. Do not correct the output by changing the input sensitivity of the recorder. Keep monitoring the mix "through" recorder. This is a guarantee that you hear what will be the result.

Now everything is connected - the console is organized, everything works - all you have to do now is listen and twist. **Warning** : Don't let the rush of the session shorten these first steps. They will save you time and free your head from problems later. Even though it seems obvious, you'd be surprised how many engineers, say, start tweaking the kick sound two steps back when it comes to console ergonomics.

Before diving into serious creativity, our next step is to prepare the subconscious mind a bit, add appetite to it.

Tuning linear levels.

It would be nice if there was a simple and productive method that has a triple advantage for your mix:

Preparation mind;

improvement vital zones channel (prevention distortion = more clean sound);

location faders V most practical And "Juicy" positions giving best control.

Here this method:

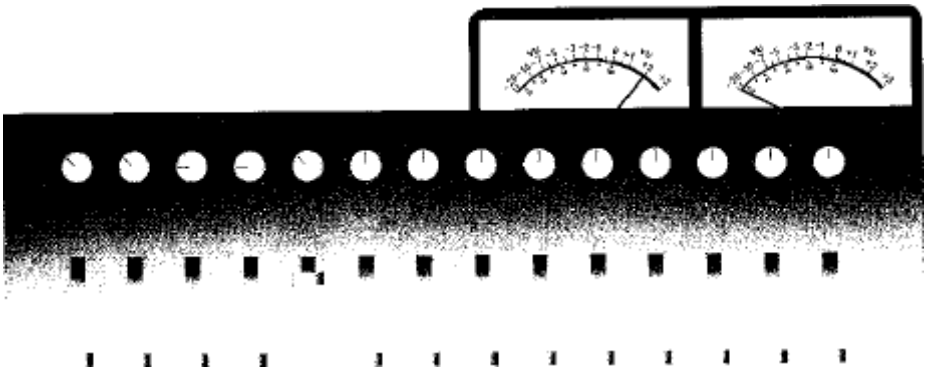
turn on signal, bringing out all faders;

turn on control monitors on adequate volume; enter the first

channel at maximum (+10dB on the fader);

lower sensitivity channel So, What would VU meter on output mix showed +2dB;

bring out fader And do That same most With the rest in turn.



Rice. 2 - Enter fader on maximum And lower sensitivity up to +2dB VU on output mix, turn off the channel and move on to the next one.

It must be a very fast process. Tuning each channel should take about 5 seconds. This will help you remember them.

I just start from left to right, push the fader all the way in, adjust the gain to +2dB on the output, then hit MUTE and move on to the next channel.

What We this do we achieve?

Your task as a sound engineer is to achieve "Maximum Illusion with Minimum Voltage". In the process, you will find that some sounds seem loud and some sounds quiet. Your subconscious mind will learn which sounds are weak and which are strong. This will help you understand which ones need more help and which ones can afford to round out to add depth to the mix.

Trying each track on a single “voltage”, the subconscious mind will remember the Illusion / Voltage ratio for each track. There is no need to make every sound jump out of the speaker with volume and presence. This approach will rob your mix of depth and create a sound made up of small, weightless sounds. As you go through the sounds, you will notice that with equal voltage, some sounds are louder and some are quieter. **Remember** : only "high-voltage" instruments limit the sound level of the mix on the final CD. The ones that shine through the mix, sitting at -10dB, are already more efficient on their own, and therefore don't raise the overall volume, as they don't need to lift themselves. All you need to do is simply lower them down just enough to achieve musical balance.

This quick view of the tracks at a single level will also put the dynamic processing into perspective. No need to compress every track, it won't bring benefit for the above reasons. Only weak signals require this.

Your subconscious mind will digest all this information until the end of the mix. While your mind is in the background digesting this seemingly insane idea with uniform levels, here is the thing you need to consciously look for. Constantly look for the "star performer" - the magic in the track. Everyone can determine who is playing the melody, but I suggest you look for the brightest track - perhaps even a member of the background “lining” that creates the groove. Try to find the one tool that makes you daydream (…that makes you daydream). This track will be very useful to you in due time.

This line-sensitivity exercise leaves technical challenges behind and prepares the mind for the fun ahead. At some point, you will be solving a complex puzzle, with dozens of sounds. They must match each other. Most engineers start Just co his beloved tool - doing his so as big as possible - and then braid everything around him. Thus, each successful sound becomes less and less, and the first one is erased with more and more hard colors. Sounds familiar?

By setting a crash course with the same level for all sounds, you are pushing your subconscious mind to create a direction that will help you avoid this common mistake. Not costs underestimate importance this small step. Your subconscious

thinking is much more powerful than you can imagine - and yet it craves action!

The extra live space and elevated fader state resulting from this exercise is just a bonus. In addition, I've been told that lowering the sensitivity offloads the power supplies of expensive consoles, so that's another penny for cleaner sound.

This approach also makes it possible to hear mix with "single voltages" -
If nominate All faders evenly, How along the line.

If your console doesn't have gain controls, pretend it does and do this anyway. Listen to each track at the same level.

Your task?

Density against transparency - cup half full, or half empty?

Have you ever noticed how the same mix can seem full and dense to someone, and dirty to someone else? Likewise, a cleaner and sparse mix for one may seem empty to someone else. I always contrast "Full
= Dirty vs. Clean = Empty. For me, this observation becomes the moment of truth at the beginning of mixing. I have to determine if the result should be full (requires cleaner sound) or somewhat empty (requires padding). Study this question, but, first of all, stop doing everything out of habit. Each setting is individual, and will give results only with a fresh look. If we don't adapt to every single task, doing everything out of habit, we will get the result completely opposite of what is required, and before you know about it, the audience will see a half-full glass, left unsatisfied.

4. Panorama

The next one is panorama. Not EQ, not compression, not reverbs (beloved ones!). It's too easy to slip into particulars, but the beauty and power of the mix is more often everything, in the Big Picture. Any drawing begins with a composition. From coarse to precise, remember? This is what panning is all about. So, before you tackle EQ, effects, automation, first play around with the panorama. This will help you visualize the Big Picture.

Bring all the instruments in a row and start playing panorama. Temporarily turn off the instruments that animate the dead center (bass, kick drum, main vocals). We need to study the effect of positioning in our mix. Without changing the balance a lot, move the pans around a bit and determine how the expression of the instruments changes relative to the neighboring ones, solely due to the position of the stereo image. Notice how the coloration changes from too hard when panned to the left, to soft and wide when panned to the center. Notice how the rhythms that are opposite each other interact differently than those that are grouped together. Notice how one setting of the panoramas makes them fuller, while another makes them cleaner.

You can spend hours tweaking the EQs to separate or mix instruments, while panorama simply solves this problem.

Симметрия против асимметрии (Полнота vs Прозрачность)

Симметричное панорамирование добавляет пустоты в полный звук, а асимметричное – грязь в чистый. Я ничего не упустил сейчас? Пример: представьте две чётко разные гитарные партии, звучащие в течение песни. Одна может быть ритмической партией (A) с множеством пауз между фразами, а другая – постоянным «чёсом» (B). Представьте, что каждая записана дабл-треком, т.е. в итоге имеем четыре дорожки. AABB. Под «симметричным» я понимаю разделение каждого дабл-трека влево и вправо AB/AB=AB(левый канал) и AB (правый канал). Это более полный звук, чем несимметричный AA/BB.

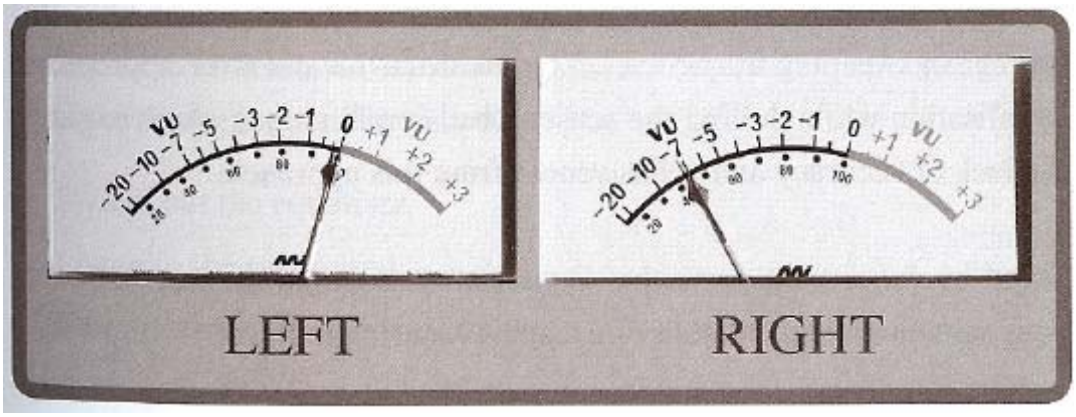
I usually spend 20 minutes turning the panorama knobs. You will be surprised how deep insight into the musical material this process gives.

Notice how one panorama rocks your body and mind differently from another. Also pay attention to the changes in the indications of the indicators of the left and right channels with different combinations.

I have always been intrigued by how differently the same picture of a panorama feels when rotated 180 degrees. This is due to the fact that the left ear has a connection with the right hemisphere, and the right ear with the left. There is more room for panning than meets the eye. The more you play with panning and observe these deep effects from different panning arrangements, the more sensitive you become to the target behind the sound, and go deeper into the Zone.

Psychoacoustics "center-left"

How You you know All sounds, located in center, will become louder V mono mode and those located deep to the left or right are quieter. Therefore, it would be reasonable to assume that sounds located clearly in the position "center-left" or "center-right" will not suffer when switching from stereo to mono. Where is this position
"psychoacoustic center-left" and how to get there? Figure 3 shows a signal balanced center-left.



Rice. 3 - Turn on signal, which want to arrange V positions center-left. Adjust panorama so that the left channel shows 0 VU and the right channel shows -7dB. Ready.

precision pan

Use the following technique to create precise symmetry between left and right tools. For example, you can place the main vocal in a special position on the left, and its second part in a mirror image on the right. This ensures that tonal balance is maintained when changing from stereo to mono.

Enter the left vocal signal (turn off all effects sends, of course). Raise its level until the indicator of the left channel touches zero. Note the position of the right channel indicator. Now you can flip this panorama with necessary accuracy for signal, displaced to the right. Raise the signal until the right indicator touches zero, and adjust the pan so that the left indicator shows the memorized figure.

5. Equalizer

Equalization - it's mystic. This is one of the most practical, and, at the same time, destructive of things, which Can do. The most big problem - keep a given perspective in the process of equalization. Up to this point, we have enjoyed creating a calm and transparent perspective. The characteristics of the ghosts between our speakers come out beautifully. They have a figure, a life and a purpose. These things are now radically changing.

Most engineers plug in an EQ, crank the gain all the way up, and start turning the knobs preoccupiedly looking for a frequency that will make the hair on the back of the neck move. The big problem with this approach is that for every correct frequency you hear, you're presented with a hundred incorrect ones, and even with a spread of 15dB. Psychoacoustically, this has the effect of dulling your hearing when trying to cope with a set of jumpy frequencies. This results in a "played but not won" effect on the EQ issue, with poor perception (bad combination), which explains the lack of accuracy and coherence resulting from such an approach.

Stay tuned, we will soon unravel this EQ mystery that will increase your sensitivity and creativity, instead of drowning them completely.

Art intelligence, Part II

Not twist at random!

Imagine the chaos of sounds from a pianist who is looking for the next sound, going up and down all the nearest - not the best picture. World-class sound engineers know their frequencies the way pianists know their keys. Fortunately, this is easy and fast to achieve. Unfortunately, this is counter-intuitive and requires self-discipline, like trying to quit smoking. But whatever you do, *don't guess!*

Должен быть способ лучше!

Решение: человеческое ухо прекрасно подходит для сравнения при переключении (A:B). Оно плохо распознает медленные изменения, т.к. быстро адаптируется и теряет перспективу. Так что вместо сравнения 50 неправильных частот при постоянной их подстройке, попробуйте сравнить две, но хороших. Этот специальный метод гарантирует сохранение перспективы и чувствительности.

Compare own imagination With dream

Give your ears the information they need by comparing just one setting to an imaginary target. Like this:

turn off the equalizer; listen raw

sound;

Imagine mine ideal sound, O which dreamed;

customize equalizer, which, How You hope embody your dreams; turn on the

equalizer.

At this point, you will be comparing the EQ settings to your imagination, and accurately You will know where you hit and where you missed. This method will quickly teach your ears how, for example, 3kHz really sounds. When you turn on the EQ, you will think: "Hehe, I thought 3kHz was higher" or "Oh shit, I meant a higher frequency!".

The moment you turned on the equalizer and realized you didn't guess right, turn it off! One second is enough to understand that the settings are not correct. Don't keep listening to bad settings and *don't twist!*

Adjust the equalizer in the off state, and turn it back on. This approach will keep your perspective clear while training your ears incredibly in a short amount of time. It's exciting and effective. The hardest part is resisting the urge to spin *but...*

NOT SPIN!

Запомните:

послушали – представили – понадеялись – сравнили – послушали – представили – понадеялись – сравнили – послушали – представили – понадеялись – сравнили.

WITH this way you Not you will look actively working But you will work carefully. Only your fear of wrong imagination can stop you from participating in this game, but you will be surprised how quickly you learn these frequencies, and learn how to achieve a goal in just a few approaches.

Remember... Not twist!

In addition to dulling the sensitivity of the ears, twisting the EQ, as a rule, introduces superfluous emphasis on frequency settings, But That, What to you need to more often Total – only a small increase in the missing frequency.

Not twist!

Frequency disguise And EQ

The outlines and coloration of the white polar bear are much clearer on a black background. If everything is too bright, you will not hear the highs - they will become hard and gray.

I spent a huge amount of time studying the famous spectrum analyzer mixes and was extremely disappointed to see that the mixes with the brightest and clearest highs had less real highs, and the deep bass mixes had less real lows. These frequencies were distributed and never bulged out.

If you EQ all your instruments solo, you usually use your favorite frequency. This creates a sound that masks that frequency on all instruments. For example, if you like the frequency of 8 kHz, and you add it everywhere, then not only will you degrade instrument separation at that frequency, but you will also create relative attenuation above 8kHz, creating a muddier, less bright overall mix.

improvident equalization

We often hear about how one tool can affect another, and it seems that this effect pops up after several hours of work, nullifying all the pleasure from the result. If we could have predicted this disguise, we could have saved our time and the frustration that turns out your phono EQ is ruining the vocals.

Predicting these psychoacoustic effects may be easier than you think. Try this: the next time you need to find the magic EQ on the lead vocal, first find an instrument that plays in front of it all the time - let's say it's a phono. Turn on a pair of phono-vocal solos and immerse the vocals in the piano. Now - the only time you can do this - slowly begin to adjust the EQ, constantly monitoring the sound of your voice. I can assure you not

it takes a lot of imagination to feel like you're actually EQing the voice, not the phono. You will quickly realize that the phono influences the voice, in the exact opposite of logic, And therein lies the whole power of this technique. It's always amazing to find that the sound behaves exactly the opposite of what you logically expect from it - you will find that you and thousands of other engineers did exactly the opposite of what they should have done to achieve the desired effect.

For example, you may find that creating a "hole" in a piano at 2kHz, to let the corresponding frequencies in the vocals peep out can reverse effect - To conspicuous decrease rigidity vote on 2kHz. Samo yourself this is only possible with the simultaneous sound of the instruments.

Moreover, if you are adjusting instruments in this way, focus on any, but not on the one you are twisting. If you're tuning guitar #1, imagine What adjust guitar #2 And At once will hear difference from this effect. Sometimes I collect four electric guitars and tune each one in turn, listening to the rest - I pursue my task until they come together. Because predict the future? Listen to it now.

Compression (auto 3D effect)

Compression ratios are multiplied, not added. I know that I'm already tired of this rule, but it should clearly lie down in the subcortex. The conclusion is: try to compress as little as possible. A small amount of gentle compression sounds more "prestigious" than too much compression or limiting. I won't go into compression theory here, but I'd like to show you a trick with a little neat compression.

It often happens that some element of the mix is too dynamic - it doesn't matter if it's vocals, percussion or bass. The jumps in dynamic range may seem a bit too wide, but that shouldn't be a problem in itself, as more often than not, the performer's dynamics match the arrangement and actual expression of the song. But how to preserve this “feeling” while preserving the sound of the element? Below next little trick automatically resize and space the performer due to their own dynamic movement

For example, you wanted to create the illusion that as you get quieter, the element gets closer, and as it gets louder, it increases in size (to bring the house down). At the same time, you do not want to see any strong fluctuations of the indicator arrow on loud fragments. For simplicity, we're using vocals in this example, but this will work with both percussion and horns.

Ingredients: All you need - compressor, reverb or effects processor, and a free-channel console with flexible stream switching.

So:

vocals send on channel 16.

Promise on echo - aux #1 With 16th channel, To your favorite reverb.

We disconnect the 16th channel from the master bus, so that the signal after the fader does not appear and does not leak to the master faders. Now you don't hear anything until you turn on Echo Send #1, in which case you'll only hear the echo without the original signal.

Feed the output of channel 16 to the 17th record bus. Submit

17th tire records on entrance your his compressor.

Compressor output to the input of the 17th channel.

Submit exit 17th channel on master bus V usual mode. Now you hear a voice.

Ready.

AND what we done?

The reverb receives an uncompressed (relatively open) signal, while the master bus receives only the compressed signal.

Indicators weekend level will stay V proper framework V moments of strong shifts textures, giving big illusion at relatively small voltage.

Rice. 3 shows this auto 3D effect How face, decaying into pieces, moving back and expanding at loud moments.

Rice. 3 - As the level drops, the echo is attenuated and the compressor pushes the main signal forward; but when the voice gets louder, the compressor pushes it back and the reverb swings open, for which thanks to the raw, uncompressed signal. Spatial illusion this effect serves to equalize the size And forms vocals, expanding expression between quiet And intense passages. When voice quiet, he advances forward, becoming larger And closer, With disappearance reverb. When vocals intense, it moves back into the mix - becoming smaller, but gaining reverb, expanding and enveloping from all sides.

More thoughts O compression

It is important to be aware of the advantages and disadvantages of deep compression before squeezing the signal too hard. Remember that compression adds distortion - by pushing the lower resolution portion of the recording forward, by increasing background noise, and/or by bringing out previously unnoticed harmonic distortion in the signal. Big, clean, signature sounds with minimal compression. Movement fader when creating a mix would be a cleaner solution. If the signal was crushed and crumbled, you may want volume. Keep in mind that your sounds in will shrink to some extent at high compression ratios and possibly pick up unwanted noise.

Eat some synergistic reasons For compression tools on stages records.

This improves ratio signal/noise.

This liberates your favorite compressors For use at intelligence V further.

This allows do track-by-track entry at correct dynamics. This improves the

mic mix already during track recording.

This Means, What less compression required at final intelligence, apart from the special effects.

Размер (восприятие и реальность)

Когда одного талантливого звукоинженера спросили «что определяет размер звука?» он ответил: «размер автобуса определяется цветом двери». Странная шифровка, я понимаю, но это стало советом, который я не забуду никогда. Конечно, понимание этого совета заняло годы. Любой звук является иллюзией, и наше восприятие может быть изменено простыми средствами, если вглядеться в восприятие, а не анализировать реальность.

7. Install balance

You are now familiar with most of the players in your mix. Their pans and EQs are fine, so it's about balance. Does it matter where to start?

The inspiration you achieve will change a lot in the process. The mix created by inspiration is usually more interesting than the mix created by logic.

Starting with bass and drums will quickly give you a relative overall level of the record and a foundation for rhythm and energy. This usually creates a good picture in your head and you will surprise other engineers with the powerful sound of drums and bass. But as for the purpose of the song - the path that led to its creation, or the heartbeat that fills it?

A less mundane and more creative approach involves finding the track with the most inspired performance. Remember, at the stage of adjusting the sensitivity of the channels, we noticed this? Now it's time to turn it on in solo mode - listen carefully - and let your imagination run wild. When the console disappears and ghosts materialize between the speakers, start playing. Add something nearby. If the ghost disappears, turn off the newbie and try something else. - slowly. Wait for the ghost to return and slowly bring in a new player. Try to capture the state of your mind in which ghosts become real and distinct.

Example: You searched and found a track with a cool saxophone. Slowly add the guitar. If it makes the ghost disappear, or doesn't blend into the sound right away, remove it. Wait for the ghost to return and try something else. If this pair does not work, then remove this tool. Find a couple that will keep the ghost alive and real. Then add more until you have a collection of the best tools to keep ghosts alive. The rest become less important. This approach creates a sometimes unusual but more magical balance.

Balancing games reason

There are many games that can be played to reach the final balance. I want to suggest some less obvious but surprisingly effective ones.

Similar to the EQ exercise, try moving the fader up and down to find the magic position. Instead of fully concentrating on the fader you are moving, listen to the nearby instrument. When, for example, trying to find the best snare level, listen to the bass drum. When you regulate guitar "A", in search of its place in the track, listen to the guitar "B". More often than not, this makes it easier to find a more accurate balance.

This is not surprising, since when you think about it, it's exactly like listening to the consumer - his (or her) mind is anywhere until you get his attention. He will not listen to the snare drum itself, or the guitar (if it does not lead), he will simply try to delve into the music, and the balance itself contributes to this more than anything else.

Я люблю перетаскивать внимание людей, периодически делая один или иной инструмент громче, а затем утаскивая его обратно в микс. (см. «Движения» далее в этой главе).

It's a common mistake to strike a balance by turning up the volume of an instrument until it's clearly audible, then moving on to the next one, and so on. After 24 channels, when you sit back and start listening, you will find that the first instrument has sunk and the last one is the clearest. This happens all the time. One way to avoid this is to listen to something other than the adjustable instrument itself.

When creating a basic balance, one of the common mistakes is the struggle for every decibel. The underlying balance must be created quickly. My best mixes are those in which I quickly created three options basic balance by writing or saving each and then comparing them. The difference between these balance sheets should be in the way they are created. For example, one can be created by listening in mono on a small Auratone. The other one can be created in stereo. A third might start with the phono, gradually add everything else, and finish with the drums. Fourth - may start with bass and drums, phono, and vocals at the end.

short With sensations

One guess as to why this is happening is that when you type fader fast, Not pondering too much for a long time, You topics most prevent

excessive intervention of the left hemisphere, using mainly the right, i.e. use sensations to make decisions. Analysis of the situation takes time, but the sensations appear instantly.

Let's say you have three static balances, but you can't decide which one is better. Difficult problem. Assuming you'll be using the fader automation later, which is better - choose a mix that can be stripped of something, or one that to which you can add something? (I usually use the penultimate option). Clear one - it is forbidden give vocals be defining factor because This usually most
mobile tool. Do you look at indicators to see which instrument most stable? Same not bad questions. Maybe, should provide a choice of gravity. The next chapter "gravity and its influence on music" will explain the essence. But in general, the instrument with the clearest rhythm creates gravity.

The choice of the best of the three should be creative, not analytical. Perhaps you will catch yourself fantasizing under some of the mixes. Pay attention, maybe this is the winner! It's taken me years to realize that I'm fantasizing in the master recording of the "underlay", so why not use this phenomenon in the mix?

Art intelligence. Part III.

In the third and final part of this trilogy, we'll plug in your ears to the best sides of the effects, move some faders, and save the mix.

Let me remind you chronology our trip:

1 Approximate outline

2 Compound cables

3 Structure Geina

4 Reduction lines

5 Settings panoramas

6 Equalization

7 Compression

8 Balance (Bye not bad. The fun continues...)

9 Reverberation And echo

10 Movement faders

11 take

In Part II we set up the panning, EQ, compression (a little bit) and started to work on the balance little by little. But before we move on to steps 8, 9, and 10, I'd like to explain some of the benefits of mono monitoring.

The present mono

Mono monitoring is more than just pressing the "mono" button on the remote. True mono only happens when you listen to only one source. (That's why any photo of a professional control room shows a small speaker with a single speaker on the console near the indicators).

In fact, two-thirds of my mixing happens in mono. Why? Easier appreciate the exact balance of two instruments when they are stacked on top of each other than when they are six feet apart. When perspective and depth are present in mono, they simply become clearer and deeper in stereo.

If you create an overall picture in stereo, some elements simply disappear or fall out in mono. I want to make it clear that once you have panned and textured the reverbs, it's best to switch to mono and listen to the amount of reverb there. Twelve hours later, your ears will still not be "blurred", and when you switch back to the mains, everything will be audible much better.

Mono V snowy snowstorm

Mono reminds me of small landscapes with snow living in a glass ball. Well, You you know. Such Can buy V any good tourist store. IN them shows the local landscape under a glass dome. Enough to shake - and you will see the Eiffel Tower, Ayer's Rock, Bing Crosby's winter cottage with small trees, horses, sleds in a snowstorm! That's what Auratone does for me. You create more accurately when the whole world fits like this in your hand. All elements; their relative sizes, colors, reverbs... A point source of mono sound creates a single stream of flame, and it's very focused and collected. Even the position of your chair will not change balance. Benefits list continues. Mono overcomes left and right channel imbalances in room acoustics, amplifiers, speakers, and even your ears!

So What, after constellations panoramas, take a look V mine glass ball, and you can work faster, with less fatigue and more accuracy.

Naturally, to you All equals needed main monitors, to control bottom.

Further We let's dip V concept reverb, including methods, which allow "connect" your ears And hear thin changes in the adjustment time of all these hidden parameters in complex reverb processing.

8. Reverberation And echo (space, glue, groove)

People love reverb! It creates a three-dimensional space around the ghosts between the speakers. This attracts the ear, allowing the sound to be heard after it has disappeared; this has an anti-gravity effect, causing the components to flutter around the main line.

For Togo, to decide With reverb, ask yourself two important question:

1. Why? 2. How?

Why reverb? It allows you to create the feeling that the instrument is in any room you want, and many modern reverbs perfectly recreate any space of your choice, or vice versa, allow you to create your own dream room. Naturally, some of the interesting reverbs are the epitome of things that don't exist in the real world. So, in addition to the question “where do I want to place the tool?” why not ask "when?".

The reason for the question "when?" in that the reverb not only gives a sense of space, it also changes the relationships in the mix. If a new character enters the mix and a new rhythm starts playing, it can affect the perception of the song more than just changing the color of the walls (the room in which it sounds).

For example, Elvis Presley's simple slap-delay brings the sound back to 50s rock 'n' roll, but putting the sound in the room where Elvis sang will make it cheap, like a terry-cloth room.

I'll try to explain more clearly. I mastered the rhythmic reverb effect. For me, the main reason for adding any echo is the rhythmic result of its appearance. If it's a rare and beautiful voice with a great tone, you can use a reverb to stretch that tone, but mostly you have to work with the rhythm. My obsession with reverb rhythms sometimes makes me think that certain instruments are just the key to controlling a good reverb.

Example: imagine you are mixing a boring song where the rhythm is nothing special. If you turn your voice off and listen to the bass and drums... you've heard it a million times (do you know this?). If so, then try this! Add a long dark, back, reverb only on barrel. (Rice 1). You will hear
"kavooompfk!" back to front at the appropriate points, in a rhythm that is radically different from the rhythm of the kick itself - the result is a kind of funky swing. This kind of movement – like someone is pushing you to the ground – suddenly changes the whole feel of the song. You have upgraded it!

Rice. 1 "Dark And mystical reverb on barrel"

If there are not enough highs in the reverb, this effect can become very hidden. Make it invisible, dark and mystical with tons of tension. The result, of course, can be debatable, but if the reverb is in the right place, it will be so strengthened in its positions that the song will no longer be listened to without it.

Music is a unique energy that can make you feel lighter, and reverb can contribute to this phenomenon. This is my secret goal when playing with reverbs - to get the listener off the floor. This is not only possible, it is vital.

Don't just play listen to movement in mix And try imagine how a form of reverb can heal lifelessness and lift your spirits. Simulating rooms is only 1 percent of the possibilities.

keep time, but Never play Time

Feel the elasticity. Elastic echo, as a rule, is more powerful than a clear metronome. Remember, precision puts you to sleep, and the groove that slurs around the beat draws attention.

essence reverb

Are you having a hard time picking up the subtleties of sound changes by turning unfamiliar knobs on a reverb?

Here's a simple and neat little method that will help you hear exactly all the rhythmic, spectral, and textural parameters of any reverb while playing with all those tricky parameters in digital complex reverbs. This approach ensures that your ears are "tuned" correctly.

It's great to have a dozen or more reverb parameters, but it's often hard to hear exactly which one does what. Here is an approach that will improve your ability to recognize. **Warning:** do this only after you have decided on the purpose of the reverb (e.g. to move the rhythm, or lengthen the notes, or just create the feeling of a room), otherwise you'll just get confused. First - the goal, the means - then.

Equipment: You will need a pink noise generator, a metronome (any click generator will do), a noise gate with a control input, and three wires.

Submit pink noise on entrance noise gate.

Send the output of the noise gate to a free channel on the console. Turn

on exit metronome on manager entrance noise gate.

Install on metronome pace songs (Later try doubled or twice as slow).

Settings noise gate:

Install maximum suppression; Attack and

release to a minimum;

Threshold triggering - So to metronome exactly activated gate.

Now hear a special sound that, when sent to any reverb, will reveal the influence of all its parameters, all at once! You will clearly hear the subtle difference between all those "Definition", "Diffusion", "Delay", "Attack" of your reverb.

Next, increase the attack of the gate, thus stretching its envelope. Maximize the attack time while lowering the threshold so as to maintain a crisp response. Slow down the decay (Decay) to create a small "tail" - enough to create pulses of some density (this is my favorite setting). This will give you the opportunity to evaluate high and low reflections, HPF, LPF, size and other parameters.

When you send this signal to a digital reverb, you can clearly hear from the pink noise channel auxes how all frequencies decay differently as you adjust filters, decay time, room sizes, pre-delay, responses, etc.

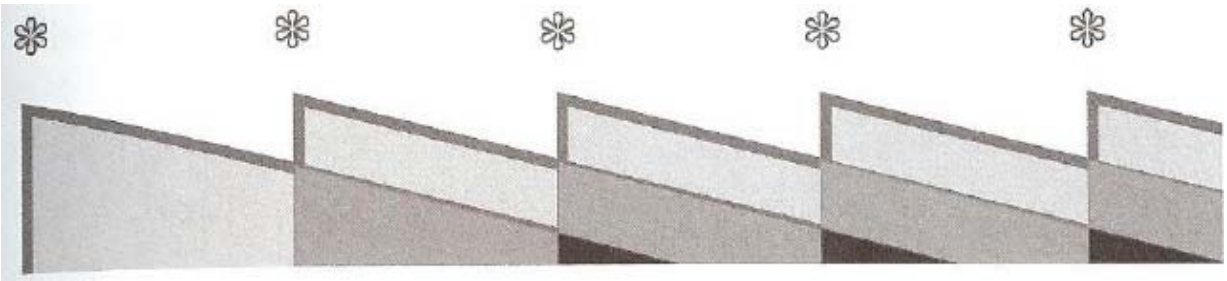
The rhythmic effect of the reverb on all your instruments will now be in front of you. Slow down and speed up the metronome twice as you like to improve the groove of your sound.

Objective test on groove (And let gravity will decide)

If you can't decide between two reverb concepts, try this test: take something heavy, like a phone book, and try shaking it to the beat of the music when listening to both options. Now determine which option gave you a feeling of lightness from the weight - step aside and let gravity decide! Sounds fantastic, I get it, but you have to give this approach a try. (More on gravity in Chapter 18).

Effect disguise standard figures reverb

Have you ever had a reverb that blended everything together, removing the space between the instruments and sinking everything into the background? This is because you are adding a "tail" to everything, which, contrary to expectations, masks the harmonics of all surrounding sounds.



Rice. 2 - General example Togo, How long reverberation kills perception harmonics. Pay Notice how each reverb tail masks the harmonics of the next sound.

One solution is to turn off the reverb as soon as your ears lose sight of the reverb tail, keeping So rest tools less blurry.

Your ears thus continue to believe that the reverb is still there, while keeping the rest of the sounds clean.

When something new catches your ear's attention, the remaining reverb tail becomes a waste of energy and space.

If the reverb has judder or vibration (i.e. the shape of the envelope not linear - see fig. 3), your ear will "hook" on it more easily, so less reverb is needed. If the shape is completely smooth, you'll need a lot of volume to hear the reverb, and a lot of volume produces more masking of the harmonics.

Fig.3 - Attach interesting form tail reverb, And she will become audible better.

Another advantage of reverbs with this shape is that they distort your perception of time - another secret ingredient we'll talk about later.

Not looking on what score reverbs is subjective, as is All in that the world, if reverb enough good, That He everyone like it. Main word here - "enough".

reverb, which shakes

Next Steps help create skeleton, on which grow up leather And reverb wings :

While listening to our pulsing pink noise, cut the reverb decay to zero and raise the return volume to compensate (to hear the early reflections). Listen for vibrations in the reverb architecture - early reflections, and especially room size vibrations. Then adjust the size of the room so that it "shakes" to the beat of the music. Use the same tempo metronome from previous experience.

Depending on your concept, you can slowly start to increase the duration of the decay. Carefully strike a balance between density and haziness.

diffuseness against clarity

Definition (definition) And diffusion (diffusion) - two parameter reverb, which difficult distinguish, even manufacturers disperse in opinions What There is What. One

affects the roughness of the end of the reverb envelope, the other - the roughness of its beginning and middle. Hear with the help of pulsating pink noise what principle your device is made of. Noise passed through the gate will allow you to fine-tune clarity and diffuseness. Ultra-short noise pulse reveals everything characteristics (the shortest attack and release on the gate).

Замечание: Чем дальше слушатель от динамиков, тем грубее должны быть диффузность и четкость. Это позволяет воздуху, к примеру, в кинотеатре, восполнить образовавшийся недостаток, дополнив картинку. Вот почему реверберация, настраиваемая в наушниках (и звучащая в них адекватно) отбрасывает звук вдаль при прослушивании через колонки.

Monitoring (headphones against the room)

Headphones give so dead Wednesday For listening, What You hear all acoustic nuances. By this reason exists trend towards creation reverbs with a very soft, not jagged decay. Meanwhile, this "peanut butter" itself creates an echo in the large room (along with everything else) and absorbs sounds in the background, along with the sounds of the original sources. Conversely, if the reverb is full
"holes" - when each individual fragment creates its own tail, the natural reverb of the room merges with the artificial one and creates a more coherent reverb picture, supporting and pushing the main sounds forward. Just try to make sure. In general, artificial reverb is an extension of natural reverb that remains invisible until you put on headphones.

AND last thing word O rhythm

When considering reverb timings, remember: "When in doubt, make them longer, not shorter." If this new rhythm is drowning in the mix, make the reverb longer to increase the overall volume. A short delay will sound clumsy and less smooth - less free energy will reach the target. Also don't forget about "Keep Time but Never Play Time" - just find what you need. For a detailed explanation of why the later echo sounds more musical than the earlier one, see The Time Machine Effect.

9. movements (dexterity hands)

We know that the listener doesn't need to know about your hands moving the faders. Your movements must be invisible and thus magical. And here's how to achieve it.

When controlling the faders, once you've managed to capture the listener's attention with the volume of the instrument, you can start to lower it - it won't be noticeable. IN result the rest Part track will become sound brighter. This something like
"dexterity hands" with sounds.

You can imagine it like this: I got up to get attention, and then sat down again, and you can still see my face, but I no longer block your view. This effect similar attracting attention bright character, incoming V crowd.

You Not you can take away from him eye, though He already And far - Bye your attention Not attract something more interesting.

Most engineers determine how loud they want things to be and leave them at that level, creating a highly static mix and a constant struggle for clarity.

I suggest you pull individual parts forward only to push them back, giving clarity and unity to the whole mix. Try alternately apply this an approach To several tools, Bye universal the sound will not become undulating - when every time the right element captures attention. With experience, you will find that for some mystical reason, everything will take on some incredible amount of presence, because at any given time one element will be a little louder than it should be, but at every moment it will be its own element. **Remember:** your job is to control the listener's attention.

The main trick to managing the internal balance of backing vocals is to make their relative balance fluctuate. Usually this suits everyone, because. each part becomes super sharp in order to move back a little.

This is the "crawling back into the mix" that gives you enough living space to re-pull back and create punctuation. If the sound is static all the time, you will not be able to create punctuation in the mix without going off scale with the associated drum dance indicators - as a result, the work ceases to be half creative.

Большие мониторы (маленькая награда)

Группе может нравиться постоянное звучание мониторов на полную мощность, но помните, что это не вечеринка. Конечно, при прослушивании я всегда перехожу на «мэйны», делаю их погромче и даю музыкантам получить удовольствие, но сам слушаю из коридора – это один из методов выживания в процессе сведения.

10. take

Grab the result V moment, When everything sounds fabulous - even If more Not everything is finished. Because the result can always get worse. (due to a technical problem or a crazy producer).

Rule #1: Feel free to kick everyone out of the test when you're alone with the track. It's for everyone's benefit, trust me.

Rule #2: After an hour of mixing with the band and producer back, grab it before they ask you to change anything. Even if you haven't had time to check the levels, even if there are too many echoes - Grab it! This first take can be a magical starting point for your original, undistorted concept.

Turn on the audition recording and write down all the wishes from the team. This is the policy, fix everything on the list ASAP, remembering that you have a version that will allow you to keep the right direction.

We treat digital overload

You push and pull the arrow to the maximum ... as you should. There is a peak that overloads the digital recorder. You play the passage dozens of times, lower the suspect by 5 dB, but can't find the culprit. If you can't quickly remove the overdrive by squeezing the causal track at that point, you can end up lowering the entire mix by a couple of dB - but that's not the best option.

Wouldn't it be great to have a way to protect your balance from overloading without having you play the same location over and over again looking for the culprit?

Try here's what:

Instead of lowering the level on suspect tracks, try simply inverting their phase! Peaks are not due to simply raising the volume of the instrument - soloist tool never Not affects on indicators individually, right? In other words, its envelope adds up to the rest of the mix, giving the resulting energy at any given time. If you invert the phase of the main instrument by 180 degrees (using the button), the envelope will be subtracted from the overall signal, and, quite possibly, the overload will disappear.

I don't mean inverting the phase of snare drums with overheads, or near and far mics, etc., it will affect their coloration. Instead, invert the entire drum section, such as guitars, orchestra, etc. This method also suppresses rare peaks in double-track vocals and other instruments.

Warning: Keeping the phase inverted can create peaks elsewhere in the mix that were fine before, so put it back in place as soon as you can.

This method works best when you only have three or four overdrives for the entire track. If there are more than 100, it's time to lower the overall level - mix really too loud.

YOUR REFERENCE CD

Have you ever worked in the studio for a week on mixing, and at the end, when everything was ready, found that the monitors were too bright, and in results of your work is not enough top? I think everyone has. Fortunately, there is a productive way to deal with such problems, which allows you to detect potential sources of trouble the moment you enter the room - and it takes only eight minutes.

Stabilize perspective

To do this, we need my most well-deserved engineering tool. I did not mention it before, tk. I use it in special cases, but if you do not mention it, the picture will be incomplete. Every time I remember that most of it no - I wonder how (or why) people manage without it - it's completely inexpensive and portable. So what am I talking about? That's right - about the reference CD.

I like to travel. I don't carry warehouses with equipment, but without a reference CD, I feel like I'm without hands. A couple of seconds after entering any control, I turn on my eight-minute reference disc, and in a short time I learn a lot more about sounding monitors than in a week of fiddling with pink noise.

So what's so special about this CD? In fact, nothing - all the difference is created by the places in which it sounds. Your reference CD is only as useful as you know it in detail in a close-to-ideal control room. And this knowledge can be improved only by listening to it in the maximum number of rooms. In all my life listened to his CD in more than 200 control rooms around the world.

Reference Stava

My reference CD won't mean anything to you until you listen to it in those 200 rooms too. If you turn it on at home, you will hear just a one-minute cut from eight different recordings.

Each engineer must develop their own reference CD. I made mine by accident. Far-sighted clients always want to hear examples of your work, but they never have time to listen to all the albums, or even just the best of the best. For this reason, when creating my "advertising" disc, I placed 50-60 second fragments of various sound options in it in sequence. Eight later minutes You listen to seven various sounds, every from which (coincidentally) perfect for bringing out the individual quality of the monitors.

IN mine option, every track separately reveals possibilities monitors according to the following parameters:

High frequencies (relative level) Hardness (total

density)

Uniformity (uniformity stereo pictures)

Bass Clarity (Bass Flame Tip Location) Accuracy working off

crooked (How are being worked out peaks waves)

Reverberation And surrounding-bass (transparency And index tactility) Color

(general color)

This is a non-binding and non-exhaustive list, there may be other qualities that deserve close monitoring that may be revealed by your favorite mixes and not mine. In my case, if I can hold test for validity, this will allow me to avoid six errors.

Listen And study

When listening to a reference CD, control only those qualities that this path should open. In my case:

The first track shows the amount of top - when she sings one special word, I remember how it sounded at AIR in London, Studio A, and is it bright enough her hoarseness, or vice versa.

The second track reveals the integrity and density of the sound of the monitors. When the guitars and drums play a tight rhythm, they show the harshness of the monitors.

Track Three is a giant wall of everything, loud as I walk around the room and listen to the uniformity.

Track 4 is a wide, open space with punchy bass and a crisp bass kick – a great way to find out where the bass in the room really focuses. Listen less to the color, and more to the integrity - the tip of the flame of your monitors. The point of maximum clarity is not necessarily the point of maximum volume. The maximum level is often where the room reflects a certain frequency in phase, and clarity - where the sound has restraint along with fullness and integrity. Walk back and forth, listening to where the bass drum sounds natural, deep, and harsh (rare coincidence).

The Fifth Track is my indicator of how sharp the signal curves are through the toms. When a powerful drum beat is playing, I can feel the wave peaks "jumping" out of the speakers clearly, if not distorted by bad amps, wobbly speakers or deformed cones.

Track Six - sounds climactic - massive bass fills the room, and the echo effect creeps back behind me.

Finally, Track 7 is a neat, nicely opened track with two acoustic guitars, colored double track vocals, fat but neat backing vocals, and violins. It's also great for checking if the left and right monitors are the same when listening in mono.

Compared to whether you like the sound of your reference CD or not, it's much more important to *compare the sound* to what you remember from other good studios.

Not do his long

Remember - a few seconds of listening to most recordings is enough to appreciate the quality of the monitor system. If it distorts, sounds flat, wide, narrow, muddy or high, you will know in the shortest test - 10 seconds - as if changing radio stations on the receiver.

Creation own CD

Create a disc with your best work. Make a selection of six or seven of your best mixes (your mixes are better because your mind remembers the depth and detail of each instrument and effect). Use original unmastered master tapes. This is important because originals retain more information. Raising the level during mastering always loses useful information at the will of the mastering engineer (how can you judge the accuracy of working out the peaks of the curve if the mastering destroyed them by 50%?). Also, bad control rooms tend to compress the sound, so don't "give" them an already compressed mix.

Now listen to your reference CD in as many world-class studio control rooms as possible :). Your goal is not to find a studio with the best sound, but rather a monitor system that "produces" the best sound - ie. where hit records come from. Go there, listen carefully, don't talk, and just let the sound sink into your subcortex.

Famous studios love to invite you to listen to reference CDs, because. proud of their monitors. If you seem like a potential client to them, no one will refuse you.

"Not" under fear execution

Don't even play your reference CD at home or at amateur acoustic systems. **Remember:** with this CD you are training your ears. Playing it at a party or just on speakers you don't intend to mix on will destroy your ears and your ability to use the disc. We have created this CD only for world class professional monitors. The experience of listening to this disc should always be useful and stored in memory. Listening to it anywhere else will dissolve your memory and lower your standards.

If You seriously come to case, That already through 20 rooms rate advantages. I don't remember people's names, but I remember sound textures and sounds well. When you remember how your disc sounded at the famous XYZ studio, and then turn it on in the local studio, You immediately determine What "tweeters" too bright or vice versa, A bass is not focused around consoles, or that the reverb is invisible or soft, or that the speakers don't pick up bursts. Armed with this experience, you may be inspired to make With something With this. If You - today's client, Not hesitate A little

move things around to re-assemble the monitors in a line. If some elements are clearly out of order, start looking for the cause, its solution, and, perhaps most importantly, do not settle for less than the maximum of their capabilities.

When his use

Use the reference CD before you get used to the coloring of any new room and its monitors. The moment you first enter a control room, your ears are at their most sensitive to the acoustics of the room. After four days, you will acclimatize, and all the nuances of the room will become familiar. Now is the time to find all the problems and fix as many of them as possible before start spend creative energy, time And money. First - your reference CD. As a rule of thumb, the first sounds you hear from your new control room should be your reference disc.

Acoustic consultant

After training your ears to find problems on many monitors, don't be surprised if you get paid to play your CD and set up your monitors. In a very short time, you will become an expert. The key to success, as in many disciplines, is obsession.

"I nothing without his reference CD"

Each engineer must create his own CD from his own mix, for his own ears. The longer you have it, the more valuable it is - so, the sooner you do it, the better. You can always add something to it, but limit yourself to six or seven tests.

Now, When at you There is idea… Not think What need to rob bank And run around around the world. Calm down. But once you've built your CD, you'll be surprised how many options you have to enhance your CD experience.

DOOR IN SKY

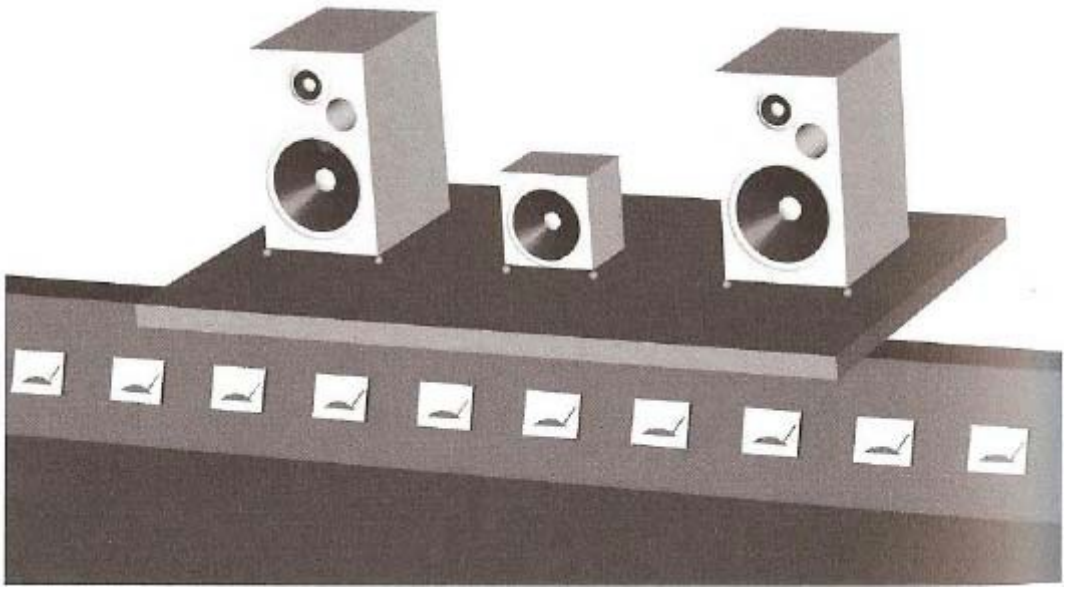
If your monitors sound good and you would like to upgrade to next level, I'll talk about a method that will further improve the sound of near-field monitors.

You probably remember one of the previous chapters - “Tuning monitors relative to center" - Where it was told How achieve from monitors undistorted "membranes" sound fields. We A little reconsider And develop this topic.

Not lose control

A control room, by definition, is a place where the creator (i.e. you) has the opportunity to reflect and have full control over the audio product. Monitoring is a big part of that, and maintaining complete and comfortable control is an issue that doesn't lose relevance as you move from one studio to another. I'd like to show you a little trick I discovered a few years ago that allows you to make the picture around near-field monitors more clear and sharp, giving more room to your ears.

One day I decided to experiment with the positioning of the near field monitors and the Auratone. Usually they are located above the remote indicators, but to make it easier to experiment, I put an old furniture door on top of the indicators, and pushed the speakers apart, placing them deeper. To my surprise, the space was filled with "ghosts". The illusory wall of sound that we are all trying to see has become more tangible. The stereo picture across the board has improved, becoming clearer, fuller, more accurate, and has gained much more depth. And all this - with the help of only a piece of wood - with approximate dimensions of 1.3m x 80cm x 40mm. Plus an added bonus is this the platform has improved bass clarity from installed "flashmount" mains.



Rice. 1 - Sound painting came to life because speakers are on his own platform. Not forget to put balls under the monitors to deepen and clear the bass.

Why does the sound get better? Well, after facing the fact face to face face, it's easy to talk. It seems to me that the board contributes to the transmission of waves from the speakers to the engineer and prevents them from deflecting and weakening. I understand that acousticians reading this may see some problems here, but in fact - it's worth trying it first to believe it. Any coloration that comes from having a platform is negligible compared to the main sound field. Wrap it in a towel or something like that to finish them off. The solid board will prevent the bass from leaking down. The sound, especially the bass, is like a flow of liquid and needs some help on its way to you. Experiment with lifting the whole structure to control the coloration.

When we have achieved what we need, we can re-adjust the distance between the monitors. You same remember What If speakers too much far Friend from friend, This can deprive you of the opportunity to set the correct panorama?

AND again phantom

The acoustic picture that lives between the speakers is a phantom, an illusion, a product of amplitude and phase coincidences between the left and right speakers. But when the sound comes from one source, the picture is always clearer - like a stone, not a rag. The distance between the speakers affects a lot. When you push them apart, the phantom between them stretches, becoming thinner in the center. In addition, pictures that should appear clearly between center and edge (center-left, center-right) are heavily influenced by And become unstable. Ideal distance between speakers - one in which a single density is created along the stereo image.

We comply balance

If the monitor system has a hole in the center of the stereo image, it is useless for mixing hits. As you'll remember from Chapter 2, if you place your monitors too far apart, which usually "looks" okay, your mix is likely to have too much level in the center, because monitors here have a reduction in output level relative to being panned left and right. And you compensate for this by raising the level in the center. This compensation is disastrous for the mix. Why? Hit records come from mono compatible mixes. If you check any stereo mix in "mono", discover, What sounds, panorama V center, become noticeably louder than moved to the side. For this reason, experienced engineers avoid overly loud center instruments when mixing in stereo. If you raise the level of the central instruments by controlling them through a monitor system with a dip in the center, you will do just the opposite of what is needed. When switched to "mono", the center instruments will rise by 3dB relative to those shifted left and right. So, in order to maintain the intended balance, one should lower the level of the central instruments, not raise it.

Of course, this leads to a puzzle - what to do with 5.1 mixing, where the center is often not a phantom, but a real speaker. Although the center speaker improves the sound in the center, do not forget that many elements, especially music, are panned through a pair - left and right channels - creating a phantom in the center. The key is to position the left and right speakers closer to each other than you initially thought. Because the membrane effect does not disappear.

One last dot

Perhaps the main reason why most studios have their monitors far apart is a false sense of breadth. It's a lie. Since the sound image decays with increasing distance, holes appear in the acoustic membrane, reducing resolution. The overall sound can seem more immersive - How from music on party - rushing on you, How Seems, from everywhere and even behind, But instead of Togo to give your ears more opportunities, This you
"blind". Instead, you should strive for a uniform stereo image, where every degree of the panorama is heard more accurately and can be controlled.

Altitude information is also easier to perceive with a narrower stereo image.

Party

At the end of the day, when the recording is over and it's time to show the result to friends, place the speakers as wide as you like. Put it on and don't worry about holes in the stereo picture - they will be filled with joyful exclamations of your guests, which will complete the picture.

In the meantime, tuck a board under the speakers to lure the ghosts in. closer.

TOO MUCH POWERFUL VOCAL?

Each vocalist is individual. When you first deal with a vocalist, you should try to have a session without the standard "presets". It's all good And Right, But What do, If singer issues so powerful And dynamic sounds that it creates a problem for the record?

Problem.

Pref Becoming,

I'm recording a fast paced rock song (many bright and dark moments) with a singer who has the most wonderful and powerful voice I've ever heard. Her quiet voice is powerful, like a loud human voice, and when she starts to sing it's like recording a jet plane, I have a big problem keeping the signal in proper range.

I use a tube preamp, then an analog compressor, and then the signal goes straight to the hard drive. When recording vocals, I usually set the compressor to decrease in the region of 3dB, at a ratio of 2.5:1, but this young lady overloads everything and everything. I lowered the threshold by -15..-20dB with the rest of the standard settings, and if I lower it further, it will simply turn off. I experimented with larger ratios, but the result is muddy sound. She's good with her mic technique and she takes a step back before 'gettin' hot' and that helps me a lot. Maybe you have some advice? Hope you can help. Alan.

Solution

You have a talented vocalist with a powerful range - in the region of 50dB, I believe. You are lucky that she has a good microphone technique, but this is certainly not enough.

Your question reminded me of when I recorded Gloria Jones at AIR Studios in London - Incredible singer with 50dB range. Unfortunately, she sounded like a drowned rat on her previous records, and I can see why. As a rule, it was necessary to compress almost everything that is possible. But there is a better way.

We achieved the best sound V her career - This was exercise on topic "Maximum illusions at minimum stress." I controlled her voice By three options:

catching flame - disposed microphone on tip flame (cm. chapters 3 And 7);

Controlling voltage - using two limiters with staggered thresholds, compression ratios and attack and release times;

By controlling "thickness" or "space". So, if she "drowns" in loud moments, let her "surface" in quiet ones. So the vocals stay the same, but the picture gets wider, and you imagine that there are more voices, not less, like a drowned rat.

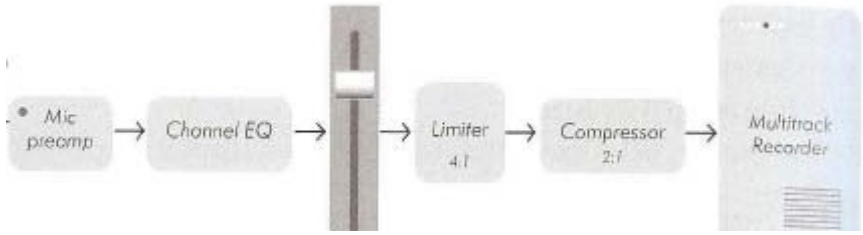
This has been one of my best-kept secrets up to this point, but your description of the problem brought back such a wave of memories...

Setup (post fade)

For start decide With microphone by factor rigidity. hard voice - soft microphone, softly - hard. Loud voice - Not Necessarily hard.

Next, determine which side of her mouth she sings. Everyone sings more on one side of their mouth than the other (at least a little), and you need to find that side - it's not hard, just listen and look at the direction the sounds are coming from. **Side note:** if it's hard to catch this, ask her to make her voice like a pirate - and determine which side of her mouth she instinctively clenched (usually it's the opposite side of the one with more wrinkles, but that's a completely different story).

Next, place the microphone on that side of the stand, or as I usually do, give her the microphone in her hand, and put a loop of the desired size on her hand. I know, I know, this looks like another crazy idea by Stav, but it makes it easier - microphone remains on the same distance, even If she will dance By room (by the way, Gloria loved to dance). This loop will not hurt to bring the microphone closer in quiet moments, but on loud fragments (when she moves her hand away), this loop will limit the movement of the hand, leaving the microphone in the tip of the flame. Here is such a cunning technique that gives freedom of movement to the vocalist, but we keep this freedom within the bounds of due.



Rice. 1 - We have compressor behind fader - V this all salt.

From this point, the microphone signal enters the channel. Set the channel gain according to the loud signal, EQ it as you wish. Rice. 1 shows the desired signal path.

Do not insert - I repeat - do not insert the compressor before fader (in pre-fade). If you do this, it will be a mistake that everyone makes, which guarantees a dead result that we are all very familiar with.

A little fidget

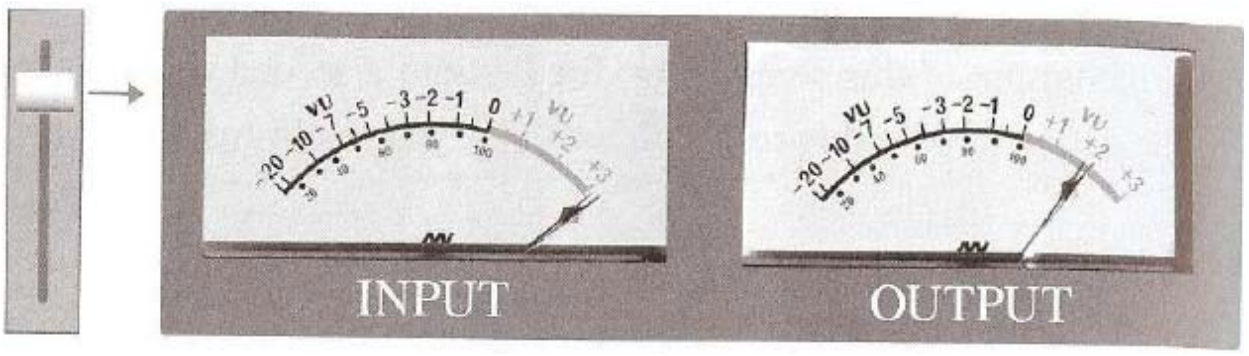
I assume that you have selected a channel with a good fader, because you will have to use it. Sorry, I understand that you were hoping not to, but you will have to adjust the signal during recording to achieve the desired effect - there is no other way. It will take a bit of practice, but I promise it will be much easier when watching the indicators. Street racers love cornering, and you have to love moving faders. The goal is preemption.

Now connect the fader's output to the first compressor, its output to the second, and its output to the recorder.

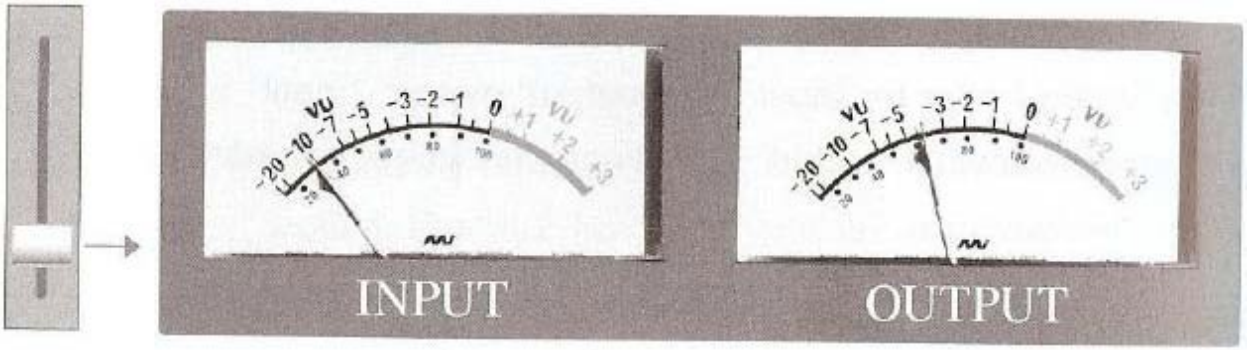
Because the solution is to control the dynamics, it is useful (but not necessary) to measure the levels of the input and output signal (input - in the sense of after the fader and before the compressors, and output - after the compressors and before the recorder).

Connection indicators

Send the post-fader signal (before the first limiter) to the input meter. You're not recording the signal, but simply watching the raw, uncompressed signal level, so you can see the effect of fader movements and performance dynamics before they're consumed by the compression. If the entry and exit indicators are next to each other, you will clearly see the effect of your work (see fig. 2).



unexpected peaks remain worked out



Even weak entrance turns out on sufficient level

Rice. 2 - Input And day off indicators allow to you navigate at work With fader

Settings compressors

Note that (Figure 1) the signal enters the limiter before the compressor. Set the limiter to a larger ratio (e.g. 4:1) than the compressor and the threshold also do higher - so way limiter will need more powerful signal, to trigger (for example, when the signal reaches 0VU instead of -4dB). Next, set the compressor to a smaller ratio, say 2:1, and to a lower threshold, say -4dB VU By indicator. Sending reference signal 0VU on entrance (By input indicator),

set the limiter output level to 0VU (by the output meter), while switching compressor in bypass. Now, including And compressor, turn it up output level up to +2dB on the output indicator.

Here Now you created this envelope Effect on voice. So What, If the vocalist will sing loudly, she will be subject to more compression, and at a lower volume, less.

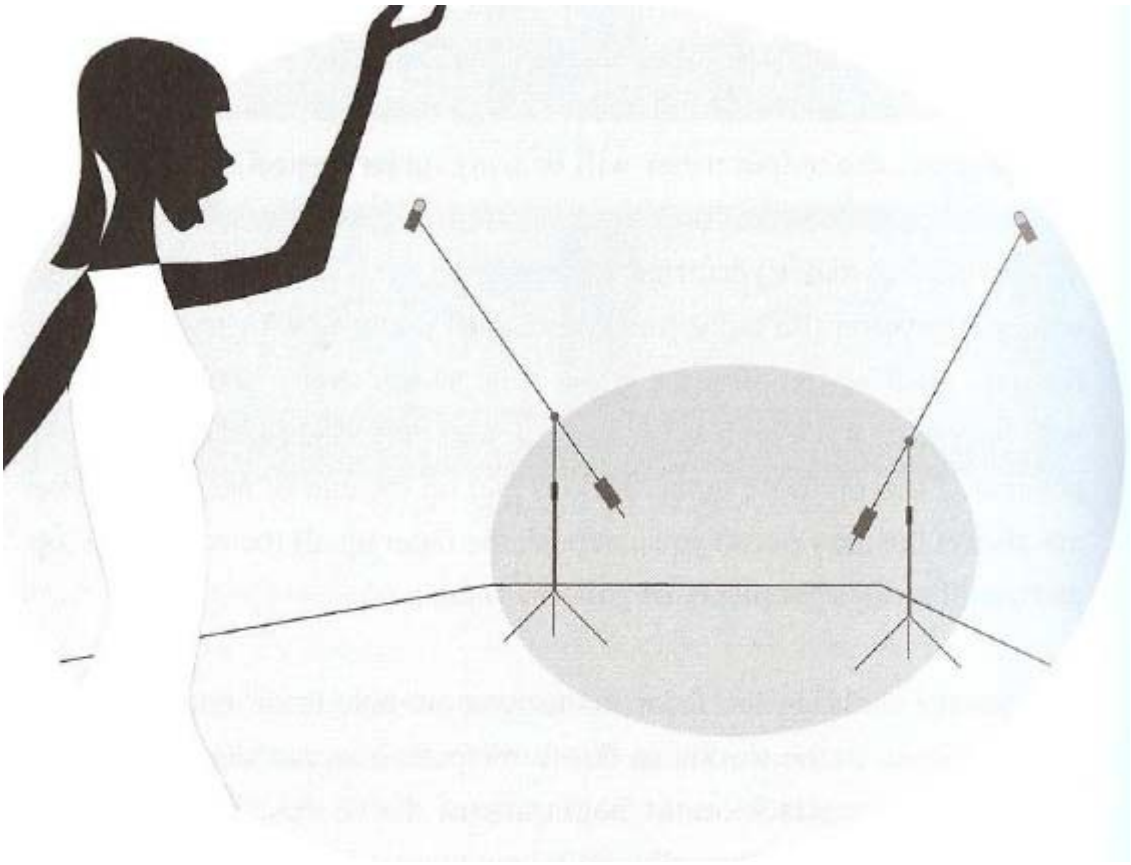
To visualize this idea, (see Fig. 2) feed the signal into the pre-fader insert, and move the fader while watching the meters. You will notice that the measured uncompressed signal level (input level) moves smoothly and clearly - up and down - following the fader. In the meantime, the compressed channel (output level) at low fader positions is larger than the input, and at high, even if the input the level will go off scale, the output will be within acceptable limits.

When working with a vocalist, you will move the fader, focusing on the input level, and trying to keep it at, say, -4dB, and if there are unexpected peaks, the compressor will save you, and you will have time to move the fader to the desired position. In quiet areas and at phrase endings (the sexiest moments, yes) you can push the fader all the way up, thus providing a good level for the recording.

The beauty of working with a fader is that you can do something that no other compressor can do - predict. When you slide the fader down before attacking, for example. If you know the song well (sometimes listening to the rhythm is enough), you can even press loud breaths, etc. With a little practice, you can even achieve a deesser effect in some cases, keep in mind that compression usually makes the "es" sound worse.

View from space

With the stress out of the way, let's see what happens to the space your vocals take up. If you're dealing with a dynamic vocalist, then the voice deserves the most room you can give it. In my case, we had at our disposal a studio that could easily accommodate orchestra from 60 Human. On opposite end rooms, on high racks, we had two additional microphones installed (Neumann U87, but any high-sensitivity microphones will work for this trick).



Rice. 3 - When voice fills room, distant microphones expand picture, topics time limiters hide the vocalist in the mix. Dynamics is completely created by her emotions.

These two microphones are recorded completely uncompressed on two separate channels. In the final mix, these two microphones can become the left and right components of the voice, with the main microphone being the center one.

Now, without any fader movement, these three tracks automatically create a cool surround effect. When she sings quietly, you hear nothing left and right, and the voice moves forward, being in the center and close. In an explosion of emotion, she will sink back in the center, and her voice will fill the room - in the mix, the sound of the room will come to you from the sides, and not like a dinosaur attack in Jurassic Park. I added some reverb to the room mics to boost the effect, and on that day, the dynamics of the voice made the hairs on my neck stand on end. I understand that not everyone can have access to a large room. If so, try recording uncompressed vocals on two microphones, and then feed these signals to good reverbs.

Compression (pre-fader And post- fader)

Personally, I think the only benefit of installing a compressor before a fader is convenience. I stuck it in the insert - and that's it. But if you're running a fader in this case, it will increase the noise in the quiet parts and the overcompression in the loud parts. If you're not touching a fader, you can cross your fingers and look away - but that won't help.

Let me try to explain: if the vocalist is singing loudly and you lower the fader (with a limiter, included V pre-fade), limiter continues compress

recompress and distort the full, not attenuated signal. When he sings softly, turning the fader up will increase the volume of the voice, but will also raise the limiter noises. The only way to fine-tune the signal level is with a mic preamp or limiter input potentiometer, both of which are usually far from handy, and are so ergonomically awkward that your reaction time suffers if you anything at all you manage to do. Post-fader technique does not create side effects like these.

What Not may headphones

Have you ever noticed what a strange feeling you get when a vocalist asks write down his somehow without headphones? This one from most scary requests that star clients can issue. And you immediately react negatively. Without thinking, most engineers switch to the "protect against damage", rolling out a long list of problems, and saying that this is a bad idea. Leaks, timing, build, more leaks, “they don't do that!”…did I mention leaks? The fact is that recording without headphones is not necessarily a minefield, and when the result promises be wonderful, it Togo costs. leaks may be minimized much more efficient how to you Seems. Certainly, vocalist Maybe hear sound on 5 milliseconds later than in headphones, but the vocalists themselves are usually 25 milliseconds ahead of the tempo.

This mystery pushed me on experiments V records vocals without headphones.

Problem (vocals without headphones!)

Hi Stav, I have a question. On Wednesday schoolchildren came to me to record a Christmas CD. There were probably 16 kids there at the same time, and I only had 6 headphones… They came last year too, so I just turned my speakers to them as monitors so they could hear the music. After recording, I reversed the phase on all tracks with children's voices relative to the musical background. I think I installed the speakers correctly - I did not like the final sound.

What I wanted to ask - maybe you have come across a similar situation? If yes, Can you suggest any corrective action. I understand that it can be done, but I don't know exactly how.

Will thankful behind any help. Trevor.

Not looking on pain (doctor), There is certain pros V failure from headphones: The

vocalist has an improved sense of tone;

More relaxed And pleasant situation;

improved communication V Not dependencies from microphone Improved

and faster communication between performers; The presence of a sense of

freedom and space

Of course, getting a good vocal sound without headphones is hard work, but don't forget that the hardest part of your overall job is always done by the vocalist. It is they who have to create magic through the limitations and obstacles of studio equipment.

For comparisons - to you need to Just press the RECORD button .

Solution

Eat three solutions: A, IN, And WITH. All of them include manipulation with phase unwanted signal, and all of them depend on the minimum compression of the vocals (so mix with soft vocals first).

Way A (at help any one microphone)

If you plan to play with the phase in the recording to minimize leakage from the loudspeakers, you have to force yourself to go against your intuition. Intuitively, you would place the speakers as far as possible. This explains why subsequent phase manipulations usually fail. If sound didn't travel so slowly, there would be less of a problem. Go against your intuition - put the speakers next to the microphone. Eliminate reflection leaks impossible, So What remains only change proportion, reducing distance from speaker to microphone as soon as possible. For example, wall to wall.

The idea is to create a direct clean leak as opposed to a spatial echo. It is important that the mix going to the speakers does not include instruments that will not be in the final mix. Use only the main themes for counting and setting the tone. Do not use pilot guitars or “TBE” (To Be Erased) signals. Do not pass the signal through any effects. Only clean sounds can be correctly subtracted by phase rotation in the final mix. Remember, the closer the speaker is to the microphone, the smaller the phase shift that prevents deduction.

Way IN (usage microphone- eight)

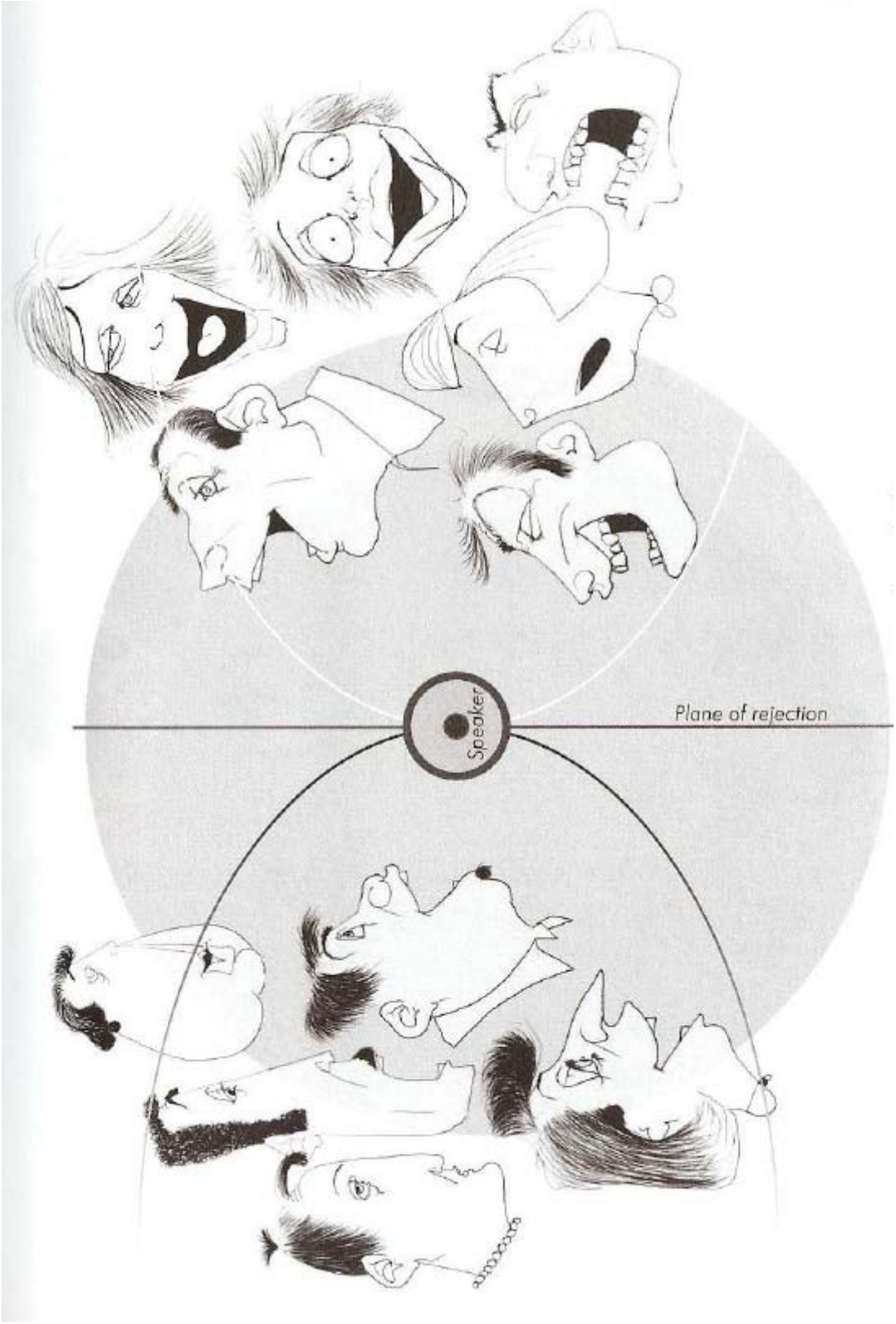
Arrange by sides from figure eight microphone By eight votes, And speaker - along the central plane - the plane of the rejection. The speaker should be small in size to simulate a point source and should not be placed next to the microphone (see Figure 1). Unlike Method A, the time delay does not play a role here, because. we do not plan to invert the phase in the future. Instead, we use the figure-of-eight mic deadzone to create on-the-fly cancellation.

Take a look How plane resections crosses speaker.

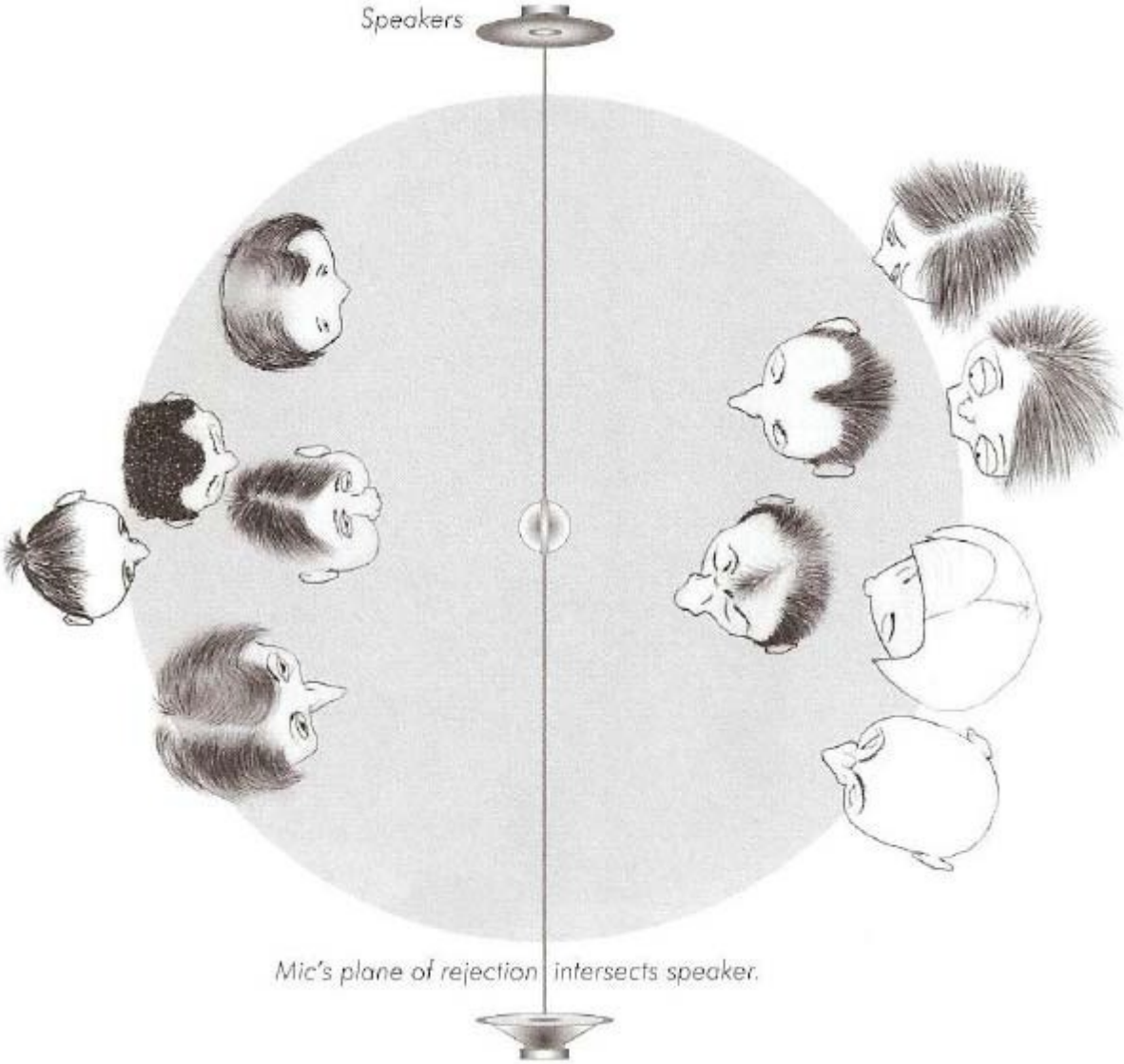
This approach uses the principle that the dead side of a figure eight is much "muffled" than the dead side of any cardioid. microphone. Remember, although the dead zone of a figure-of-eight mic is quite "deaf", it is also quite thin. Position the speaker so that the axis of its coil passes through the plane of the dead zone. It is not difficult do, listening signal from microphone through headphones And moving loudspeaker in search of a dead zone. **Note:** use pink noise for these searches. Or just serve the mix you want to use.

Rice. 1 and 2 show side and top views for this method. Anything on the notch line will sound like hundreds of miles away.

Of course, you need to set the microphone first in relation to the performers, and then look for a dead center, otherwise you will spend a lot of time (moved the speaker, moved the microphone, moved the speaker again ...). And one such circle may not be enough. Install the speaker approx. Check the sound of the vocals. Adjust speaker position for minimal leakage. **Note:** This is best done in a dead room.



Rice. 1 - All, What located on plane resections (plane od rejection), will sound How behind hundreds miles.



Rice. 2 - View way IN With heights avian flight. Microphone eight - in the center.

Way WITH (cardioid microphone)

Lay the speaker on the floor, with the speaker facing up. Position the vocalists around the loudspeaker, facing each other, and the cardioid mics facing the speaker. I suggest a minimum of four mics with a maximum of four vocalists per mic. Position each microphone at the same distance from our single point speaker.

Result

Success! Yohho! I used second way (microphone-eight) And was struck by the almost complete absence of leakage of the phonogram. And not only - I also established myself as a professional among clients!

Логика, страх, и анти-интуиция

Не бойтесь. Экспериментируйте. Вслушивайтесь в результат. Меняйте что-то. Снова слушайте.

Я никогда был быстрым, потому что любил экспериментировать. Это научило меня не заменять полностью вслушивание логикой. Не важно, насколько логичным кажется что-то, часто анти-логичное порождает великолепный результат.

So remember - the dead zone of the figure-eight is more muffled than the dead zone of any cardioid. Move the speaker, looking for a dead zone by hearing through the headphones. It can be quite narrow, so try to use a small speaker. Reduce room reflections as much as you can.

Afterword

We as engineers are always looking for a way to give freedom to the child inside the artist. Each superstar is different, but when the red light turns on, they all have one thing in common: at such moments they become uninhibited, like children. If working without headphones helps you get rid of your tightness, it's worth it.

Reversible mixing

Here's a unique way to bring out the best - in terms of intelligence - in all of us. Have you ever wanted to mix like someone else - more experienced, famous? Here's how you can do it!

There are many characters hidden in each of us, like Dr. Jekyll and Mr. Hyde. And this way is to turn into Mr. Hyde, which can be very useful! But first, a little story that led to such an unexpected conclusion.

Search ideal copies

If a serious client sends their master tape overseas, our studio can create a precision copy of a 2" 24-track master film, in case of a plane crash. After some experimentation, we found that tapes recorded backwards sounded better than straight-forward tapes!

Actually, the reason is simple. The envelope of a percussion signal begins with a sharp attack in the form of a momentary peak or dip in the sound. Analog copies possess infamous famous property "eating" peaks And subsequent envelope distortion. If you play the tape backwards, you get a void after the attack, and there is nothing to distort further in the signal - there are only “stubs” of the beginning of the sound. Think about it this way: we often think of recording on tape as printing. Now imagine trying to leave a clear footprint on a sandy beach. Copying a film "straight" is like pushing hard and then - gradually pulling the foot out of the sand. Reverse playback is like pushing in slowly and pulling out quickly. The latter option will leave a more accurate print. This is the advantage of back-to-front copies. Now back to history.

I copy the whole album backwards and have nothing to do with this time. And of course, I go to the console. It's just that I love mixing and I can't resist the urge to move the fedyer, turn the pan and equalizers while “umar scarlet amaM” rushes from the monitors.

I noticed that when you listen to the audio in reverse, there are no sharp bursts, And much better heard sound all tools. Usually You hear
"Snap!" which disables you from sound on some shares seconds - A after all there Same there is a sound instrument, But You from him thrown back attack. A in time auditions backwards, every sound seems to be approaching from hundreds of miles away, and then, in the very at the end, when you caught it, there is a slight click, which was the attack. The sharpness of the picture is completely replaced. Instead of EQing the attack, you switch specifically to the resonance, ringing, tone of the instrument - which, when you imagine it in your mind, still includes the entire envelope.

To cut a long story short, I compared and found that when played backwards, you can balance backing vocals, horns, strings much better, tk. nothing will put you off from their sound. All you hear is the sustain of the note being played, i.e. Stretched notes are easier to mix.

During this listening session, it was not difficult to connect a quarter-inch stereo recorder to record the mix sounding backwards. And then, at the end, I turned the tape over again to finally hear the result. To my surprise, the sound turned out to be completely balanced, plus there was an attack that was torn right in the face. I wanted to say something like “Wow! Did I do that?!"
"I could not create such a bulging sound with all the desire!". And it happened only because I was able to work with the sound going in reverse - from silence to peak. The drums and all the picks were more powerful than expected, but the tonal balance of the kit was just right. Equalizing it was a real pleasure. It felt like you could hear every frequency in the drum as they said "mmmmmm zzzzzzt ahhhhhh" instead of "tyts-tyts-tyts".

Reversible reproduction - something What O switching

Of course, when you flip a 24-track 2-inch tape to play it backwards, the first track, which is usually on top, ends up on the bottom. So when you reverse any analog multitrack tape, remember to flip all the wires corresponding to the channels in reverse. After finishing work in this mode and re-turning the film, this will have to be done again. If you're doing a serious mix this way, reverse should only be used for balance and EQ. After that you can turn the tape over and continue mixing in the normal mode, to be able to creating echo effects. Any background effects can be recorded on free tracks for reverse playback.

Strike against reverse

Unfortunately, I've found that some modern multitrack systems don't allow you to reverse multitrack playback, and I'm sorry if you find yourself in that situation. Perhaps there will come a day in digital audio when it will be as easy as switching wires on a cassette deck.

If you try mixing this way, I can promise you one thing - it will sound like someone else, with different ears, did the mix.

Removal musical language

It's amazing how reversing an audio track removes musical language. Instead of distracting you with the familiar sound of your own language, or the familiar sound of the saxophone, the musical language becomes, it seems, devoid of everything, leaving you with only form and a set of harmonics. Even the rhythm disappears. This is a completely new psychoacoustic space that you can plunge into. The removal of the musical language sharpens the unconditioned perception of sounds.

I Not I will become speak, What reverse mixing necessary And recommended everyone - This quicker bizarre reception For seeking opportunity refresh creative forces.

Warning: you probably don't want to work this way in front of a producer - he'll think you're crazy. Invite him after the work is completed. Let the beautiful result hide our little secret.

CAR TIME

Phenomenon "Once in a blue moon"

This phenomenon is so incredible that I risk losing your trust. But despite the risk, I can't help but tell you about this wonderful effect.

Have you ever had situations where the instrument is constantly out of time? And how would you react to the fact that today you will receive the ability to correct this? Slow down or speed up a performer using a standard device that is already under your fingers?

What if I told you that you can change the rhythm of an instrument using its fader?! If the musician is straying, you can slow him down by turning up the fader. And vice versa - to accelerate, lowering. I don't expect you to believe this until you try it, but after that you will be stunned, just like I once was. Two simple rules: slow down – fader up, acceleration – fader down.

This method helps to establish an overall balance, improving the groove and timing of the components, and corrects uneven phrasing. For example, imagine a vocal phrase with the third syllable clearly out of rhythm. A quick rise on the third syllable will make it swing better, make it more true, and make the overall sound more focused.

intuitive movements All spoil

Many engineers think they need to tone down problem areas. According to this theory, the proper effect of this will only be if the passage is played late. If the opposite is true, then by muffling it, you will only exacerbate the situation.

Knowing this effect was most helpful to me when I was mixing material with the worst performance I've ever heard - absolutely everyone was out of rhythm with each other. I've tried everything in my store of tricks so that "seat" This together. I began With bass and drums, added guitars, then keyboards, brass, percussion, and then vocals - everything sounded dirty. Starting over, I started from the voice, then brass, percussion, guitars and drums - it's still dirty. I found the best artist and built the world around him - the groove never came. There was no inspiration. Then I remembered the time machine effect:

I looked for the tracks that hit the beat the best - phono. I set the phono fader to medium and turned everything else off.

I added a kick - it was out of rhythm with the phono. I asked myself - "she is ahead of or lagging behind?" She ahead of And I began lift up her, Bye she Not slowed down to hit the phono beat. It sounded too loud, but already the result! Then I turned it off again.

I turned on small drum, analyzed And his, compensated And turned off. I added a

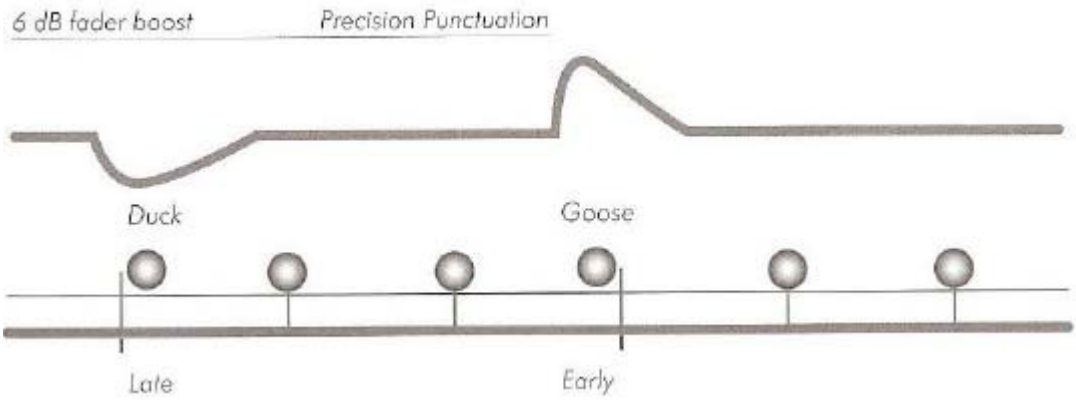
rhythm guitar to the phono, and did the same as with the rest.

The sensations from such a process were very strange - without any musical taste, just adjusting the timing of the instrument and then turning it off.

Twenty four channel later, I arranged fingers By channels consoles And turned on all the previously muted channels to hear for the first time what happened. I didn't believe my ears! The balance was strange, but the groove appeared! Remember our proverb: "If it's magic, then it's better not to argue with it." After that, I completed the balance only with the help of an equalizer.

Advanced tricks cars time

Here's an example of how to use the Time Machine effect to fix bumps in a vocal track. Her first syllable was late, and her fourth came too early. The fader can be lowered to the first syllable, then slowly returned, and raise sharply on the fourth syllable to make the sound error intentional and add a groove to it.



How works car time

To clarify for myself how it works, it took me two years. I have excluded this explanation from Stav's published world. If I had to wait two years, you can wait at least a couple of months. Now the time has come. Here it is.

A natural echo or reflection is always quieter than the sound itself. This is accepted as normal. If the delay is louder than the original sound, it sounds annoying, and sometimes even seems to be backwards. Thus, if two instruments are aiming at the same measure, but one of them is late, it will be better if the sound of the latter is quieter. If the first is earlier, and the second is normal, then vice versa. If you have such a grace note and you don't want to create any special effect, the second hit should be quieter.

In most cases, you will use the time machine effect to fix uneven play. It is unlikely that you will strive to create a strange special effect, if everything already sounds strange. I think you'll be happy if you can just make everything sound normal.

gravity, And her influence on music

All because of free energy

The main difference between an ordinary sound engineer and the best of the best is that V works recent felt significant energy, V sound, V mix - in everything. This does not mean that the sound is louder - the SPL meters show the same numbers. But good sound seems much more powerful.

It seems that the best sound engineers and producers put the result of their work into some secret source of energy that most of us do not have, which we are not even aware of, which allows the sound to easily "get" to the listener .

What This maybe ?

And if I say that there is some kind of energy field that surrounds any speaker, and which affects the creation of music and its perception? And even more - what if this energy is so great that we can topple upside down without it, but at the same time it is so constant that we do not notice it?

This powerful universal constant is used by those who are aware of its existence in order to improve the results of their creativity.

I I say about Gravity.

Gravity does not affect the mix "physically" - the sound does not fall to the floor, nothing like that, but we - fall! Gravity affects every molecule in our body, pulling it down, while music plays the same forces, lifting us off the ground.

Let's push moon

Are you familiar with this: Gravity affects our perception of music, because. in music There is a bunch of elements, "pushing" body on interaction gravity. For example, music tells you when to push off the floor, when to stop doing it, and it tells you when *to wait for* a new push, but when it actually happens is determined by gravity. If these musical "shocks" are not in sync with gravity, then the music becomes sluggish...there will not be a flow of energy from it, because. in fact, it will take energy from you, and gravity will have to be fought.

Not earnestly? Let's let's analyze on example.

Let's say you're on a dance floor and you have the feeling that someone has spilled something sweet on the floor because your feet seem to be sticking, albeit just a little, to the floor. And you will leave with the thought - "I don’t dance tonight ...". But you will be surprised - it is worth sounding Stevie Wonder, and your legs will rush into the dance as if nothing had happened. Effect sweet gender disappeared! Legs dance themselves By yourself And You think -
"What This, Yes I whether This?"

So what happened? The extra "lift" comes from the fact that musicians like Stevie Wonder feel gravity so instinctively that they give you updrafts when you need to get off the floor, when you need to stop, when you need to move forward. And it's all in sync (or, as Stevie says, "pocketed!"), so it feels like it's meant to be. In fact, not only does it feel like it should, but the sound becomes more energetic and more people like it.

Hit records are made that way. People all over the world, of any musical taste and of any nationality, get the same gravity and notice when they start to be “nudged”. Simply put, when you catch Gravity lifting you up, you can use the same laws to go down.

How use this force

Gravity is always trying to kill you - knock you to the floor. Music is one of the few elements in this world that uses gravity to lift itself. It has to do with inertia and timing. What do I mean by inertia? Each musical composition has "weight". The bass is heavier than the ringing of the triangle, and each of them has characteristic movement. If This heavy sound With big inertia He must start earlier to be in time with the main beat. If it's very light and a weightless sound (such as a shaker), it will react differently according to its weight. So you need to think about the weight of each sound in the mix and make gravity work for you. Use gravity as your guide when adjusting echo delays, or even instrument balance. If you bend gravity in your favor, it can work like a "weight", making you and your energy move, and, in the end Finally, increase the "blow" of the mix. Good engineers know how to release the inner energy, causing it to spring and sway in the gravitational field, collecting its energy at every opportunity.

Sometimes there are bands that sound like a thug, trampling their head into the ground with each blow. In such cases, I have to work hard to achieve swing by adapting the philosophy that the only the purpose of a musician in life is to provide a clue to the effect that creates *a gravitational wave* in a song.

Weight ghost

I know I probably haven't convinced a lot of people reading this yet, so let's talk about an experiment I ran several times in a room full of students. This is my main argument, if you will.

First, I play the basic rhythm track to the students without any effects. Then I imposing digital delay for snare drum, sounding at the same volume. The delay is adjustable from a millisecond to a second, and each student can adjust it as he sees fit. None of them are about balance, EQ or anything else, just the timing of the snare delay. Then we write down 10 seconds of each option and another stereo track. After ten students, we have a 100 second mix with a different delay sound every ten seconds.

I play the tape and ask the students to vote on which option they like best. The votes always cluster around a few options - usually no more than four favorites, and the rest are ignored.

People mostly choose based on math (how well the echo hits the beat) and context on what color the snare should be. Great, no surprises.

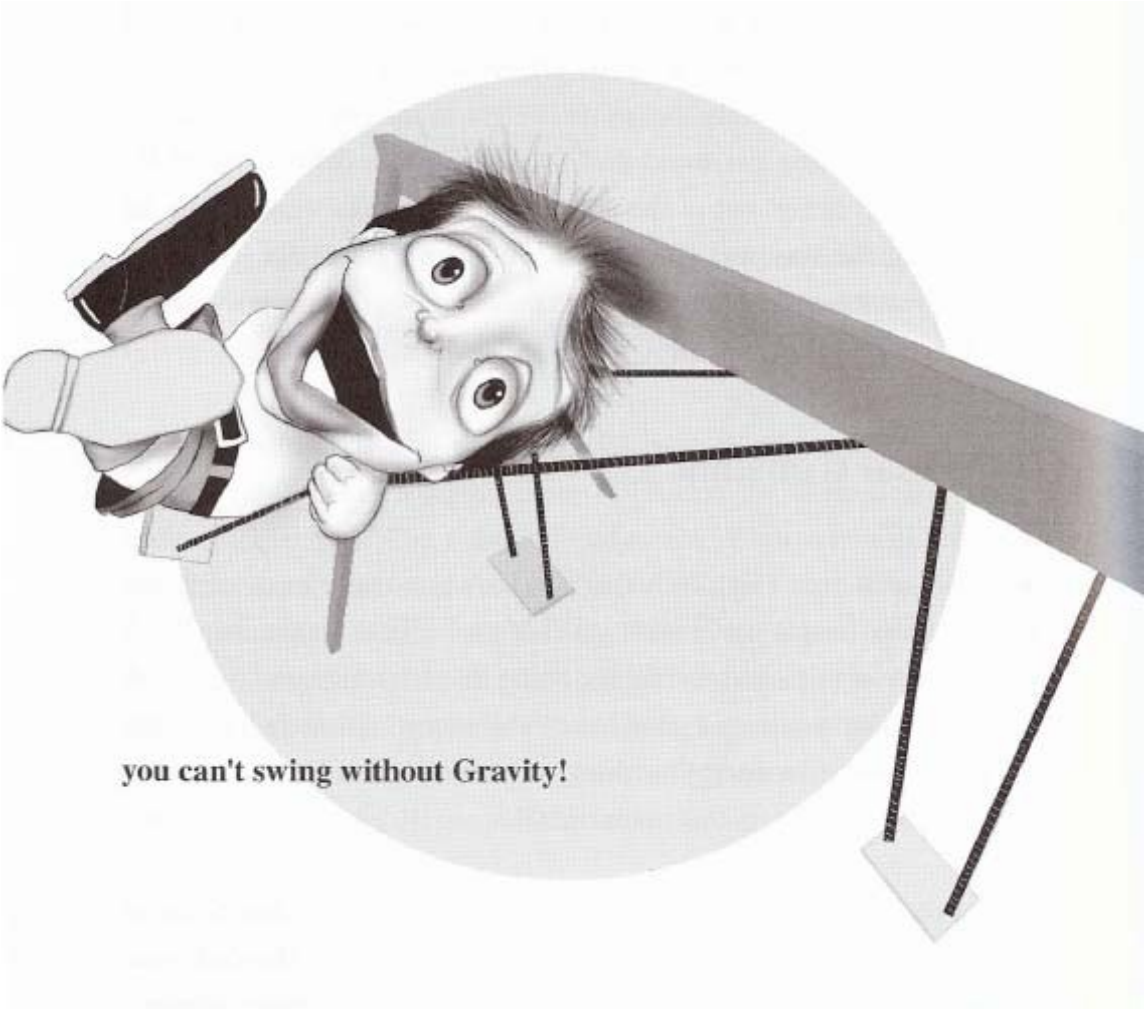
After the first round of "intelligent" voting, it's time to show students how gravity works. I ask everyone in the class to take something heavy, like a telephone book, or a two-inch tape, or the like. - and try to move these objects in the air to the beat. Then I ask them to forget about personal preference and instead just tell me which delay setting makes the item seem lighter in their hands. Suddenly, the vote comes down to one or two options. By turning off the left hemisphere, students are allowed to engage in the work of Gravity and get the opportunity to "hear like a producer." The result surprises many, but no doubt. That's what it means to "hear like a producer."

Universal constant

George Martin once said, "If you have good taste, it doesn't matter how many people agree with you." Great rule. Think about it: if you set up the delay so that people are off the floor, it means you have good taste - the phone book becomes easier, and you have free energy that works for you on the record. Now you no longer think “Ha, yes, this is mathematics!”.

OK, You, Maybe, Not work With music, which must be similar "Yesterday" by the Beatles or the theme from "Titanic" by Celine Dion, but regardless of the genre, if you put this free energy to work for you, your products will be purchased by a large part of the target audience.

Subjective taste is a matter of chance. But if you let Gravity decide for you, then you are speaking the language of humanity.


you can't swing without Gravity!

We take care ears

Where would you be if you didn't have hearing? It doesn't hurt to take a little care of him so that he never has to ask himself that question.

Hearing care is a very important point, although many do not like to discuss it, due to some embarrassment. For example, using earplugs is “not cool!”. Unfortunately, I know a lot of great engineers who have gone deaf at the moment because they didn't follow some simple precautions. If you don't take care of your ears, all the secrets in this book will be worthless. Listen carefully: there is care for the ears at home, in the studio, at a concert.

Care about ears Houses

Your ear canals are constantly producing deposits. Deposits are the body's way of dealing with loud music, i.e. trying to protect yourself.

Ears should be washed once every two years. For those who are not familiar with this process: it is not as painful as it may seem. If audio is your profession, consult a competent physician and, if necessary, flush channels.

You may or may not feel a difference. But if the perception of highs rises, it is due to working with too loud sound, and you should now be extremely careful. Deposit protection is removed and you have a second chance to achieve the best results.

Treatment pain V ears

If you have only hints of ear pain, take the smelliest onion you can find, cut off a piece, wrap it in a thin layer of tissue paper, and stick it in your ear before bed. When you wake up, the pain will go away. Why? Substances secreted by onions kill bacteria (this is also effective when you catch an ear infection early). The napkin absorbs the liquid secreted by the onion, and prevent him from moving.

Care about ears V studios

When I look for the tip of the flame of a sounding electric guitar, I plug my ear with my finger. I noticed that this does not interfere with finding the correct microphone position. This is very good, because with no ear protection, you experience threshold shifting at the very beginning of a recording session and can end up in trouble until the very end of the day, and beyond.

Что есть «смещение порога»

Смещение порога – это реакция организма, направленная на снижение чувствительности к шуму. Это временная глухота в какой-то степени, и лучше её избегать. Она всегда требует от нескольких часов до дней для восстановления, а иногда слух так и не восстанавливается. Когда вы сталкиваетесь со звуком значительной громкости, ваши уши «притупляются» быстро, но восстанавливаются – медленно. Даже длительное прослушивание на не очень большой громкости всё равно вызывает смещение, просто из-за усталости.

When You achieve necessary sound, usual case - do monitors louder to control the lows on bass, kick, etc. You hit the bottom right, and then turned the volume down again - right at this time you most clearly hear the top. Ears hear beats more accurately when the sound is quiet, and bass when it is loud. So if you have long mixing session - work at a moderate volume. I work surprisingly long hours (in proportion) on the small Auratone.

Care about ears on concerts

Always use self-retracting earplugs at any concert. If you ended up at a concert without them, it doesn't matter if you mix or just passing by - ask a friend for a cigarette. Break off the filter, remove the paper from it, and break it lengthwise. Then fold each of the halves in half and insert into your ear. This will attenuate the level by about 20 dB, which is necessary to relax the muscle in the middle ear responsible for the threshold shift.

As a bonus, you will be able to track the sound more clearly, because. ears will stop constantly compressing. The sound envelopes of human speech will no longer be blurry, and you will hear human speech more clearly with earplugs than without them.

If you're working on a console, your best bet is to work on the sound until it's all right. sit down, And Then apply plugs. Just check All well, before topics how to plug your ears. Without plugs, you can do much more damage - it's not worth it. I think someone wrote a rule a couple of decades ago: "if something sounds bad, just turn it up louder and no one will notice." Perhaps this is partly true, but that's a completely different story.

Не присоединяйтесь к вечеринке

Никогда не обращали внимание, что рентгенологи всегда уходят за свинцовую перегородку? Это потому, что они ежедневно подвергаются излучению, а пациент – только изредка. Звукоинженеры – так же. Так что не спешите присоединиться к вечеринке в студии. Посетители «дорываются» до этого раз в неделю, а ваша работа с оборудованием – ежедневна, как у рентгенолога.

Games For fortifications hearing

Ask a trusted friend to set up an equalizer, and turn it on/off while you sit in front of the monitors and listen to vocals or phono through this equalizer. Your task is to accurately name the frequency and the level of boost/cut in dB required to bring the EQ back to zero.

That's how the game is played. He says “Enter”, waits five seconds, “Exit”, again five seconds, and so on again. Meanwhile, you give him instructions on how to rebuild equalizer so that the difference between input and output gets smaller and smaller. Change places after 5-10 minutes. I usually do this for hours with my best friend in my first multi-channel studio - Eastern Recording Studio, Richmond, Virginia. We started by simply raising one frequency on the vocal to maximum (using the 10-band graphic equalizer). After we both learned how to bring the vocal back to its original state, we tried to cut the frequency as much as possible. It turned out to be much more difficult! As you can imagine, it sounded like someone turned up a bunch of frequencies at once. Our goal was to eventually achieve the same effect on the phono. But, alas, we have not achieved this.

Formula hit

I have spent many years looking for a common denominator for a huge number of hit records. I was a radio DJ at school. When an interesting entry appeared, I could bet anything that it would be one of the first in the rankings.

I soon began to imagine that all these records had unique combinations of secret ingredients that made them different from others. We've all heard theories about what it could be, but there are always exceptions to the rule.

Discussion hit records - taboo among eminent producers. Most fast way stop session - start talking O formula hit. Maybe, This worsens their "scent" on good result.

Please don't be offended - I really don't know how to tell you this - but political pressure prevents me from revealing the answer to this question. But I can suggest the way by which you yourself will find the answers to all questions. It's like giving a fishing rod instead of a fish. If you think this chapter is too "teasing" - rip out this page and don't read any further.

In general, here is a simple, step-by-step path that, although not quickly, will reveal the answers to a person worthy of it. What should be done:

The basis

At everyone hit Always There is potential become hit.

Other records don't have it.

Perhaps there are many hits that will never see the light of day due to ineffective promotion, but we do not consider them here.

From a scientific point of view, we organize the search according to two principles: A and B. One is honest advantages, and the other is honest disadvantages, or “B-sides”. Why are these "B-sides" so important? To find commonalities, we must cut through false paths in the process. "B-sides" were created, recorded and performed by the same people, but they are not hits. This is amazing! We've just cut off those factors - engineers, studios, producers, performers. With these "B-sides" the same happens on the radio stations - in open competition with the "A-sides" - which excludes any marketing component. This is also good, because I'm more interested in psychoacoustic elements than in politics and promotion.

If we can somehow subtract the qualities of the "B-sides" from the "A-sides", we will see the necessary elements that lead to success. So how do you put the "B-sides" out of phase with the hits? Ask any question to the A-side group, and every time you get a "Yes" answer, ask it to the B-side group as well. In the end, you will formulate a list of ingredients that are unique to the "A-sides", this will be the formula!

Big question

Are all these rules - or secret ingredients - contained in the hits? The rule, by definition, has no exceptions. So, to rephrase, are these ingredients so essential that if one is lost, the likelihood of success will be significantly reduced? By "significantly" I mean that these ingredients are so essential that you won't find a single Top 10 track that doesn't have them. And not even the first. If you find an exception, then the corresponding criterion from the formula can be safely excluded.

The following example will help you understand the difference between "required" element records, And element, which "not bad" was would have.

I, How researcher, believe What There is indestructible rules, though artists and hate herself opportunity existence rules (Bye any from them Not will be "creative").

there are general rules for making hits , they may not be in the "language" you expect them to be, after all, musicians have been looking for them for hundreds of years. I mean that the general rules cannot be expressed in musical terms, that explains why professional musicians don't "see" them either.

My opinion is that these general factors are not due to musical reasons, but to psychoacoustic ones, and music is a tool for the consciousness of these subconscious states.

This explains why such a wide range of musical styles has gone through the top of the charts over the years. This also explains the existence of the “one-hit” nature of some artists and why, despite the knowledge of the “rules of music” of some groups, they stop making hits until they make a big difference.

Enough theories! (Where answers?)

Create a list of the top 10 entries of the year, for the last 10 years. This will be our "A" set. The December issues of magazines are usually full of such selections. You need 10 examples - one for each year.

The goal is to find common ground in all these entries. It's actually easier than it looks, but the trick is to find elements that are unique to these things - B-sides shouldn't have them. For example, they all have a component in the area 1kHz. At all There is Start And end. Everyone has a pace - but not the same as the rest (despite some theories).

We need a set "B" - to weed out false general factors. We have to find things that are in all the A-sides but not in any of the B-sides. (**Note:** there are at least seven of them!).

Remember that in the "B" set there should be records that failed despite even the best attempts to promote them to hits. My selection of examples "B" - it's a mirror image of the "A" set: recorded, produced, performed, mixed, brought to the market by the same people and played on the same radio stations.

Getting Started To comparison

Listen to your collection of A-sides for two hours daily. Over time, they will sound similar, despite the musical difference. You may begin to notice how they use textual devices, such as the word "You." “Without you”, “She loves you”, “You are the best”, “You are you are you”. By God, I think I've groped something! But then an instrumental comes in, and your whole theory about the text flies out the window. The only option when everything converges is when there are no exceptions in the "A-sides". If V some from "A-sides" meets something interesting, A V others - no, it means that this element can be useful, interesting, convenient, but not necessary. Your attention to the text may be correct, but do not elevate the text to the rank of a necessary component in the formula.

In the end (if you haven't given up yet), the qualitative factors that come mind, exclusive to A-sides and other big hits, and even to Fifth Beethoven's symphonies. To weed out false hopes, I test the theory on two or three powerful hits of completely opposite styles, like the same Beethoven's Fifth Symphony, "Let It Be" and the absurd "Sugar Sugar" by Archies.

When you hit a match, it feels like a shot of a champagne bottle. Have a good hunting!

Why London?

When my friends found out that I was going to be a sound engineer, they said that I should work in New York.

It was clear to me that if it takes ten years to become a good engineer at the right studio, you should first get to know a couple of engineers to understand who they are. Are they happy? If you know that it takes 10 years to become a firefighter, it doesn't hurt to get to know one of them and find out what and how it really is. It may not be as interesting as they show on TV. It might get boring.

So one summer vacation I went to New York to see the big studios. The conversations that took place there boiled down to the scheme:

"This is the control room." And they called the models of devices installed in it. Then they called the names of celebrities who recorded here. Then they talked about how much money they earn. Then they convinced that this is the best studio in the city. But, unfortunately, not a word was said about "sounds". I returned from New York with a feeling of confusion.

This bothered me because I believed that you need to become a person like those with whom you work. Adapt to their attitudes, ethics, humor, even accent.

The next holiday I happened to be in Britain. I've been to all the best studios in London and the conversations there have been noticeably different. They took me to the control room and said: “This is the control room, you know all the equipment - the same as everyone else.” They didn't name the stars they recorded there - that's not common in Britain. They were not proud of their profit - it is less than in America or somewhere else, and they do it not for money, but for the love of sound. And then they said - "Don't forget visit studio further down this same road, she Same good." AND V end, looking at his watch: "So much fuss and advertising, but about the sound - only talk?"

Eh, I would wanted be one of them!

So And came out - I got caught!

So I spent the next three years trying to work there (but that's another story). After going around all the studios in London, I chose two of them. AIR Studios And Trident studios. They to me liked more everyone. I would gave finger behind That, to work for any of them. They had a great, professional yet friendly culture and the best sounds I've ever heard. It wasn't easy And had to three times fly V London, to achieve success. This was long story, which I will not fool you, but I came up with a couple of rules that I think helped me.

rule #1 (Not beg V AIR on job).

It may seem strange, but it was the main key to success. You see if You ask their O employment, And They refuse, absolutely No reasons

come back here again. Instead, I made friends with a couple of engineers from there. We used to meet in a pub nearby, talk about sound, and sometimes I was invited to sessions. I could sit and be silent, help where I could, turn it on, turn it off, take notes - no more. Never asking for a job, I was safe. I think they knew that I was looking for a job at other studios, but I never asked for it on AIR.

My plan was this: I walked around the city in search of a job at the maximum many studios. If the studio offered me a job, I said: “Thank you, I'll think about it.” If I failed, I asked "Why?" and so I learned from my mistakes. Of course, I started at the bottom and worked my way up. Going directly to the studio of my dreams would be a deadly mistake. I spent a lot of time interviewing in the studios I didn't want to work for, studying what questions people like to ask, and preparing for my main interview. As with everything, the more you work, the less nervous you become.

Hard deadline

I flew to London on a 21 day sightseeing ticket. This meant that if I did not return after 21 days, my return ticket would expire and I would have to buy an expensive new one. I never forget my thoughts in morning last day. At I still didn't have a job in London and had to come back in a few hours or buy a new ticket.

I thought: “If I return home, I know exactly what will happen next. I'll live in my apartment at 111 Grove Ave, drive my little sports car, maybe grow up with a job, date X's girlfriend, and stay a big fish in a small pond - all very safe, predictable, with little room for growth. On the other hand, if I stay, I'll either break down or get a job at AIR! In general, it's no secret how I did it. I stayed. Four days later, AIR offered me a job. It was the best day of my life.

I remember everything like it was yesterday. I went into the reception room of AIR, 214 Oxford Street and saw the chief engineer there. He asked me how I was doing. I said, “Great. I had an interview at the DeLane Lea Music Center on Tuesday." He replied in a nonchalant manner, "Well, if you want to work here, I think we'll arrange it." I almost died on the spot.
“Um… yeah… of course… that would be great!” He said, "Talk to Dave," the studio manager, "We were talking about you." I ran to the receptionist to find out where Dave was. He was in the toilet. I ran up to the toilet, and there, in the process of washing hands, my interview took place.

Enough nostalgia.

Let's get back To next secrets.

Often given questions. Letters in Stav's personal e-mail.

Since Stav's first publications in AudioTechnology magazine, readers have been invited to ask questions and receive answers via e-mail. Of course, there are a lot of letters. Some generate new articles, some shed a ray of light on issues that have not been covered before - but all, to some extent, involve a unique and often "anti-logical" imagination Stav. How he himself exactly noticed
"I live For important questions from obsessed engineers."

We now invite you to join in and ponder some of the questions encountered in Stav's correspondence.

Monitors - the best on Your sight

Stav, I'm going to order a pair of near field monitors. What do you recommend?

First of all, don't spend a lot of money. Listen before you buy. Money and quality rarely intersect. The magic is in the location and in listening to the material as close as the room allows.

Do your mixes sound a bit "hard" or a little "soft" now? Take speakers with a stiffness that is the opposite of the desired result. I use soft speakers to get hard sound. A lot of people make too hard sound, so I can recommend hard speakers.

The goal is to listen to your reference CD through all the potential speakers and decide which ones bring out the most of the track's qualities. In order not to be confused by the acoustics of the listening room, listen to one channel (one speaker, not stereo). This focuses the sound on it and spreads the sound less around the room. Close your eyes and move back and forth, left and right and decide which one creates a more familiar sound - this one will give more depth.

Depth is the rarest quality of a speaker. I am very picky about depth. Its presence in the dynamics allows you to capture it in your recordings, the depth is what most modern CD production lacks.

Avoid cutting tops - look for deep bottoms. The hardest part is finding solid bass, say, two feet from the speaker. He's usually farther away. Don't be surprised if you find that a cheaper speaker gives you more depth. This is amazing, and it's a good find. Expensive speakers usually use hard plastic cones. Theoretically, such cones do not deform at high volumes. Unfortunately, this usually results in harmonic synthesis at lower frequencies and quiet bass. And this is very bad news! Personally, I prefer light paper cones that move easily - in the near field you will not use high volume anyway.

high level film And number

Dear Stav, we use Quentegy GP9 film. Is the 0VU signal of a tape recorder corresponding to -14dB peaks at the input to a digital device during transferring a signal from an analog source to a hard drive? At this point, getting a good level is trial and error, but I do not believe that there is no formula by which you can accurately calculate. I would be grateful for any advice.

Yes! The key to success is experimentation (I believe I've discovered the formula we're all looking for). Because your GP9 is set to 520nWbm, setting 0VU to -14dB on the transducer's digital peak detector should give the maximum safe ratio (marginal but not overload) of the signals.

Sometimes such indicators are glued LEDs, which make it difficult to accurately judge where the required bar is in dB, but you are right - aim for -14. In my experience, the film saturates before -15dB, which prevents overshoot. The beauty of GP9 and other extreme high end tapes is that they saturate later but are faster and stiffer - more like a brick wall than a rubber membrane... well, not How, How Ampex 456. This makes her the most predictable So tune peaks to zero is not difficult.

The trick is not to reproduce without overload, but to find the limit. Having found the limit in this way, measure the channel gain at the reference level in order to be able to repeat the conversion later.

If you're having trouble with a bad LED bar (like on many digital recorders), I suggest ordering one or two quality VU meters that can be plugged anywhere in case of such a problem. Connect a VU meter to the output of the digital track, tune it to 0VU=-14dB, and then just use that setting for the rest of the tracks.

Another way to tune in to -14dB in systems with poor indications is to create a -14dB digital signal, feed it into the VU meter, and tune it to 0.

Panorama.

Stav, I didn't really understand the benefits of the order in which you pan and equalize in the Art of Mixing articles. Wouldn't it be better to equalize before panning? Is it possible to more effectively create the depth of the mix by EQing the guitars, making them non-overlapping, and then spreading them around the panorama? After all, this is logical?

Yes, it sounds logical, and that's why many are wrong - precisely because it seems logical. Think about it: you can always switch to mono and EQ the masking effect, no matter where they are panned afterwards. It is much more important to analyze the intersections of the guitars before twisting the EQ. - and it is best to do this by first setting the panorama. For example, you may find What from five recorded guitars better Total sound: 1 And 5 left, 2 u3 By center, 4 to the right. The gain from this is simple - now you should only care about masking pairs of guitars - 1 and 5, 2 and 3. guitar 4 is pulled to the right, it doesn't matter how it reacts to, say, guitar 5.

arrangement speakers

What say O constellation speakers? Not far away whether They Friend from friend?

Yes. You should read the chapter on monitor placement, and I think you should move them. Don't make the mistake of arranging them the way they look best. It is difficult to judge how far the back wall is, but I think this arrangement (see photo) will be more accurate. After completing the exercises described in the said chapter, you will come to an arrangement that is closer to my version.

Rice. 1 - Proposed location speakers.

Also, depending on the location of the standing waves, I suspect that you are much closer to the speakers, and it is worth moving the console a couple of feet at least. I understand that this can create a problem with visual contact, but if there is an opportunity to better place the monitors, you need to use it. Bringing monitors closer together improves signal reproduction between monitors. It may seem counterintuitive, but sound is not photography that benefits from large sizes. If you are not all right with the acoustics, the old arrangement may not be suitable for balancing.

Hard disk

Tell me Please, How Right choose recorder?

Well, if your chosen recorder supports 24bit, you can't go wrong. Just take the cheapest and most convenient. Go to the dealer and play with the models you like, and you will understand what you need.

Here is a good test. Which recorder will allow you to turn on the recording "on the fly", And then (by pressing one buttons) Maybe repeat exactly this operation? If

the recorder can't do that, you'll be distracting your right brain just to jump back to record the guitar solo takes.

Having a tube at the input is a popular solution these days and is better than using cheap console mic preamps. Unfortunately, this is a sure way to the problem! Your goal is to get the analog signal through a high quality path before it touches the ADC. I'm not just talking about preamps and EQ, I'm talking about the entire path. This is the only way to get the best resolution. It takes more confidence and time in the recording stage, but it's worth it and you'll save time in the mixing stage. So don't sell your advanced analog equipment, digital devices will take years to "catch up" with them.

Warm analog sound

At you There is advice, How get That most warm analog sound V digit? equalizers, plugins or more something ?

I guess you didn't make the common mistake of adding too much high end to the sound - it's a holdover from the analogue tape/vinyl days. When writing in digital, this is no longer necessary, because. there is no loss of "sound" signals. We added highs at every stage in analog times, as they were lost with tape wear and during the mastering process. IN the present time problem - bass. problems With riding at us more Not will be, now the high ones “stick out” from any recording, and twenty years ago the sound became muffled.

The danger is that often destroys mixes - the fear of not getting the maximum apparent volume. This quiet panic makes the knobs turn further than would like the creator who lives in each of us. A few years ago I recorded and mixed an album by Eric Burden. His voice taught me an important lesson: signal level is nothing, but its texture is everything. When he sang softly, the signal level was the same as when he sang loudly in a choir. The loud parts were not higher in level, they were more
fuzzy And With greater hoarsely V voice.

Remember: if the mix is pleasant, the listener will add an extra 20dB to us. If it gets harsh and annoying, your listeners will turn down the volume, if not turn it off at all. One of the producers I respect once told me the difference between a "pleasant" and "pressing" mix. Pleasant always wins in sales.

Advantages mono

In the photos of large studios, I often see a small mono speaker above the console indicators. In addition to checking for mono compatibility of stereo material, there are Are there any other benefits of working in mono?

I'm a big fan of mixing when monitoring in mono. Of course, you need to work in stereo when panning and evaluating the width of the stereo image, but when it comes to relative balance, EQs, etc. – mono mode is the most advantageous. Why? You are automatically drawn into more attentive listening - as if becomes less distracting factors And hearing becomes more
"focused". I explain advantages mono next by analogy: Imagine that you need to compare the heights of spirit objects that are six feet apart.

Wouldn't it be easier to put them side by side? That's exactly what mono does: it shifts everything together for easier analysis.

And don't forget: true mono only happens with one loudspeaker. Emulating “mono” with two speakers has no effect, so if you decide to work in mono, turn off one speaker.

Dense vocals By center

Having become, some records have incredibly dense vocals in the center, it seems that the singer himself is standing there, only twice as large! My question is: how do you achieve such an effect? I assumed that everything that is panned in the center itself has such a dense picture, but in practice, simply placing the vocals in the center did not achieve this. It has always been somewhere "between the speakers", but as far as density is concerned, it remains as it was. Maybe it all comes down to the quality of the microphone / pre? Or is there something I don't know about?

Technically, you are absolutely right - any signal panned to the center should sound clear in the center. Its width in the stereo picture depends on the quality of the monitor system. But I understand that this is not about that. You may have noticed that on your speakers some of the center signals are crisper and tighter than in other mixes.

For what I like this question - for accuracy. Many people ask how to make the sound wider, and you concentrated on the location in the center. To create a clear center, more elements of the mix must be of good quality. The first topics that should be taken care of are panning the remaining instruments, equalization, reverb.

One of the most common mistakes I see in mixes is a messy center Part. Logics simple - If Not can decide, Where arrange instrument, why not shove it into the center? If you haven't read the chapter on panning in this book, you'll most likely create a "chaotic landscape." What I mean here: ear better perceives relative the difference how absolute. Thus, if you remove the instruments from the center, according to the musical style, leaving a "hole" in the middle, whatever you put in it - the same voice - will be read much better. I think big mistake pushing tools V center, When You Not you know Where it is better to move them. Not only does this “clutter up” the center, it also worsens the mono-compatibility of the mix.

The fewer frequencies you use, the better. Because no two speakers sound perfectly matched, if you feed them pink noise in the center, the center picture will be very wide. Is it because you are trying force the speakers to compromise on thousands of frequencies at the same time. Naturally, if a vocal contains three fundamental frequencies, it will sound narrower.

Remember: narrowband equalizer sounds more transparent A broadband - fuller.

Because most people EQ instruments in solo mode, they try provide Effect presence For everyone. This All equals, What do

all components in the photograph are "in focus", with a lens of infinite depth. The best 3D effect is achieved with thick lenses with a shallow depth that blurs everything out of your area of interest. Why is this forgotten when mixing? I can never understand. Remember that the quieter elements of the mix don't require too much level. In fact, you make them more cloudy, and then you raise the level to compensate for this. And it will make your center vocals stand out even more. If the reverb setting separates its sound in time from the original source, this will prevent smearing while pushing the signal forward.

vocals Not fit V mix

Having become, I make my first record - an eight-track project. It has two guitars (acoustic and electric), acoustic bass, congas, lead vocals and two backs. I have already recorded acoustic and electric guitars, bass and lead vocals. I started doing the pre-mix, and when I started putting the vocals into it, it felt like it wasn't interacting with the instruments. I did what you suggested in the mixing chapter and it improved the situation, but I still didn't get the sound I wanted. If I add a little reverb, it sounds unnatural - like someone is singing karaoke. If I do it more, the voice is lost. I would like to know what the advanced ones do engineers With vocals. For example, When I I listen CD cat Stevens, I I feel that the vocals sit in their place, among other instruments.

When you mentioned Cat Stevens, I realized that the reverb was not the problem, because he didn't use echo on his greatest hits. Great singers like Cat Stevens or Paul McCartney don't need reverbs because they themselves perfectly control the timbre, duration and body of their voice.

I know this isn't the answer you were expecting, but it sounds like your singer is a little out of rhythm and a little out of tune. This can create the effect of separation from the overall sound (very karaoke). The reverb puts the voice in his own space, and that makes him sound even more separate!

If timing is the main problem, create a stretched echo effect that will create a temporary blur. Add it to the voice (pre-fader) and bring the vocal fader out all the way - leaving only the echo return in the mix. (Don't worry, this is only part of the process, not the end result.) Now take everything out of the mix temporarily, except for the hard rhythm instruments, and listen to how they sound together with vocal echo effect.

Now comes the fun part: start tweaking the echo effect parameters (pre-delay/room size/shape/slope/ etc.) to get the groove going. It's not scary if the echo lags behind by a few beats, "smearing" along the rhythm. *Don't* make the echo fall right into the rhythm. It should “envelop” the beats, while not clearly out of rhythm. Well, in extreme cases, change the vocalist (joke).

Prevention better, how treatment, And better, than punishment.

I can't figure out what causes such inaccuracies. Maybe the problem is with the sound of the mix in the headphones - I always blame myself if the vocalist has problems.

Remember: the main mix of the day is the mix in the vocalist's headphones, not the one in your monitors. I I suspect What You submit to headphones vocalist that same mix, What And in your monitors - this mistake is often made by inexperienced engineers, but professionals do not do this. Always make an individual mix for the artists. For this and invented prefade sends.

You *must* wear the same headphones, connected to the same amplifier, on the same level, as at the artist. Give them an excess of bass and rhythm (the basis of rhythm and harmonies) and listen carefully to the effect of the time machine while listening to the rest of the rhythm. If the vocalist cannot hear the entire mix, he or she will have few reference points to sing against. If the vocalist is in a hurry, raise his voice in rhythm, and vice versa. If it sings too sharp on the top notes and flat on the bottom - lower it (or the whole mix) in the headphones in level - this is the "headphone mix" syndrome that is too loud. The auditory canal of the ear is filled with small hairs, each tuned to its own frequency. If the volume of the signal exceeds the allowable one, those hairs that should not begin to oscillate, which causes a feeling that you are not hitting the notes.

Recording concert space

How to correctly position the microphones when recording the concert space at live performances (for more depth and recording the reaction of the audience) so as to avoid phase problems?

Use two figure-of-eight mics away from the stage, with the dead zone facing the portals. Or cardioid microphones, located similarly. The phase will not create a problem since you want a sense of space, not direct sound from speakers, right?

Your problem (or vice versa - an advantage, depending on how much control you have over this issue) - the choice of delay of ambient microphones by means of their offset from the loudspeakers. The further you take them, the greater the delay in echo effect. I believe that the main signal will be from the "close" microphones of the performers. The way to make the space sound is not difficult - take the ambient microphones as far away as you can. Then turn them so as to squeeze the leakage from the speakers to the maximum. Now you can add some highs to bring out the volume, audience reaction and reverb without the embellished portal signal. I assume that you have instinctively reversed the phase of the amp mics in an attempt to give a "round" sound. This will not give any effect, because. these mics don't have a close-in-phase "neighbor" in your mix.

The overall key to the sound of your live recordings is the placement of the ambient mic. **Remember:** their goal is not to add a little applause, but to recreate the overall size of the picture, the overall balance and vibration of the room. Vibration is directly related to the delay created by distance. **Note:** the sound travels at an approximate speed of 1 foot per 1 millisecond, so if the distance is 50 feet, then the delay will be 50 milliseconds. Installing a figure-eight microphone with a dead zone to the speaker system will increase this time, because. only the signal that arrived from the walls in the form of a reflection will remain unsuppressed.

Result: As always, Becoming... excellent explanation! I have looked at a couple PZM microphones, and since the room is not big (only holds about 75 people), do they need to be set up differently?

Strange choice - PZMs. I Can offer two approaches:

1. Hang them on the wall. They like to sit on large surfaces, picking up the bass. Otherwise, the lack of bass is felt very clearly. They are made to eliminate phase mismatches from direct and reflected sound from the wall. So when you place them on a wall or floor, they are so close to that surface that nothing can bounce off of it towards the microphone. When you just put them on a stand, they work like normal microphones with a strong low cut. Their back side has more suppression than the cardioid.

2. Place them vertically, "back to back", in the middle of the crowd, pointing left and right. This will create an almost perfect sphere for collecting stereo sound. The panorama of the stereo picture at the border of the transition is very smooth. Listening to the signal from them, coupled with portal signal, reverse the phase on both mics and find the position that will have the maximum amount of bass - this can be done, for example, when detuning the sound of a kick or bass guitar. I would also recommend putting a large solid square board between the mics to improve bass response and improve stereo separation.

Both approaches can have an interesting effect. The decisive factor for me when choosing is the location of the listeners relative to the microphones. Ask yourself - which option will give the best ratio of audience and portal sound levels? And which one will give the most uniform distance to each listener? I understand that none of these options have microphones on the “mute” side to the speakers, but only figure-of-eight microphones can do this. In general, each approach has drawbacks.

If the portals are in the center, according to method No. 2, placing 2 PZM microphones in antiphase, you will reduce the leakage from the portals to some extent, while expanding the stereo effect of the audience sound. Warning: this method may shift the sound of the leak from the portals to one side. You will hear all the described effects immediately. Ideally, I'd rather have a pair of assistants dragging the mics in a straight line from the center of the stage to the center of the back wall while I listen to the bass and kick drums in some sort of constant riff. This would give more flexibility in choosing the delay. As they move, I would constantly click the phase button to check the sound for sub-harmonic integrity - the most uplifting, anti-gravity sound. With enough preparation, this process would take minutes! The overall sound changes incredibly as you move.

Reverb before delay

After becoming, I experimented with delays before reverbs. It really opens up new possibilities, and I would say that I didn't pay enough attention to this concept earlier, especially in the case of vocals. I have found that with enough pure stereo delay, very little or no reverb is needed. It would be very interesting to hear your opinion on this matter.

After years of watching my sound disappear into a haze of reverb, I discovered delay. That's right! The smoother the transition from the main sound to the reverb, the more reverb you need. In other words, a soft-tailed plate reverb may require a three-second decay to reveal itself to the right extent. Add a 50ms pre-delay and you only need 2 seconds. In The Art of Mixing, Part III, I explained how uneven reverbs create a more visible effect with less masking. If the reverb has shape and texture, your ear will pick it up easily and not much reverb is needed.

Compression back-to- front

Mike, I read chapter O intelligence "back to front". use whether You at this compression?

Yes, Sometimes. She softens sound, How peanut oil.

Wouldn't this completely change the sound of compression? With this compression, there may be no attack time. Which in itself can make the sound more punchy, right ?

Almost right. The attack becomes a release and the release becomes an attack, but the sound does not become more punchy because she cannot catch the beginning of the sound. The start is now in the tail, so there is no way for you to catch it. The sound is soft, but not punchy. This is similar to the effect of reducing attack with preemption. It's a nice effect, but more like oil than a boxing glove.

If you send the reversed version to another machine with the same EQ settings and everything, you can mix as usual, but enjoy the benefits of reverse information.

Bingo! Exactly! This idea!

Nice see, What at someone Same There is crazy ideas.

Gate back to front

Are there any other cool effects you can get with reverse mixing and processing?

Set the gate and echo effect to the snare drum. How? Send the snare signal to the gate during reverse playback. Set attack to slow and release to fast. Write the result to a free track. After turning everything around again, returning it to normal, listen to the result - notice how the gate slowly opens right before the attack and closes in a rather interesting place. Add echo processing.

Too much loud small drum

I'm mixing a project that combines rock instrumental (drums, bass, guitar) with orchestral elements like clarinet, saxophone, flutes And classical guitars.

faced With problem - small drum too much loud. I Not Can achieve acceptable volume balance.

If the only reason the snare jumps out of the overall sound is because other instruments use voltage less efficiently, that's fine. The lesson here is that next time you give some of the other instruments more depth, more bass, more dynamics, while doing little to affect the overall volume of the recording. Remember that it is the weakest link in a recording that determines the overall volume, and instruments can become over-processed—flattened—if you try to remove all the weak links. Once I understood the principle of the weak link, I quickly learned to find the culprit, which allowed me to achieve more groove space on other sounds. My mixes ended up being rounder than before and with more depth.

It also seems to me that another possible reason for this situation is the probable excessive compression of all other instruments.

Principle "What Not may headphones: monitor on semi"

In the third option, discussed in the chapter "What Headphones Can't", why do you put the monitor on the floor? Because of the number or direction of the microphones upwards?

This is solely due to the fact that the backs of cardioid microphones must look at one point, and the floor is practically the only place where this can be done. From the floor, the sound of the speaker will reach everyone who is located around. Also, the human ear is more sensitive to sounds coming from below. Perhaps due to the fact that our ancestors listened to predators creeping up from below.

Are planning time

How do you plan your work on recording and mixing (and in other necessary areas)? Are there any books that cover this topic from the perspective of a sound engineer?

I once noticed that many clients who come in for a small two-hour session actually take five hours - until I accidentally look at the clock and realize that I spent the rest of the day on the well-known “well, let's A little!" - especially at the editing stage. What will be the ideas?

Before a session, even before it's scheduled, I try to work on scheduling my work time. I have a simple formula that usually works well, something like this: take the studio time that the client has, and divide into three equal parts. The first third is dedicated to recording the arrangement. The second is for composition, and the third is for mixing. Then you can break each third into three more parts, and, having counted the time due to each, ask the client: “What do you think, what will we spend these N hours on?”. This is a key point in the matter of planning - make the client imagine how he will spend this time, because he soon realizes that time is really short. Well, for example, imagine that the first third is given 10 hours, during which you need to record 10 arrangements. The client might say, “That looks good. I think we can record one song an hour." Each song is 5 minutes long, so we have time for 12 takes. At this point you insert: "Don't forget some time on discussion between doubles, to every exactly knew What do".

Client believes What costs add 5 minutes between every takes on detuning, listening, etc. In short, we roll up to 6 takes.

Then you ask if he would like to listen to the recording. you can write everything, but musicians will want to listen to what they have recorded. Here's another 10 minutes to listen to two takes. You may have to listen to some duplicates 2 times. So it is necessary to lay double time for listening. Five minutes to perform, five minutes to listen, five minutes to discuss - a total of 15 minutes per track. Bottom line - in reality, you will get 4 takes per hour, not 12. The client may not like this answer, but you can be sure that he will not be happy about getting out of the temporary control session, which will end in time on mixing. As you understand, the problem is not to start, but to finish. At the beginning of the session, everyone is extremely relaxed - as if they own all the time in the world. Coffee is slowly drunk, people are talking about what they did last night. The engineer takes his time setting up the microphone on the drum. And all this until the moment when, after 16 hours, everyone in a panic starts to fight for every second.

It is correct to ask the client about his idea regarding the time spent on the appointment. How long can you sing without getting tired? But only you can decide the timing of the information. Engineers are different. Some mix fast, some all night. I divide the summary into three parts. The first part is dedicated to the balance of EQ, pans and levels. Second - fully on selection effects And reverbs. AND third - on trendy fader movements. But it happens differently. The main thing is your understanding of what you need. Dividing time into thirds means that you can work in any schedule. If someone asks me, "Can you mix this song in 2 hours?" I'll say, "So, 40 for balance, 40 minutes for reverbs, 40 minutes for faders and getting the final result. And will it be the first take? No! Maybe the second one? Don't think. Usually the fifth. Will I play the first take for clients? Yes. Then I remove clients from the control until I get a mix. Then I call them, reproduce and I write comments. And then I do another take.

Key To result - to plan time So, to This suited And you, And client. We love it when everything turns out easily and naturally, as if by magic, but in order to avoid subsequent disappointment - sketch out a work plan.

breaks on dinner And downtime because of breakdowns

What do you spend the time allotted for rest, and does the client pay for this time? And one more thing - if the studio equipment breaks down, what do you do when it comes time to issue an invoice?

A break is a general rule: yes, I charge for breakfast and lunch. It is never too late to give up this money.

Breakdown - I never charge for downtime due to breakdowns. If I'm working in a studio that might have technical problems, I take a chess watch with me - the one with two buttons, you know. And at the end of work one dial shows the time of work, and the other - the time for breaks due to breakdowns.

Mastering

Mastering engineers always tell me that this mastering should be done by them and only them, but the truth is that more and more people are getting good results. Even if it's a simple demo. I understand that mastering starts well. mix and style... but what is your approach to mastering? Is there a general formula that should be followed? Frequencies to be raised or lowered, Q values, etc. – what is important about this? Is multiband compression necessary for mastering? It would be cool to know your opinion

By tradition, mastering is associated with replication. No copy can compare to the original, and the job of the mastering engineer is to correct the recording in such a way as to minimize losses during duplication.

If You you know What usually does mastering engineer With record, to "lay" it on the final carrier, then you should make your mixes so that these operations are the "finishing touches" in your recording. For example, if you know it will add highs, don't make the recording too bright right away. If you know he's going to put it through a compressor, worry about your compression. It is very important to take this into account. This determines the difference between the results of mastering - the degradation of the recording or its final touches.

Multiband Compression: After years of working with multiband compression, I've come to the conclusion that it's not necessary. In fact, it presents your mix as if through a stained-glass window - like a three-way speaker system with a crossover. If a normal compressor does not compress your track properly At least a multi-band compressor will do it. In fact, multi-channel compression was invented For broadcasting, to achieve maximum modulation on all frequencies. But How And any another tool, she Maybe be used as for good and for evil.

Trick question: I always point out a few key points in a mastering engineer's signal chain that reveal his understanding of compression. When two compressor installed sequentially, compressor With higher ratio should be "upstream" to protect the next compressor from overcompression. Usually the opposite is true, the limiter is the last in the chain. It is better for him to be at the beginning - try the opposite and compare.

side chain (For what And When)

The chapter “Opening the Compressors” seemed to open my eyes, but I still feel like I can’t squeeze everything I want out of my side-chain compressor.

If you've ever used a noise gate's control input, you'll recognize the resemblance to midrange compression. Unlike a noise gate, which rarely uses another instrument to control it, a side chain can use the same instrument while still using radically tuned equalization. The main advantage of the midrange compressor is the ability to make the compressor “hear” a completely different sound due to the equalizer - not the one you hear. The most common example is the creation of a de-esser to reduce the imbalance in highs. My Favorite Disser Effect

is obtained by limiting the sidechain with a specially tuned equalizer for the voice, and not by passing the voice through a multi-band compressor that cuts only high. multiband compressor, every once at pronunciation "WITH", "F" or
"T" makes the voice muddy for a moment. When you use a sidechain, it lowers the level of the entire voice at once, and it doesn't become muddy. The voice remains full, it simply steps back when the "S" is pronounced. This can be achieved by sending to the manager entrance specially equalized version vote. IN case excess high side-chain trigger is suitable for any frequency except 14kHz.

Secret scheme: cut everything below 1 or 2kHz, then max out a thin band at 14kHz, and send that signal to the limiter's sidechain input. Set the minimum attack, release around 0.05 seconds (usually a click slower than the fastest release), and a ratio of 20:1 (or just maximum). Adjust the amount of limiting with threshold. Now the compression works on "F" and "T" (not only on "C"), but on nothing else. Lower the threshold and it will only catch "C". Once you've got a crisp response, you can drop the ratio down to 10:1 if you feel like it.

Other interesting sidechain applications: in theory, you can use other tools to manage the side chain. You can use this to mute the entire mix as the kick kicks.